

# Discontinuous Galerkin formulations for thin bending problems



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PROEFSCHRIFT

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*To my family*



# Foreword

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Nguyen Tien Dung  
Delft, May 2008



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# Chapter 1

## Introduction

Thin bending problems arise when structures with one dimension, the thickness, relatively small compared to the other dimensions, undergo bending. Examples for these types of structures are copious. They range from natural constructions, such as fragile egg shells, shells of sea species and tree leaves, to man-made constructions, such as house roofs, spacious vaults and aircraft bodies; see Figure 1.1 for an illustration.

Thin structures have numerous attractive properties. Their functional flexibility with diverse shapes is valued and often exploited by architects. When designed properly, thin structures can have a high load-carrying capacity, passing over a large span while requiring an economical amount of material. Other properties, such as their light weight and simplicity in shape-forming processes, are also important.



(a)



(b)

Figure 1.1: Optimisations of nature and men: (a) An extant leaf-mimic insect, the family Phylliidae; (b) An Airbus A380, the current world's largest passenger airliner. Source: <http://wikimedia.org>.

Over the years, a large number of thin structures have been constructed, and so many efforts have been made to analyse their behaviour. It is well-known that analyses of thin structures are important and challenging, especially for thin bending problems.

## 1.1 Overview

In structural mechanics, the classification of thin structures is based on the shape of their *mid-surfaces*, which are the surfaces that halve the thickness dimension of the structures. A structure belongs to the beam class when one dimension of its mid-surface is small compared to another. It is a plate when the mid-surface is flat and it is a shell if the mid-surface is curved.

This section gives an overview of significant approaches for solving thin bending problems.

### 1.1.1 Beams

A straight and thin cantilever subjected to gravity loading might be the simplest case of thin bending problems. This problem is known as the Galileo's enquiry from the seventeenth century, see the treatise of Love (1934) for an interesting historical introduction. The best-known and simplest model is based on classical, or Bernoulli-Euler, beam theory. A key assumption in the theory is *normality* of the cross section. It implies that a plane section, initially normal to the beam axis, remains plane and normal to the axis. See Figure 1.2 for an illustration.

Adopting the above-mentioned assumption, the rotation  $\phi$  of a plane normal to the axis coincides with the first derivative (or slope) of the deflection  $u$  with respect to the coordinate  $s$  of the plane,  $\phi = du/ds$ . After some manipulations, the bending problem can be described by a fourth-order differential governing equation in terms of the deflection. To solve the equation, it can be transformed to a variational problem that involves only second-order derivatives of the displacement. The displacement field is said to belong the  $H^2$ , or  $C^1$  continuity, space that means both the displacement and its first derivative are continuous. Here,  $H^2$  denotes a standard Sobolev space of order two, which will be defined precisely in the next chapter.

When the finite element method is used for beam bending problems, conventionally Hermitian shape functions are applied. A conforming  $C^1$  continuity

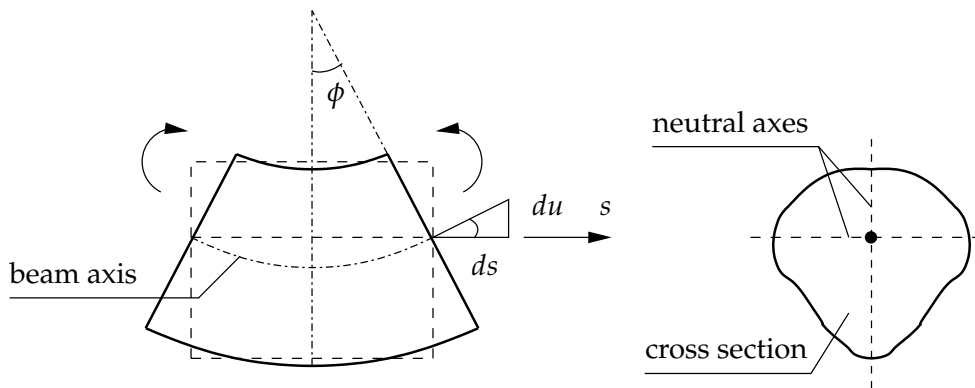


Figure 1.2: Beam bending.

across element boundaries can be easily obtained since both nodal displacement and rotation degrees-of-freedom involve. This approach has become standard in finite element analysis of beam bending problems (Zienkiewicz, 1977; Reddy, 1984; Hughes, 2000).

Another well-known model for beam bending problems is based on the Timoshenko theory. The assumption of normality is now relaxed. The formulation can deal with shear deformable problems by introducing the displacement and the rotation as independent fields. Using this theory,  $C^0$  finite elements can be applied and low-order polynomials can be used. However, when the thickness becomes small, there appears the phenomenon of shear locking. The beam becomes spuriously stiff and the finite element solution is no longer reliable. Selective integration methods are often applied to remedy the shear locking problem, see Zienkiewicz (1977), Hughes et al. (1977) and Cook (1981) for further discussions.

### 1.1.2 Plates and shells

Compared to the case of beams, thin plate and shell problems are more interesting but also more difficult. Within the concept of the finite element method, formulations for thin plates and shells can be built upon on various approaches using a number of different finite elements.

### Classical $C^1$ finite elements for bending

Since the work of Kirchhoff (1897) for thin plates and the extension by Love (1934) to thin shells were introduced, various finite element formulations have been proposed and developed. There exist a number of formulations which have been built on the Kirchhoff-Love assumption. This assumption for thin plates and shells is extended from the assumption for Bernoulli-Euler beams and includes a bi-axial bending. It states that a material line, initially straight and normal to the mid-surface, remains straight and normal to the mid-surface during deformation. Similar to the case of thin beam bending, thin plate and shell bending problems are associated with fourth-order partial differential equations. Following the finite element concept, a weak form involving the displacement functions can be derived. This demands the use of a  $C^1$  conforming basis. Both the displacements and their first-derivatives are continuous while the second-derivatives (curvatures) are square integrable.

Unlike the case of beam elements,  $C^1$  conforming elements for thin plate and shell problems are not easy to construct. It is known that elements are  $C^1$  compatible across an inter-element edge if the displacement itself and its first normal derivative over the edge are completely specified by degrees-of-freedom on that edge (Felippa, 2004). Many efforts have been spent to construct  $C^1$  conforming elements involving displacement and rotation degrees-of-freedom, but these efforts have been limited in success. In a short but important paper by Irons and Draper (1965), it was shown that it is impossible to construct  $C^1$  conforming plate elements by using degrees-of-freedom with nodal displacements and rotations only. Some elements have been introduced in the literature that fulfill  $C^1$  continuity requirements, but they involve the use of second-order derivatives as degrees-of-freedom. These elements are complicated to construct and may introduce excessive continuity. The reader can refer to a number of works, including those by Bernadou et al. (1992), Bernadou (1996) and Chapelle and Bathe (2003), for more discussions.

### $C^0$ finite elements

An approach for solving difficulties involving continuity requirements is to consider the Reissner-Mindlin theory that allows an occurrence of transverse shear deformations (Mindlin and Cooper, 1960; Mindlin, 2006). Finite element formulations involve the displacement and rotation degrees-of-freedom as indepen-

dent unknowns.  $C^0$  elements therefore can be applied. The use of these elements has gained favour since they allow the use of simple basis functions. These elements however exhibit other difficulties, primarily related to shear locking and membrane locking in the thin bending limit. Numerous works have proposed remedies to the locking problems, including the use of high-order polynomials or selective integration (Zienkiewicz et al., 1977; Hughes and Tezduyar, 1981; Brezzi and Fortin, 1991). Furthermore, various shell formulations and finite elements have been proposed to overcome the locking phenomena as well as to deal with a range of problems such as large strain elasticity, taking thickness change into account and applying directly three-dimensional constitutive laws to thin plate and shell problems. Successful approaches include assumed strain methods (Bathe and Dvorkin, 1986; Park and Stanley, 1986; Park et al., 1995), enhanced strain methods (Simo et al., 1990; Büchter et al., 1994) and mixed and hybrid methods (Chroscielewski et al., 1992; Sansour and Bufler, 1992; Sansour and Bednarczyk, 1995). The reader can refer to the works of Başar and Ding (1997), Bischoff and Ramm (2000), and Sansour and Kollmann (2000) for more discussions.

### **Spline-based finite elements**

More recently, thin plate and shell bending theories have been revisited with finite element formulations which utilise displacement-only degrees-of-freedom to build  $C^1$  continuity. These methods typically introduce a degree of element non-locality to reconstruct a basis with the necessary regularity. An approach based on subdivision surfaces was developed in the work of Cirak et al. (2000) to describe the undeformed geometry and generate a  $H^2$  displacement field for thin shell problems. In the element-free Galerkin (EFG) method, the satisfaction of  $C^1$  continuity requirements are met by using a non-local test function and the moving least-squares technique (Krysl and Belytschko, 1995; Noguchi et al., 2000). Hughes et al. (2005) proposed the non-uniform rational B-splines (NURBS) method in which basis functions with an arbitrary degree of continuity can be generated easily without the introduction of rotation degrees-of-freedom.

### **Rotation free finite elements for bending**

Another novel approach is the use of low-order rotation-free finite elements for thin plate and shell bending problems. A sample of which can be found in Phaal

and Calladine (1992), Oñate and Zarate (2000) and Oñate and Flores (2005) and a review of a number of formulations can be found in Gärdsback and Tibert (2007).

### **Classical non-conforming finite elements for bending**

A promising avenue is the use of non-conforming elements within the concept of discontinuous Galerkin methods in order to relax the continuity requirements and maintain consistency of finite elements. Babuška and Zlámal (1973) presented a penalty formulation to solve the biharmonic equation using standard  $C^0$  basis functions and imposing weak continuity of the normal derivative across element boundaries. However, as with conventional penalty methods in general, consistency of the formulation is not satisfied. Generalised variational principles, applied to nonlinear elasticity, were presented in the work of Bufler (1979) and different versions of the work theorems were unified later (Bufler, 1983). In these works, continuity requirements of admissible functions across external and internal boundaries can be relaxed while consistency of the formulation is satisfied. Building on the *interior penalty* method for second-order elliptic problems to restore consistency (see Ref. Arnold et al. (2002) for an overview), Engel et al. (2002) presented a  $C^0$  interior-penalty formulation for Kirchhoff plates in which standard  $C^0$  Lagrange finite element basis functions are used. Several element boundary integrals are introduced to the variational formulation to weakly enforce continuity of normal derivatives across element boundaries, while preserving consistency and maintaining stability. The method performed well for both Bernoulli-Euler beam and Kirchhoff plate bending problems using displacement unknowns only. A drawback of the interior penalty method is the presence of a dimensionless penalty parameter which must be chosen suitably large to guarantee stability. The difficulty lies in how large the term must be as it cannot be easily quantified *a priori*.

## **1.2 Aims, scope and approach**

The aim of this work is to develop discontinuous Galerkin formulations for thin bending problems, with the focus on challenging Kirchhoff plate and Koiter shell models with similar kinematic assumptions, both in geometrically linear and geometrically nonlinear regimes. We shall restrict our consideration to elastic problems.

Formulations shall utilise standard  $C^0$  Lagrange finite element basis functions, with displacement degrees-of-freedom only. In order to obtain an accurate, robust, simple, and efficient formulation, the proposed finite elements are required to be:

- *Weakly compatible*, weak  $C^1$  continuity across element edges is enforced;
- *Consistent*, they are able to capture the analytical solution in the limit of finite element refinements; and
- *Stable*, ideally the stability condition can be precisely quantified.

These properties are essential for the convergence of the formulations.

Our approach builds upon recent developments for discontinuous Galerkin methods for elliptic problems.

## 1.3 Outline

The content of each chapter is briefly described as follows. Chapter 2 serves as an introduction to discontinuous Galerkin methods. The introduction is given by means of the application of discontinuous Galerkin methods for elasticity problems. Some preliminary definitions within the discontinuous Galerkin concept are introduced and a framework to derive discontinuous Galerkin formulations is presented for the purpose of later use. We shall focus on some particular methods which provide a basis for our work in the following chapters.

A discontinuous Galerkin finite element formulation for Kirchhoff plates is proposed in Chapter 3. The formulation exploits conventional  $C^0$  finite element basis functions with respect to displacement unknowns. The formulation is analysed in terms of stability and convergence.

In Chapters 4 to 6, thin shells are considered. The governing equations for the geometrically nonlinear Koiter shell model are derived in Chapter 4. The variational formulation for the considered problems appears in the form of the virtual work in the reference configuration and in terms of the Green-Lagrange strain and the second Piola-Kirchhoff stress. A  $C^0$  finite element formulation for linear Koiter shell models is presented in Chapter 5. The formulation is an extension of the discontinuous Galerkin formulation presented in Chapter 3 for Kirchhoff plates to the case of curved surfaces. In Chapter 6, a geometrically nonlinear

formulation for thin shells without rotation degrees-of-freedom is presented. An exact linearisation of the proposed formulation is employed. Numerical results are presented for a range of benchmark problems.

The final chapter, Chapter 7, presents conclusions of this work.

## **1.4 Notation**

We will mostly use tensor notation in order to write formulations in compact forms. In some parts involving implementation aspects, matrices and vectors may also be used. Indices are used extensively in the presentation, with Latin indices running from one to three and Greek indices running from one to two. Frequently, Einstein summation convention is employed. When both covariant and contravariant bases exist, subscripts denote covariant components and superscripts denote contravariant components.

## Chapter 2

# Discontinuous Galerkin methods for elasticity

### 2.1 Background

A Galerkin method is a method for finding approximate solution to differential equations. The method is built on a weak form of the considered problem. An approximate solution is given by a linear combination of functions from a chosen basis. Depending on the particular choice of the basis functions, a class of Galerkin methods is defined. In the conventional Galerkin method (Bubnov-Galerkin method), test functions are chosen from the space of the approximation solution. In contrast, when the test and trial functions are not taken from the same basis functions, the method is known as a weighted-residual method or Petrov-Galerkin method. Galerkin methods in fact form the foundation of the finite element method. The reader can refer to many text books, including the works by Zienkiewicz (1974), Ciarlet (1978), Reddy (1984) and Hughes (2000), for more details.

A discontinuous Galerkin method involves piecewise discontinuous functions for constructing the approximation solution of the considered problem. Within the finite element concept, discontinuities arising across element boundaries are controlled in order to capture closely the exact solution. In other words, the method allows the continuity requirements across element boundaries to be relaxed. Figure 2.1 illustrates finite elements for a bending problem using the discontinuous Galerkin approach.

Over the years, many discontinuous Galerkin methods have been proposed

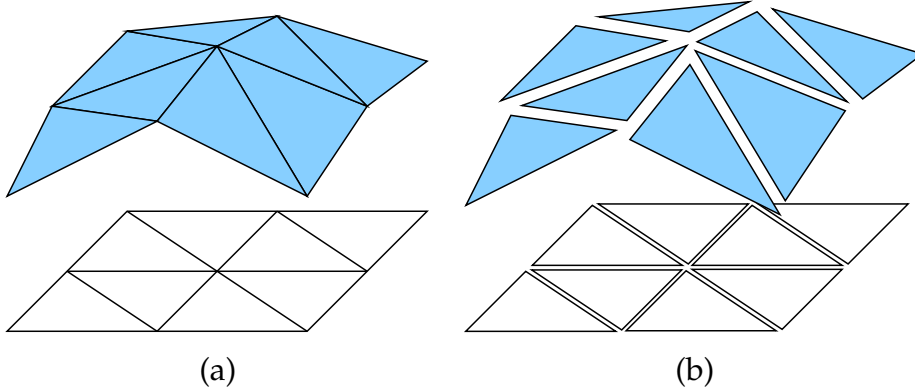


Figure 2.1: Illustration of finite elements for a bending problem: (a)  $C^0$  continuous elements; (b) Purely discontinuous elements.

and developed for solving numerous problems of mechanics. There exist two main directions in the early development of discontinuous Galerkin methods. First, that is the development of interior penalty methods for parabolic and elliptic problems. Second, that is the development of discontinuous Galerkin methods for hyperbolic problems. Recently, these methods have been unified. An historical overview can be found in a number of works, including those by Cockburn (2001) and Arnold et al. (2002).

### 2.1.1 Interior penalty methods

In the late 1960s, works on penalty methods for parabolic and elliptic problems were introduced. In these works, the Dirichlet boundary conditions were weakly imposed by introducing integrals on the exterior boundary. A number, so-called penalty parameter, was used as a stabilisation parameter. See the work of Arnold et al. (2002) for more details.

Babuška and Zlámal (1973) developed a penalty method to deal with interior element boundaries for plate bending problems. The method built upon nonconforming finite elements with piecewise cubic polynomials. The method is stable, however, it lacks consistency. For the biharmonic equation, Baker (1977) introduced a finite element method that is stable and consistent. The formulation is built on  $L^2$  finite element spaces, and weak  $C^1$  continuity across element boundaries is imposed weakly. This method belongs to the class of interior penalty methods. The interior penalty methods were also presented for second-order elliptic and parabolic problems in a number of works, including those of Dou-

glas and Dupont (1976) and Wheeler (1978). Arnold (1982) analysed on interior penalty methods and applied them to nonlinear parabolic problems. Investigations into the interior penalty approach were then not carried forward.

### 2.1.2 Discontinuous Galerkin methods for hyperbolic problems

A number of efforts have been directed to the development of discontinuous Galerkin methods for hyperbolic problems. The first work in this direction was by Reed and Hill (1973) for solving the linear equations of neutron transportation. In this work, the space of the approximation solution was constructed element-wise and a *numerical flux* was defined on element boundaries. Later, Jamet (1978) introduced a formulation for linear parabolic equations in which the discontinuity in time was considered. The approach was then developed for nonlinear hyperbolic systems to construct high-order accurate methods (Shu and Osher, 1988; Cockburn and Shu, 1989; Cockburn et al., 1989).

In 1997, a significant work was presented by Bassi and Rebay for convection-diffusion equations. In this approach, the second-order hyperbolic equation was written as a system of two first-order equations. Then, a discontinuous Galerkin discretisation technique was applied for both convection and diffusion parts. The method was generalised for nonlinear time-dependent convection-diffusion problems in the work by Cockburn and Shu (1998), which is called the local discontinuous Galerkin (LDG) method. Some developments of this method for second-order problems were presented by Baumann and Oden (1999), Brezzi et al. (2000) and Bassi and Rebay (2002).

### 2.1.3 Recent developments

Some efforts have been made to unify families of discontinuous Galerkin methods, including the works by Cockburn et al. (2000), Cockburn (2001) and Arnold et al. (2002). It has been shown that a critical point in deriving discontinuous Galerkin formulations is the choice of numerical fluxes. These fluxes initially appear in the concept of finite volume methods are then employed in finite element processes. Discontinuous Galerkin methods can therefore be considered to unify finite volume and finite element methods. The interior penalty method is a special case of the discontinuous Galerkin methods by choosing proper numerical fluxes.

Inspired by some new techniques from discontinuous Galerkin methods for hyperbolic problems, interior penalty methods recently have been revisited with great attention. Numerous formulations have been proposed and developed for a range of hyperbolic, parabolic and elliptic problems, including fourth-order problems (Engel et al., 2002; Brenner and Sung, 2005; Mozolevski et al., 2006), elliptic problems in non-convex polygonal domains (Wihler, 2002), strain gradient-dependent damage (Wells et al., 2004; Molari et al., 2006), wet-chemical etching (Sudirham, 2005), the Cahn-Hilliard equation for phase separation (Wells et al., 2006), the flow of fluids with different thermo-dynamical properties (Pesch, 2007).

## 2.2 Elasticity problem

Consider a body  $\Omega$  in the space of dimension  $d$  with the boundary  $\partial\Omega$  (see Figure 2.2). The boundary is partitioned into the Dirichlet boundary  $\Gamma^u$  and the Neumann boundary  $\Gamma^h$ , such that  $\overline{\Gamma^u \cup \Gamma^h} = \partial\Omega$  and  $\Gamma^u \cap \Gamma^h = \emptyset$ . We denote the components of the prescribed body force by  $f_i$ , the unknown displacement field by  $u_i$ , and the unknown stress field by  $\sigma_{ij}$ ,  $i, j = 1 \rightarrow d$ . Components of the prescribed displacement on the displacement boundary  $\Gamma^u$  are denoted by  $g_i^u$  and components of the prescribed traction on the force boundary  $\Gamma^h$  are denoted by  $g_i^h$ . Note that forces and displacements are vector quantities, and stresses and strains are second-order tensor quantities of dimension  $d$ . The strain-stress relation is assumed to be linear. We consider the following boundary value problem: find  $u_i$  such that

$$-\sigma_{ij,j} = f_i \quad \text{in } \Omega, \quad (2.1)$$

$$\sigma_{ij} = c_{ijkl} u_{(k,l)} \quad \text{in } \Omega, \quad (2.2)$$

$$u_i = g_i^u \quad \text{on } \Gamma^u, \quad (2.3)$$

$$\sigma_{ij} n_j = g_i^h \quad \text{on } \Gamma^h, \quad (2.4)$$

where  $c_{ijkl}$  is a fourth-order tensor which describes the linear relationship between the stress and strain tensors, and  $n_j$  is the outward unit normal vector to  $\partial\Omega$ . Here, following the notation presented by Hughes (2000), we have denoted  $u_{(k,l)}$  as the symmetric part of the second-order tensor  $u_{k,l}$ ,  $u_{(k,l)} = (u_{k,l} + u_{l,k}) / 2$ .

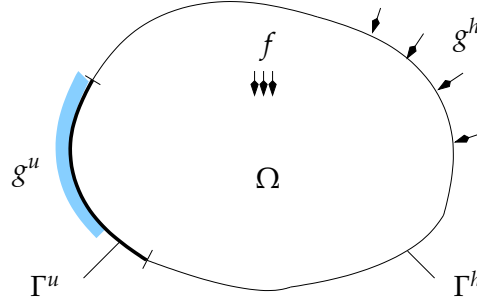


Figure 2.2: Elasticity problem

## 2.3 Finite element basis

We recall here the standard definition of Sobolev spaces, which are important to construct finite elements. A Sobolev space  $H^m(\Omega)$  of order  $m$  is the space consisting of functions in  $\Omega$  that themselves together all their weak partial derivatives up to and including those of order  $m$  belong to the square-integrable space  $L^2(\Omega)$  (Reddy, 1998). Mathematically, it is defined by

$$H^m(\Omega) = \left\{ \varphi \in L^2(\Omega) : \partial^\alpha \varphi \in L^2(\Omega), \forall \alpha \text{ such that } |\alpha| \leq m \right\}, \quad (2.5)$$

where  $m$  is a non-negative integer and  $\alpha$  is a multi-index number that represents non-negative integers  $\alpha_i$ ,  $i = 1 \rightarrow d$ . Here,  $|\alpha|$  denotes the length of  $\alpha$ ,  $|\alpha| = \sum_{i=1}^d \alpha_i$ . Note that when  $m = 0$  hence  $H^0(\Omega) \equiv L^2(\Omega)$ . Whereas, the notation  $H_0^m(\Omega)$  is used when a homogeneous zero boundary condition is employed.

Sobolev spaces are equipped with Sobolev norms and semi-norms as follows: for  $m = 0$ , the norm (which is often called as  $L^2$  norm) is defined by

$$\|\varphi\|_{0,\Omega} = \left( \int_{\Omega} \varphi^2 d\Omega \right)^{\frac{1}{2}}, \quad (2.6)$$

and for  $m > 0$ , the norm is defined by

$$\|\varphi\|_{m,\Omega} = \left( \sum_{|\alpha| \leq m} \|\partial^\alpha \varphi\|_{0,\Omega}^2 \right)^{\frac{1}{2}} \quad (2.7)$$

and the semi-norm is defined by

$$|\varphi|_{m,\Omega} = \left( \sum_{|\alpha|=m} \|\partial^\alpha \varphi\|_{0,\Omega}^2 \right)^{\frac{1}{2}}. \quad (2.8)$$

Consider a partition  $\mathcal{P}^h$  of the domain  $\Omega$  into  $N$  elements  $K_i$ , with element boundaries  $\partial K_i$ ,  $i = 1 \rightarrow N$ . Let  $\bar{K}_i$  and  $\bar{\Omega}$  denote the closures of  $K_i$  and  $\Omega$ , respectively, and  $\tilde{\Omega}$  denotes the interior discretised domain of elements. Here,  $\bar{K}_i = K_i \cup \partial K_i$ ,  $\bar{\Omega} = \Omega \cup \partial\Omega$ ,  $\bigcup_{i=1}^N \bar{K}_i = \bar{\Omega}$ , and  $\bigcup_{i=1}^N K_i = \tilde{\Omega}$ . Note that  $\tilde{\Omega}$  does not contain element edges. We denote the union of all element boundaries by  $\Gamma$ ,  $\Gamma = \bigcup_{i=1}^N \partial K_i$ , and the interior part of this union by  $\tilde{\Gamma}$ ,  $\tilde{\Gamma} = \Gamma \setminus \partial\Omega$ . Hereinafter, we use the superscript  $h$  to denote the discretised (approximate) counterpart of a quantity or field. For the sake of simplicity, we restrict  $\Omega$  to be a polyhedral domain. Therefore, we can assume that discretised boundaries can represent the original boundaries exactly. Mathematically, the partition  $\mathcal{P}^h$  is defined as follows

$$\mathcal{P}^h := \left\{ K_i \mid \bigcup_{i=1}^N K_i = \tilde{\Omega} \text{ and } K_i \cap K_j = \emptyset \text{ if } i \neq j, 1 \leq i, j \leq N \right\}. \quad (2.9)$$

The displacement  $u_i$  and stress  $\sigma_{ij}$  are approximated by polynomial functions defined on the elements  $K \in \mathcal{P}^h$ . We denote the space of polynomial functions of degree  $k$  on  $K$  by  $P^k(K)$ . It is required that  $k \geq 1$  in order to validate the approximation functions of unknowns in equations (2.1) to (2.2). The finite element spaces are defined as

$$V^h := \left\{ v_i^h \in L^2(\Omega) : v_i^h|_K \in P^k(K), \forall K \in \mathcal{P}^h \right\}, \quad (2.10)$$

$$\Sigma^h := \left\{ \sigma_{ij}^h \in L^2(\Omega) : \sigma_{ij}^h|_K \in P^k(K), \sigma_{ij}^h = \sigma_{ji}^h, \forall K \in \mathcal{P}^h \right\}. \quad (2.11)$$

## 2.4 Jump and average operations

Particular operations are introduced here since the finite element spaces are discontinuous. We consider a single element  $K \in \mathcal{P}^h$  with the outward normal vector  $\mathbf{n}^K$  to the element boundary  $\partial K$ . The trace on  $\partial K$  of a function  $\varphi$  which is defined on  $K$  reads

$$\varphi^K = \lim_{0 \leftarrow \epsilon} \varphi(x - \epsilon \mathbf{n}^K), \quad (2.12)$$

where  $x$  denotes the position vector of a point on  $\partial K$ , and  $\epsilon$  is a small positive value. Two neighbouring elements  $K^+$  and  $K^-$  with a sharing edge  $e$  are shown in Figure 2.3. As the function  $\varphi$  can be discontinuous across element boundaries, there exist two values denoted by  $\varphi^\pm$  of the function  $\varphi$  on  $e$ . For an edge  $e$  on the interior boundary  $\tilde{\Gamma}$ , we define the jump  $[\![\cdot]\!]$  and the average  $\langle \cdot \rangle$  operators for a

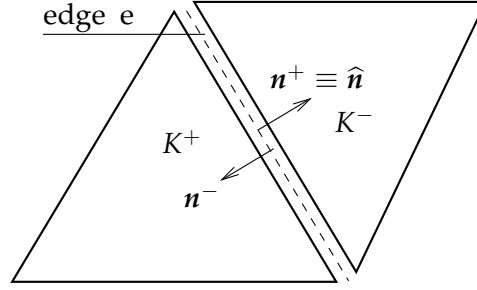


Figure 2.3: Two neighbouring elements and their sharing edge

scalar function  $\varphi$  as follows

$$[\![\varphi]\!] = \varphi^+ - \varphi^-, \quad (2.13)$$

$$\langle \varphi \rangle = \frac{1}{2} (\varphi^+ + \varphi^-). \quad (2.14)$$

For an edge  $e$  on the exterior boundary  $\partial\Omega$ , the jump and average are defined as

$$[\![\varphi]\!] = \langle \varphi \rangle = \varphi. \quad (2.15)$$

We introduce a sign function  $\Xi(e, K)$  on  $e \cap \partial K$  by

$$\Xi = \begin{cases} \Xi^+ = 1 & e \in \tilde{\Gamma}, K = K^+, \\ \Xi^- = -1 & e \in \tilde{\Gamma}, K = K^-, \\ \Xi^+ = 1 & e \in \partial\Omega. \end{cases} \quad (2.16)$$

It is straightforward to prove that, for any scalar functions  $\varphi$  and  $\phi$  in  $L^2(\partial K)$ , the following identity holds true

$$\sum_{K \in \mathcal{P}^h} \int_{\partial K} \varphi \phi \, ds = \int_{\Gamma} \langle \varphi \Xi \rangle [\![\phi]\!] \, d\Gamma + \int_{\tilde{\Gamma}} [\![\varphi \Xi]\!] \langle \phi \rangle \, d\Gamma. \quad (2.17)$$

Applying the above relation to the case shown in Figure 2.3 when  $n_i^+ = -n_i^-$  thus  $n_i^+ \Xi^+ = n_i^- \Xi^- = \hat{n}_i$ , we have

$$\sum_{K \in \mathcal{P}^h} \int_{\partial K} \varphi \phi n_i^K \, ds = \int_{\Gamma} \langle \varphi \rangle [\![\phi]\!] \hat{n}_i \, d\Gamma + \int_{\tilde{\Gamma}} [\![\varphi]\!] \langle \phi \rangle \hat{n}_i \, d\Gamma, \quad (2.18)$$

where  $\hat{n}_i$  are components of the sharing normal vector  $\hat{\mathbf{n}}$  on the edge  $e \in \Gamma$  (see Figure 2.3). The above operations, defined for a scalar, can be applied to components of vector or tensor quantities to define the average and jump for these quantities. The identities (2.17) and (2.18) are utilised throughout the thesis.

## 2.5 Discontinuous Galerkin weak formulation

Following the framework presented by Arnold et al. (2002), a discontinuous Galerkin weak form for the problem in equations (2.1) to (2.4) is derived. Properties of the formulation are then discussed.

### 2.5.1 Derivation of weak formulation

#### Balance of momentum

We first consider the equilibrium equation (2.1). Within finite element concept, the stress tensor  $\sigma_{ij}$  is replaced by the approximation solution  $\sigma_{ij}^h \in \Sigma^h$ . Multiplication of both sides of equation (2.1) by a test functions  $w_i^h \in V^h$  and integration over a single element  $K$  lead to

$$-\int_K w_i^h \sigma_{ij,j}^h d\Omega = \int_K w_i^h f_i d\Omega \quad \forall w_i^h \in V^h. \quad (2.19)$$

Integration by parts of the first term of the above equation results in

$$\int_K w_{(i,j)}^h \sigma_{ij}^h d\Omega - \int_{\partial K} w_i^h \sigma_{ij}^h n_j^K ds = \int_K w_i^h f_i d\Omega \quad \forall w_i^h \in V^h, \quad (2.20)$$

where  $n_j^K$  are components of the outward unit normal vector to the boundary  $\partial K$  of element  $K$ .

A concept in discontinuous Galerkin methods is adopted here: the unknown  $\sigma_{ij}^h$  on the boundary integral is now replaced by  $\hat{\sigma}_{ij}^h$  which is a stress numerical flux. It represents a numerical stress of the stress  $\sigma_{ij}^h$  on both sides of the sharing edge. The definition for  $\hat{\sigma}_{ij}^h$  will be introduced later in more details. Now, we have

$$\int_K w_{(i,j)}^h \sigma_{ij}^h d\Omega - \int_{\partial K} w_i^h \hat{\sigma}_{ij}^h n_j^K ds = \int_K w_i^h f_i d\Omega \quad \forall w_i^h \in V^h, \quad (2.21)$$

which is the weak formulation of the balance of momentum for a single element.

#### Constitutive equation

The constitutive equation (2.2) is now considered. The weak form of this equation, obtained by multiplying both sides of equation (2.2) by a test function  $\tau_{ij}^h \in \Sigma^h$  and integrating by parts over  $K$ , reads as follows

$$\int_K \tau_{ij}^h \sigma_{ij}^h d\Omega = - \int_K \left\{ \tau_{ij}^h c_{ijkl} \right\}_{,l} u_k^h d\Omega + \int_{\partial K} \tau_{ij}^h c_{ijkl} \hat{u}_k^h n_l^K ds \quad \forall \tau_{ij}^h \in \Sigma^h. \quad (2.22)$$

Here,  $u_k^h$  is the approximation displacement,  $u_k^h \in V^h$ ,  $\hat{u}_k^h$  is a displacement numerical flux which will be defined later on the element boundary. Integration by parts for the second term of equation (2.22) leads to

$$\int_K \tau_{ij}^h \sigma_{ij}^h d\Omega = \int_K \tau_{ij}^h c_{ijkl} u_{(k,l)}^h d\Omega - \int_{\partial K} \tau_{ij}^h c_{ijkl} (u_k^h - \hat{u}_k^h) n_l^K ds \quad \forall \tau_{ij}^h \in \Sigma^h. \quad (2.23)$$

**Remark 2.5.1** To derive equation (2.23), integration by parts has been used twice. The first time is to obtain the weak form (2.22) within the concept of discontinuous Galerkin methods. The second time is in equation (2.23) with a purely mathematical meaning. These two steps produce a contribution of the difference between  $u_k^h$  and  $\hat{u}_k^h$  on  $\partial K$ , to the formulation. When this difference is zero, the formulation (2.23) returns the weak form of the constitutive relation for continuous Galerkin formulations. In following chapters, we will use these two steps to derive discontinuous Galerkin formulations for considered problems. Note that  $\hat{u}_k^h$  and  $\hat{\sigma}_{ij}^h$  have not been defined yet. ■

### Weak formulation

Our goal is to build a weak formulation which is expressed in terms of displacement components. To obtain this, the constitutive relation in equation (2.23) is adopted in equation (2.21). As  $w_{(i,j)}^h \in \Sigma^h$ ,  $\tau_{ij}^h$  can be replaced by  $w_{(i,j)}^h$  in equation (2.23). Substituting equation (2.23) into equation (2.21) yields

$$\begin{aligned} \int_K w_{(i,j)}^h c_{ijkl} u_{(k,l)}^h d\Omega - \int_{\partial K} w_{(i,j)}^h c_{ijkl} (u_k^h - \hat{u}_k^h) n_l^K ds \\ - \int_{\partial K} w_i^h \hat{\sigma}_{ij}^h n_j^K ds = \int_K w_i^h f_i d\Omega \quad \forall w_i^h \in V^h. \end{aligned} \quad (2.24)$$

Summing equations (2.23) and (2.24) over all elements  $K \in \mathcal{P}_h$ , we obtain

$$\begin{aligned} \int_{\tilde{\Omega}} \tau_{ij}^h \sigma_{ij}^h d\Omega = \int_{\tilde{\Omega}} \tau_{ij}^h c_{ijkl} u_{(k,l)}^h d\Omega - \sum_{K \in \mathcal{P}_h} \int_{\partial K} \tau_{ij}^h c_{ijkl} (u_k^h - \hat{u}_k^h) n_l^K ds \\ \forall \tau_{ij}^h \in \Sigma^h \end{aligned} \quad (2.25)$$

and

$$\begin{aligned} \int_{\tilde{\Omega}} w_{(i,j)}^h c_{ijkl} u_{(k,l)}^h d\Omega - \sum_{K \in \mathcal{P}_h} \int_{\partial K} w_{(i,j)}^h c_{ijkl} (u_k^h - \hat{u}_k^h) n_l^K ds \\ - \sum_{K \in \mathcal{P}_h} \int_{\partial K} w_i^h \hat{\sigma}_{ij}^h n_j^K ds = \int_{\tilde{\Omega}} w_i^h f_i d\Omega \quad \forall w_i^h \in V^h. \end{aligned} \quad (2.26)$$

Taking average and jump operators (2.14) to (2.15) and the identity (2.18) into account, we can rewrite equations (2.25) and (2.26) as

$$\begin{aligned} \int_{\tilde{\Omega}} \tau_{ij}^h \sigma_{ij}^h d\Omega &= \int_{\tilde{\Omega}} \tau_{ij}^h c_{ijkl} u_{(k,l)}^h d\Omega - \int_{\Gamma} \langle \tau_{ij}^h c_{ijkl} \rangle \left[ u_k^h - \hat{u}_k^h \right] \hat{n}_l d\Gamma \\ &\quad - \int_{\tilde{\Gamma}} \left[ \tau_{ij}^h c_{ijkl} \right] \langle u_k^h - \hat{u}_k^h \rangle \hat{n}_l d\Gamma \quad \forall \tau_{ij}^h \in \Sigma^h \end{aligned} \quad (2.27)$$

and

$$\begin{aligned} \int_{\tilde{\Omega}} w_{(i,j)}^h c_{ijkl} u_{(k,l)}^h d\Omega &- \int_{\Gamma} \langle w_{(i,j)}^h c_{ijkl} \rangle \left[ u_k^h - \hat{u}_k^h \right] \hat{n}_l d\Gamma \\ &- \int_{\tilde{\Gamma}} \left[ w_{(i,j)}^h c_{ijkl} \right] \langle u_k^h - \hat{u}_k^h \rangle \hat{n}_l d\Gamma - \int_{\Gamma} \langle w_i^h \rangle \left[ \hat{\sigma}_{ij}^h \right] \hat{n}_j d\Gamma \\ &- \int_{\tilde{\Gamma}} \left[ w_i^h \right] \langle \hat{\sigma}_{ij}^h \rangle \hat{n}_j d\Gamma = \int_{\tilde{\Omega}} w_i^h f_i d\Omega \quad \forall w_i^h \in V^h. \end{aligned} \quad (2.28)$$

Recall that  $\hat{n}_i$  are components of the sharing normal vector  $\hat{\mathbf{n}}$  on the edge  $e \in \Gamma$ .

We intend to express the stress components  $\sigma_{ij}^h$  appearing in equation (2.27) in terms of  $u_i^h$  and  $\hat{u}_k^h$  explicitly. To this end, two lifting operators  $R : L^2(\Gamma) \rightarrow \Sigma^h$  and  $L : L^2(\tilde{\Gamma}) \rightarrow \Sigma^h$  are introduced. These operators will be used to transform the contribution of terms on element boundaries into fields defined in element interiors. Consider a function vector  $\boldsymbol{\varphi}$  that its components are square-integrable defined on edges  $e \in \tilde{\Gamma} \cup \Gamma^u$ ,  $\varphi_i \in L^2(e)$ ,  $i = 1 \rightarrow d$ . The lifting operations are defined as: find  $R_{kl}(\boldsymbol{\varphi}) \in \Sigma^h$ ,  $L_{kl}(\boldsymbol{\varphi}) \in \Sigma^h$  such that

$$\int_{\tilde{\Omega}} \tau_{kl}^h R_{kl}(\boldsymbol{\varphi}) d\Omega = - \int_{\Gamma} \langle \tau_{kl}^h \rangle \varphi_k \hat{n}_l d\Gamma \quad \forall \tau_{ij}^h \in \Sigma^h, \quad (2.29)$$

$$\int_{\tilde{\Omega}} \tau_{kl}^h L_{kl}(\boldsymbol{\varphi}) d\Omega = - \int_{\tilde{\Gamma}} \left[ \tau_{kl}^h \right] \varphi_k \hat{n}_l d\Gamma \quad \forall \tau_{ij}^h \in \Sigma^h. \quad (2.30)$$

Taking equations (2.29) and (2.30) into equation (2.27), we get

$$\sigma_{ij}^h = c_{ijkl} \left\{ u_{(k,l)}^h + R_{kl} \left( \left[ u^h - \hat{u}^h \right] \right) + L_{kl} \left( \langle u^h - \hat{u}^h \rangle \right) \right\}. \quad (2.31)$$

From equation (2.28), the variational weak form for the considered problem is defined as: find  $u_i^h \in V^h$  such that

$$B(\mathbf{w}^h, \mathbf{u}^h) = L(\mathbf{w}^h) \quad \forall w_i^h \in V^h, \quad (2.32)$$

where  $B(\mathbf{w}^h, \mathbf{u}^h)$  denotes the bilinear form,

$$\begin{aligned} B(\mathbf{w}^h, \mathbf{u}^h) &= \int_{\tilde{\Omega}} w_{(i,j)}^h c_{ijkl} u_{(k,l)}^h d\Omega - \int_{\Gamma} \langle w_{(i,j)}^h c_{ijkl} \rangle \left[ u_k^h - \hat{u}_k^h \right] \hat{n}_l d\Gamma \\ &- \int_{\tilde{\Gamma}} \left[ w_{(i,j)}^h c_{ijkl} \right] \langle u_k^h - \hat{u}_k^h \rangle \hat{n}_l d\Gamma - \int_{\Gamma} \langle w_i^h \rangle \left[ \hat{\sigma}_{ij}^h \right] \hat{n}_j d\Gamma \\ &- \int_{\tilde{\Gamma}} \left[ w_i^h \right] \langle \hat{\sigma}_{ij}^h \rangle \hat{n}_j d\Gamma, \end{aligned} \quad (2.33)$$

and  $L(\mathbf{w}^h)$  denotes the linear form,

$$L(\mathbf{w}^h) = \int_{\tilde{\Omega}} w_i^h f_i d\Omega. \quad (2.34)$$

Equation (2.31) for the stress  $\sigma_{ij}^h$  together with the variational form in equations (2.32) to (2.34) provide the basis for constructing discontinuous Galerkin methods later. Note that, equations (2.3) and (2.4) are being used to describe boundary conditions for the considered problem.

### 2.5.2 Consistency of the weak formulation

The numerical displacement  $\hat{u}_i^h$  and the numerical stress  $\hat{\sigma}_{ij}^h$  on an element boundary  $e \in \Gamma$  are defined as functions of the displacement unknown  $\mathbf{u}^h$ , see Arnold et al. (2002). Consistency of these numerical fluxes require that if there exists the exact solution  $\mathbf{u}$  for our problem, the follow relations hold true

$$\hat{u}_i^h(\mathbf{u})|_e = u_i|_e, \quad (2.35)$$

$$\hat{\sigma}_{ij}^h(\mathbf{u})|_e = \sigma_{ij}(\mathbf{u})|_e, \quad (2.36)$$

where  $i, j = 1 \rightarrow d$ . This means that when the exact displacement is taken into account, the numerical solution and the exact solution coincide on element edges.

Integrating by parts the first term on the right-hand side of the bilinear form (2.33) and replacing  $u_i^h$  by the exact solution  $u_i$ , we have

$$\begin{aligned} B(\mathbf{w}^h, \mathbf{u}) - L(\mathbf{w}^h) &= - \int_{\tilde{\Omega}} w_i^h \left\{ (c_{ijkl} u_{(k,l)})_{,j} + f_i \right\} d\Omega \\ &\quad + \int_{\Gamma} \langle w_i^h \rangle \llbracket c_{ijkl} u_{(k,l)} - \hat{\sigma}_{ij} \rrbracket \hat{n}_j d\Gamma + \int_{\tilde{\Gamma}} \llbracket w_i^h \rrbracket \langle c_{ijkl} u_{(k,l)} - \hat{\sigma}_{ij} \rangle \hat{n}_j d\Gamma \\ &\quad - \int_{\Gamma} \langle w_{(i,j)}^h c_{ijkl} \rangle \llbracket u_k - \hat{u}_k \rrbracket \hat{n}_l ds - \int_{\tilde{\Gamma}} \llbracket w_{(i,j)}^h c_{ijkl} \rrbracket \langle u_k - \hat{u}_k \rangle \hat{n}_l d\Gamma. \end{aligned} \quad (2.37)$$

Since  $u_i$  are components of the exact solution,  $\llbracket c_{ijkl} u_{(k,l)} \rrbracket = 0$  and  $\llbracket u_k \rrbracket = 0$  on  $\tilde{\Gamma}$  and  $(c_{ijkl} u_{(k,l)})_{,j} + f_i = 0$  in  $\tilde{\Omega}$ . Therefore, we may rewrite (2.37) as

$$\begin{aligned} B(\mathbf{w}^h, \mathbf{u}) - L(\mathbf{w}^h) &= - \int_{\tilde{\Gamma}} \langle w_i^h \rangle \llbracket \hat{\sigma}_{ij} \rrbracket \hat{n}_j d\Gamma + \int_{\Gamma^h} w_i^h (g_i^h - \hat{\sigma}_{ij} \hat{n}_j) d\Gamma \\ &\quad + \int_{\tilde{\Gamma} \cup \Gamma^u} \llbracket w_i^h \rrbracket \langle c_{ijkl} u_{(k,l)} - \hat{\sigma}_{ij} \rangle \hat{n}_j d\Gamma + \int_{\tilde{\Gamma}} \langle w_{(i,j)}^h c_{ijkl} \rangle \llbracket \hat{u}_k \rrbracket \hat{n}_l d\Gamma \\ &\quad - \int_{\Gamma^u} w_{(i,j)}^h c_{ijkl} (g_k^u - \hat{u}_k) \hat{n}_l d\Gamma - \int_{\tilde{\Gamma} \cup \Gamma^h} \llbracket w_{(i,j)}^h c_{ijkl} \rrbracket \langle u_k - \hat{u}_k \rangle \hat{n}_l d\Gamma. \end{aligned} \quad (2.38)$$

If numerical fluxes are consistent, equations (2.35) to (2.36) together with boundary conditions (2.3) and (2.4) hold true. Hence, expression (2.38) equals to zero for all  $w_i^h \in V^h$ . This means that the discontinuous Galerkin weak formulation (2.32) is consistent. In the opposite statement, if  $B(w^h, u) - L(w^h) = 0$  holds true for all  $w_i^h \in V^h$ , from the above expression we have

$$[[\hat{u}_k]] = 0 \quad \text{on } \tilde{\Gamma}, \quad (2.39)$$

$$\hat{u}_k = u_k \quad \text{on } \tilde{\Gamma} \cup \Gamma^h, \quad (2.40)$$

$$\hat{u}_k = g_k^u \quad \text{on } \Gamma^u, \quad (2.41)$$

$$[[\hat{\sigma}_{ij}]] = 0 \quad \text{on } \tilde{\Gamma}, \quad (2.42)$$

$$\hat{\sigma}_{ij} = c_{ijkl} u_{(k,l)} \quad \text{on } \tilde{\Gamma} \cup \Gamma^u, \quad (2.43)$$

$$\hat{\sigma}_{ij} \hat{n}_j = g_i^h \quad \text{on } \Gamma^h. \quad (2.44)$$

This means that numerical fluxes are consistent and the boundary conditions are fulfilled.

In summary, the consistency of the numerical displacement  $\hat{u}_i^h$  and numerical stress  $\hat{\sigma}_{ij}^h$  ensures the consistency of the weak formulation and vice versa.

## 2.6 Interior penalty formulation for elasticity

Bases on the weak formulation presented in the previous section, a discontinuous Galerkin formulation can be derived by using a particular choice of the numerical displacement  $\hat{u}_i^h$  and numerical stress  $\hat{\sigma}_{ij}^h$ . Following the work presented by Arnold et al. (2002) for second-order problems, we now derive the interior penalty method for elasticity. In this method,  $\hat{u}_i^h$  and  $\hat{\sigma}_{ij}^h$  are defined as

$$\hat{u}_i^h = \langle u_i^h \rangle \quad \text{on } \tilde{\Gamma} \cup \Gamma^h, \quad (2.45)$$

$$\hat{u}_i^h = g_i^u \quad \text{on } \Gamma^u, \quad (2.46)$$

$$\hat{\sigma}_{ij}^h \hat{n}_j = \langle c_{ijkl} u_{(k,l)}^h \rangle \hat{n}_j - \frac{\eta d_{ik}}{h} [[u_k^h]] \quad \text{on } \tilde{\Gamma}, \quad (2.47)$$

$$\hat{\sigma}_{ij}^h \hat{n}_j = c_{ijkl} u_{(k,l)}^h \hat{n}_j - \frac{\eta d_{ik}}{h} (u_k^h - g_k^u) \quad \text{on } \Gamma^u, \quad (2.48)$$

$$\hat{\sigma}_{ij}^h \hat{n}_j = g_i^h \quad \text{on } \Gamma^h, \quad (2.49)$$

where  $\eta$  is a positive number which is called the penalty parameter,  $h$  represents the size of elements, and  $d_{ik}$  is a stiffness tensor which depends on the material

properties. Within the discussion in the previous section, it is clear that the numerical displacement  $\hat{u}_i^h$  and numerical stress  $\hat{\sigma}_{ij}^h$  are consistent. Substituting this choice of numerical fluxes into the equations (2.31) to (2.33) and then rearranging them, we may now redefine the variational formulation for the considered problem as: find  $u_i^h \in V^h$  such that

$$\begin{aligned}
& \int_{\tilde{\Omega}} w_{(i,j)}^h c_{ijkl} u_{(k,l)}^h d\Omega - \int_{\tilde{\Gamma} \cup \Gamma^u} \langle w_{(i,j)}^h c_{ijkl} \rangle \llbracket u_k^h \rrbracket \hat{n}_l ds \\
& - \int_{\tilde{\Gamma} \cup \Gamma^u} \llbracket w_i^h \rrbracket \langle c_{ijkl} u_{(k,l)}^h \rangle \hat{n}_j d\Gamma + \int_{\tilde{\Gamma} \cup \Gamma^u} \frac{\eta}{h} \llbracket w_i^h \rrbracket d_{ik} \llbracket u_k^h \rrbracket d\Gamma \\
& = \int_{\tilde{\Omega}} w_i^h f_i d\Omega + \int_{\Gamma^h} w_i^h g_i^h d\Gamma - \int_{\Gamma^u} w_{(i,j)}^h c_{ijkl} g_k^u \hat{n}_l d\Gamma \\
& + \int_{\Gamma^u} \frac{\eta}{h} w_i^h d_{ik} g_k^u d\Gamma \quad \forall w_i^h \in V_h. \quad (2.50)
\end{aligned}$$

Equation (2.50) is identified as the interior penalty formulation for second-order elliptic problems (Douglas and Dupont, 1976; Arnold et al., 2002) which is applied for elasticity. The first term of the above equation is the standard term within the conventional continuous Galerkin method. The second term ensures *consistency* of the formulation. The third term, which directly comes from the choice of the numerical stress  $\hat{\sigma}_{ij}^h$ , restores the *symmetry* of the formulation. The fourth term, the *stabilisation* term, ensures the stability of the formulation if the penalty parameter is big enough. The fifth and the sixth terms are standard terms of force contributions in volume and on boundary. The seventh term represents the work on the displacement boundary. The last term in the above equation is the stabilisation term on the displacement boundary.

The presented formulation is expressed in term of the displacement unknown only. The values of  $h$  and  $d_{ij}$  are evaluated for each edge  $e$ . In practice, they can be defined in various ways. For instance,  $h$  on  $e \in \tilde{\Gamma}$  is defined as the distance between two centroids of neighboring elements (Engel et al., 2002). See Figure 2.4 for an illustration. On  $e \in \Gamma^u$ ,  $h$  is set to be the distance from the centroid of the element containing  $e$  to the boundary. Components of the stiffness tensor on  $e$  can be defined by

$$d_{ik} = \langle c_{ijkl} \rangle \hat{n}_j \hat{n}_l. \quad (2.51)$$

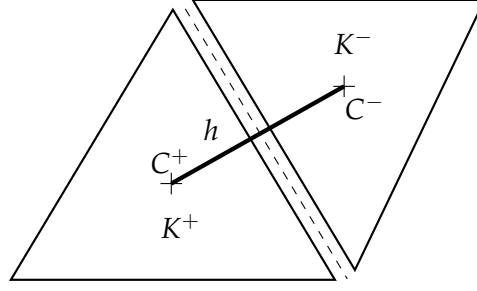


Figure 2.4: Mesh-size representation on a sharing edge

## 2.7 Lifting-type formulations for elasticity

Following the approaches presented by Bassi et al. (1997), Brezzi et al. (2000) and Bassi and Rebay (2002) for second-order problems, we now introduce two lifting-type formulations for elasticity. These formulations are built upon the lifting operator  $R : L^2(\Gamma) \rightarrow \Sigma^h$  defined in equation (2.29). As we will see later, the lifting operator  $L : L^2(\tilde{\Gamma}) \rightarrow \Sigma^h$  defined in equation (2.30) is no longer necessary with particular choices of numerical fluxes.

In practice, the lifting operations will be employed locally. Consider a function vector  $\boldsymbol{\varphi}$ . Its components are square-integrable functions defined on edges  $e \in \tilde{\Gamma} \cup \Gamma^u$ ,  $\varphi_i \in L^2(e)$ ,  $i = 1 \rightarrow d$ . For each edge  $e$ , a local lifting operation  $r^e : L^2(e) \rightarrow \Sigma^h$  is defined as: find  $r_{kl}(\boldsymbol{\varphi}) \in \Sigma^h$  such that

$$\int_{K^+ \cup K^-} \tau_{kl}^h r_{kl}^e(\boldsymbol{\varphi}) d\Omega = - \int_e \langle \tau_{kl}^h \rangle \varphi_k \hat{n}_l ds \quad e \in \tilde{\Gamma}, \forall \boldsymbol{\tau}_h \in \Sigma_h, \quad (2.52)$$

$$\int_{K^+} \tau_{kl}^h r_{kl}^e(\boldsymbol{\varphi}) d\Omega = - \int_e \tau_{kl}^h \varphi_k \hat{n}_l ds \quad e \in \Gamma^u, \forall \boldsymbol{\tau}_h \in \Sigma_h. \quad (2.53)$$

Here,  $K^+$  and  $K^-$  are elements sharing the edge  $e$  (revisit Figure 2.3 for an illustration),  $r_{kl}^e(\boldsymbol{\varphi})$  is a second-order tensor defined on finite element interiors. Note that  $r_{kl}^e(\boldsymbol{\varphi})$  vanishes outside  $K^+ \cup K^-$ . This is an important property for the practical point of view. From the definition of the lifting operator  $R_{kl}(\boldsymbol{\varphi})$  in equation (2.29), it is clear that

$$R_{kl}(\boldsymbol{\varphi}) = \sum_{e \in \tilde{\Gamma} \cup \Gamma^u} r_{kl}^e(\boldsymbol{\varphi}) \quad \text{in } \tilde{\Omega}. \quad (2.54)$$

### 2.7.1 First lifting formulation

Within the concept behind the discontinuous Galerkin method presented by Brezzi et al. (2000) for elliptic problems, we now consider the following choice

of numerical fluxes  $\hat{u}_i^h$  and  $\hat{\sigma}_{ij}^h$ :

$$\hat{u}_i^h = \langle u_i^h \rangle \quad \text{on } \tilde{\Gamma} \cup \Gamma^h, \quad (2.55)$$

$$\hat{u}_i^h = g_i^u \quad \text{on } \Gamma^u, \quad (2.56)$$

$$\hat{\sigma}_{ij}^h = \langle c_{ijkl} \{ u_{(k,l)}^h + R_{kl}^g(\llbracket \mathbf{u}^h \rrbracket) \} \rangle + \eta \langle c_{ijkl} r_{kl}^e(\llbracket \mathbf{u}^h \rrbracket) \rangle \quad \text{on } \tilde{\Gamma}, \quad (2.57)$$

$$\hat{\sigma}_{ij}^h = c_{ijkl} \{ u_{(k,l)}^h + R_{kl}^g(\llbracket \mathbf{u}^h \rrbracket) \} + \eta c_{ijkl} r_{kl}^e(\mathbf{u}^h - \mathbf{g}^u) \quad \text{on } \Gamma^u, \quad (2.58)$$

$$\hat{\sigma}_{ij}^h \hat{n}_j = g_i^h \quad \text{on } \Gamma^h, \quad (2.59)$$

where,  $\eta$  is the penalty parameter,  $r_{ij}^e(\llbracket \mathbf{u}^h \rrbracket)$  and  $r_{ij}^e(\mathbf{u}^h - \mathbf{g}^u)$  are defined by equations (2.52) and (2.53),  $R_{kl}^g(\llbracket \mathbf{u}^h \rrbracket) \in \Sigma^h$  denotes the second-order tensor which includes the contribution of the boundary value  $\mathbf{g}^u$  by

$$R_{kl}^g(\llbracket \mathbf{u}^h \rrbracket) = R_{kl}(\llbracket \mathbf{u}^h \rrbracket) - R_{kl}(\mathbf{g}^u) \quad \text{in } \tilde{\Omega}. \quad (2.60)$$

For a later use, a local lifting function  $r_{kl}^{e,g}(\llbracket \mathbf{u}^h \rrbracket) \in \Sigma^h$  which includes the contribution of the boundary value  $\mathbf{g}^u$  is also defined,

$$r_{kl}^{e,g}(\llbracket \mathbf{u}^h \rrbracket) = r_{kl}^e(\llbracket \mathbf{u}^h \rrbracket) - r_{kl}^e(\mathbf{g}^u) \quad \text{in } \tilde{\Omega}. \quad (2.61)$$

From the above equations,  $r_{ij}^e$  and  $r_{ij}^{e,g}$  have the dimension of the strain and represent the enhanced strains due to jump in displacements across an edge  $e$ . Second-order tensors  $R_{ij}$  and  $R_{ij}^g$  therefore have the meaning of total strain-like terms in  $\tilde{\Omega}$  which are contributed by  $\llbracket \mathbf{u}^h \rrbracket$  and  $\mathbf{g}^u$  from all the edges  $e \in \tilde{\Gamma} \cup \Gamma^u$ . Substituting equations (2.55) to (2.59) into equations (2.32) to (2.34), we get

$$\begin{aligned} & \int_{\tilde{\Omega}} w_{(i,j)}^h c_{ijkl} u_{(k,l)}^h d\Omega - \int_{\tilde{\Gamma} \cup \Gamma^u} \langle w_{(i,j)}^h c_{ijkl} \rangle \llbracket u_k^h \rrbracket \hat{n}_l d\Gamma \\ & - \int_{\tilde{\Gamma} \cup \Gamma^u} \llbracket w_i^h \rrbracket \langle c_{ijkl} u_{(k,l)}^h \rangle \hat{n}_j d\Gamma - \int_{\tilde{\Gamma} \cup \Gamma^u} \llbracket w_i^h \rrbracket \langle c_{ijkl} R_{kl}^g(\llbracket \mathbf{u}^h \rrbracket) \rangle \hat{n}_j d\Gamma \\ & - \int_{\tilde{\Gamma} \cup \Gamma^u} \eta \llbracket w_i^h \rrbracket \langle c_{ijkl} r_{kl}^e(\llbracket \mathbf{u}^h \rrbracket) \rangle \hat{n}_j d\Gamma = \int_{\tilde{\Omega}} w_i^h f_i d\Omega + \int_{\Gamma^h} w_i^h g_i^h d\Gamma \\ & - \int_{\Gamma^u} w_{(i,j)}^h c_{ijkl} g_k^u \hat{n}_l d\Gamma - \int_{\Gamma^u} \eta w_i^h c_{ijkl} r_{kl}^e(\mathbf{g}^u) \hat{n}_j d\Gamma \quad \forall w_i^h \in V_h. \end{aligned} \quad (2.62)$$

Taking lifting operations in equations (2.52) to (2.54), (2.60) and (2.61) into equation (2.62) and then rearranging it, we can define a variational formulation for

the considered problem as: find  $u_i^h \in V^h$  such that

$$\begin{aligned} \int_{\tilde{\Omega}} \left\{ w_{(i,j)}^h + R_{ij} \left( \llbracket \mathbf{w}^h \rrbracket \right) \right\} c_{ijkl} \left\{ u_{(k,l)}^h + R_{kl}^g \left( \llbracket \mathbf{u}^h \rrbracket \right) \right\} d\Omega \\ + \sum_{e \in \tilde{\Gamma} \cup \Gamma^u} \int_{\tilde{\Omega}} \eta r_{ij}^e \left( \llbracket \mathbf{w}^h \rrbracket \right) c_{ijkl} r_{kl}^{e,g} \left( \llbracket \mathbf{u}^h \rrbracket \right) d\Omega \\ = \int_{\tilde{\Omega}} w_i^h f_i d\Omega + \int_{\Gamma^h} w_i^h g_i^h d\Gamma \quad \forall w_i^h \in V_h. \end{aligned} \quad (2.63)$$

This is a lifting-type formulation utilising the approach of Brezzi et al. (2000) for elasticity elliptic problems. It does not require the introduction of a stiffness term at element boundaries and the element size parameter. The method utilises lifting operations to transform the displacement jump across element boundaries into the field defined on element interior. This field contributes as an extra strain-like field in the body due to the discontinuity of displacements across element edges. The method is thus particularly attractive to nonlinear problems.

### 2.7.2 Second lifting formulation

Following the approach presented by Bassi et al. (1997) and Bassi and Rebay (2002) for second-order problems, we now consider the following choice of numerical fluxes  $\hat{u}_i^h$  and  $\hat{\sigma}_{ij}^h$  defined on an edge  $e$  by

$$\hat{u}_i^h = \langle u_i^h \rangle \quad \text{on } \tilde{\Gamma} \cup \Gamma^h, \quad (2.64)$$

$$\hat{u}_i^h = g_i^u \quad \text{on } \Gamma^u, \quad (2.65)$$

$$\hat{\sigma}_{ij}^h = \langle c_{ijkl} u_{(k,l)}^h \rangle + \eta \langle c_{ijkl} r_{kl}^e \left( \llbracket \mathbf{u}^h \rrbracket \right) \rangle \quad \text{on } \tilde{\Gamma}, \quad (2.66)$$

$$\hat{\sigma}_{ij}^h = c_{ijkl} u_{(k,l)}^h + \eta c_{ijkl} r_{kl}^e \left( \mathbf{u}^h - \mathbf{g}^u \right) \quad \text{on } \Gamma^u, \quad (2.67)$$

$$\hat{\sigma}_{ij}^h \hat{n}_j = g_i^h \quad \text{on } \Gamma^h, \quad (2.68)$$

Substituting equations (2.64) to (2.68) into equations (2.32) to (2.34), we obtain

$$\begin{aligned} \int_{\tilde{\Omega}} w_{(i,j)}^h c_{ijkl} u_{(k,l)}^h d\Omega - \int_{\tilde{\Gamma} \cup \Gamma^u} \langle w_{(i,j)}^h c_{ijkl} \rangle \llbracket u_k^h \rrbracket \hat{n}_l d\Gamma \\ - \int_{\tilde{\Gamma} \cup \Gamma^u} \llbracket w_i^h \rrbracket \langle c_{ijkl} u_{(k,l)}^h \rangle \hat{n}_j d\Gamma - \int_{\tilde{\Gamma} \cup \Gamma^u} \eta \llbracket w_i^h \rrbracket \langle c_{ijkl} r_{kl}^e \left( \llbracket \mathbf{u}^h \rrbracket \right) \rangle \hat{n}_j d\Gamma \\ = \int_{\tilde{\Omega}} w_i^h f_i d\Omega + \int_{\Gamma^h} w_i^h g_i^h d\Gamma - \int_{\Gamma^u} w_{(i,j)}^h c_{ijkl} g_k^u \hat{n}_l d\Gamma \\ - \int_{\Gamma^u} \eta w_i^h c_{ijkl} r_{kl}^e \left( \mathbf{g}^u \right) \hat{n}_j d\Gamma \quad \forall w_i^h \in V_h. \end{aligned} \quad (2.69)$$

Taking the lifting operations defined in equations (2.52) to (2.54) into the above equation, we can define a variational formulation for the considered problem as: find  $u_i^h \in V^h$  such that

$$\begin{aligned}
& \int_{\tilde{\Omega}} w_{(i,j)}^h c_{ijkl} u_{(k,l)}^h d\Omega + \int_{\tilde{\Omega}} w_{(i,j)}^h c_{ijkl} R_{kl} \left( \llbracket u^h \rrbracket \right) d\Gamma \\
& + \int_{\tilde{\Omega}} R_{ij} \left( \llbracket w^h \rrbracket \right) c_{ijkl} u_{(k,l)}^h d\Gamma + \sum_{e \in \tilde{\Gamma} \cup \Gamma^u} \int_{\tilde{\Omega}} \eta r_{ij}^e \left( \llbracket w^h \rrbracket \right) c_{ijkl} r_{kl}^e \left( \llbracket u^h \rrbracket \right) d\Omega \\
& = \int_{\tilde{\Omega}} w_i^h f_i d\Omega + \int_{\Gamma^h} w_i^h g_i^h d\Gamma - \int_{\Gamma^u} w_{(i,j)}^h c_{ijkl} g_k^u \hat{n}_l d\Gamma \\
& + \sum_{e \in \Gamma^u} \int_{\tilde{\Omega}} \eta r_{ij}^e \left( w^h \right) c_{ijkl} r_{kl}^e \left( g^u \right) d\Omega \quad \forall w_i^h \in V_h. \quad (2.70)
\end{aligned}$$

Equation (2.70) appears as a general form of the discontinuous Galerkin formulations presented by Bassi et al. (1997) and Bassi and Rebay (2002) for second-order problems. Note that in the formulation of Bassi et al. (1997) there is no penalty term ( $\eta = 0$ ). Whereas, the formulations introduced by Bassi and Rebay (2002) utilise  $\eta = 1$ . Without the product term  $R_{ij} \left( \llbracket w^h \rrbracket \right) R_{ij}^s \left( \llbracket w^h \rrbracket \right)$ , it leads to a sparser stiffness matrix compared to the case of the lifting formulation in equation (2.63).

Note that, in all presented formulations, interior and exterior boundaries have been dealt with in the same fashion. This permits not only the weak enforcement of continuity on interior edges between elements, but also the weak imposition of Dirichlet boundary conditions.

### 2.7.3 Analysis of discontinuous Galerkin formulations

As the chosen numerical fluxes are consistent, consistency of the presented formulations is ensured. The formulations are stable using a large enough penalty parameter  $\eta$ . In the numerical analysis, this condition leads to a positive-definite stiffness matrix. In the interior penalty method, the critical value of the penalty parameter for stability however is not straightforward to quantify. Note that, a too large value of  $\eta$  should not be used as an ill-conditional stiffness matrix may appear. Whereas, the stability condition for the lifting formulations can be concretely quantified. It can be proven that the first lifting formulation is stable when  $\eta \geq 0$ . For the second lifting formulation,  $\eta \geq \sup_{K \in \mathcal{P}^h} (N_K)$  is required,  $N_K$  is the number of edges on element  $K$ . Note that  $N_K = 3$  when triangular elements are used.

For further analyses of the formulations, the reader can refer to a number of works for second-order problems, including those of Douglas and Dupont (1976), Bassi and Rebay (1997), Bassi et al. (1997), Brezzi et al. (2000), Castillo et al. (2000), Arnold et al. (2002) and Brezzi et al. (2006).

## Chapter 3

### $C^0$ formulations for thin plates\*

The numerical simulation of thin plates involves the solution of a fourth-order differential equation. Within the finite element concept, such a problem demands the use of  $C^1$  basis functions and leads to significant challenges and difficulties. Over the years, numerous finite elements for thin plate bending problems have been proposed and developed, such as  $C^1$  conforming elements (Clough and Tocher, 1965; Zlámal, 1968; Zienkiewicz, 1971; Ciarlet and Raviart, 1972), non-conforming elements (Bazeley et al., 1965; Babuška and Zlámal, 1973),  $C^0$  elements using mixed methods (Mindlin and Cooper, 1960; Brezzi et al., 1989; Arnold and Falk, 1989) or more recently  $C^1$  elements with specially constructed basis functions (Krysl and Belytschko, 1995; Cirak et al., 2000; Oñate and Zarate, 2000; Hughes et al., 2005). Conforming or conventional non-conforming finite element formulations can perform well, but are problematic to construct. Elements for mixed formulations can circumvent the need for  $C^1$  basis functions, although at expense of other difficulties, primarily related to shear locking.

This chapter is devoted to the development of  $C^0$  formulations for linear elastic Kirchhoff plates. The approach builds on developments in discontinuous Galerkin methods for second-order elliptic problems, particularly on the works of Bassi and Rebay (1997), Bassi et al. (1997) and Brezzi et al. (2000). The method relies on a lifting operation to transform jumps in the normal derivative across element boundaries to a field defined on element interiors.

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\*This chapter is based on Dung and Wells (2006) and Wells and Dung (2007).

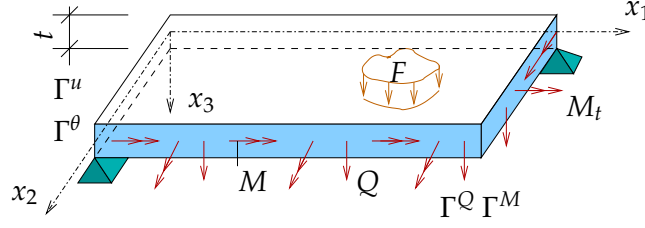


Figure 3.1: Plate bending problem.

### 3.1 Governing equation for thin plate bending

For the sake of simplicity, a polygonal plate is considered. The plate has a mid-surface denoted by  $\Omega$  and thickness  $t$ , see Figure 3.1. The boundary of  $\Omega$  is partitioned such that  $\overline{\Gamma^u} \cup \overline{\Gamma^Q} = \overline{\Gamma^\theta} \cup \overline{\Gamma^M} = \partial\Omega$  and  $\Gamma^u \cap \Gamma^Q = \Gamma^\theta \cap \Gamma^M = \emptyset$ . A fixed Cartesian system  $(x_1, x_2, x_3)$  in which the axis in the  $x_3$  direction is normal to the mid-surface is adopted. Hereinafter,  $\alpha, \beta, \delta$  and  $\gamma$  are indices denoting the  $x_1$  and  $x_2$  directions, and  $n_\alpha$  and  $s_\alpha$  denote the components of the unit outward normal and the unit tangent vectors to the plate boundary, respectively.

The strong form of the plate bending problem is stated as follows: given the distributed force  $F$ , and the transverse displacement, the normal rotation, the normal moment and the transverse force on parts of the boundary, denoted  $g^u$ ,  $g^\theta$ ,  $M$ , and  $Q$ , respectively, find the transverse displacement  $u$  such that

$$m_{\alpha\beta,\alpha\beta} = F \quad \text{in } \Omega, \quad (3.1)$$

$$u = g^u \quad \text{on } \Gamma^u, \quad (3.2)$$

$$u_{,\alpha} n_\alpha = g^\theta \quad \text{on } \Gamma^\theta, \quad (3.3)$$

$$m_{\alpha\beta} n_\alpha n_\beta = M \quad \text{on } \Gamma^M, \quad (3.4)$$

$$-m_{\alpha\beta,\beta} n_\alpha - (m_{\alpha\beta} n_\beta s_\alpha)_{,s} = Q \quad \text{on } \Gamma^Q, \quad (3.5)$$

$$m_{\alpha\beta} n_\alpha s_\beta|_{c^+} - m_{\alpha\beta} n_\alpha s_\beta|_{c^-} = 0 \quad \forall c \in \Gamma^Q, \quad (3.6)$$

where  $c^+$  and  $c^-$  are two nearest-neighbour points of the corner  $c$  on the shear boundary  $\Gamma^Q$ . The last equation is related to equilibrium of plate corner forces. In the above equations, the moment tensor is given by

$$m_{\alpha\beta} = \mathcal{C}_{\alpha\beta\gamma\delta} u_{,\gamma\delta}, \quad (3.7)$$

where  $\mathcal{C}_{\alpha\beta\gamma\delta}$  is a fourth order constitutive tensor. For an isotropic plate,

$$\mathcal{C}_{\alpha\beta\gamma\delta} = \frac{t^3}{12} \{ \mu (\delta_{\alpha\gamma} \delta_{\beta\delta} + \delta_{\alpha\delta} \delta_{\beta\gamma}) + \lambda \delta_{\alpha\beta} \delta_{\gamma\delta} \}, \quad (3.8)$$

where  $\delta_{\alpha\beta}$  is the Kronecker delta and the Lamé parameters are defined in terms of Young's modulus  $E$  and Poisson's ratio  $\nu$ ,

$$\lambda = \frac{E\nu}{1-\nu^2}, \quad (3.9)$$

$$\mu = \frac{E}{2(1+\nu)}. \quad (3.10)$$

Plane stress problems are considered, as implied by equation (3.8).

**Remark 3.1.1** Equation (3.5) is a modified shear boundary condition. It was proposed by Kirchhoff, see Timoshenko and Woinowsky-Krieger (1959, pages 83–92), in order to solve the so-called Poisson paradox in describing boundary conditions for thin plates. This paradox comes from the fact that there exist three components of kinetic quantities on the boundary: normal moment, twist moment and shear force while there are only two independent boundary conditions. ■

## 3.2 Galerkin formulations

### 3.2.1 Finite element discretisation

Consider a partition  $\mathcal{P}^h$  of the domain  $\Omega$  into  $N$  elements  $K_i$ ,  $i = 1 \rightarrow N$ , such that  $\bigcup_{i=1}^N \bar{K}_i = \bar{\Omega}$  and  $\bigcup_{i=1}^N K_i = \tilde{\Omega}$ . Here,  $\tilde{\Omega}$  is the union of all element interiors and does not contain element edges. The union of all element edges is denoted by  $\Gamma = \bigcup_{i=1}^N \partial K_i$ , and the union of all internal element edges is denoted by  $\tilde{\Gamma} = \Gamma \setminus \partial\Omega$ . Two function spaces are defined as follows:

$$V^h = \left\{ u^h \in H^1(\Omega) : u^h|_{K_i} \in P^k(K_i) \forall K_i \in \mathcal{P}^h, u^h|_{\Gamma^u} = g^u \right\}, \quad (3.11)$$

$$W^h = \left\{ w^h \in H^1(\Omega) : w^h|_{K_i} \in P^k(K_i) \forall K_i \in \mathcal{P}^h, w^h|_{\Gamma^u} = 0 \right\}, \quad (3.12)$$

where  $P^k(K_i)$  are standard Lagrange finite element shape functions of polynomial order  $k$ . Note that the space  $V^h$  satisfies the displacement boundary condition, and it is assumed that the displacement boundary condition can be satisfied exactly by functions in the space  $V^h$ . Note that the rotation boundary conditions are not contained in the definition of  $V^h$ .

For two elements,  $K^+, K^- \in \mathcal{P}^h$ , which share an interior edge  $e$  (see Figure 3.2),  $n_\alpha^+$  denotes the components of the normal vector  $\mathbf{n}^+$  on  $e$  that points from  $K^+$  to  $K^-$  and  $n_\alpha^-$  denotes the components of the normal vector  $\mathbf{n}^-$  on  $e$  that

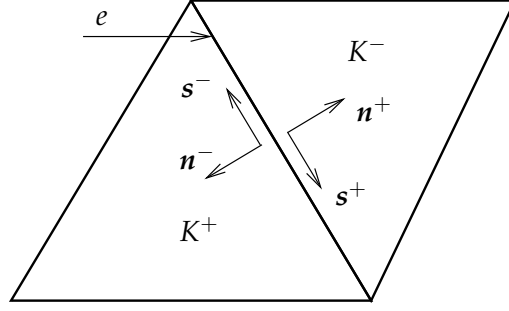


Figure 3.2: Neighboring elements and their shared edge.

points from  $K^-$  to  $K^+$ . For a scalar function  $\varphi$  on an interior edge  $e$ , let denote  $\varphi_{,n} = \varphi_{,\alpha} n_\alpha \Xi$ , where  $\Xi$  is the sign function defined in equation (2.16). Recalling the definition of the jump and average terms in equations (2.14) to (2.15), we now define the following jump and average terms of  $\varphi_{,n}$ :

$$[[\varphi_{,n}]] = \varphi_{,\alpha}^+ n_\alpha^+ + \varphi_{,\alpha}^- n_\alpha^-, \quad (3.13)$$

$$\langle \varphi_{,n} \rangle = \frac{1}{2} (\varphi_{,\alpha}^+ n_\alpha^+ - \varphi_{,\alpha}^- n_\alpha^-), \quad (3.14)$$

where  $\varphi^\pm$  denotes the value of the function on the side of element  $K^\pm$ , see equation (2.12) and Figure 2.3. On an exterior edge,  $n_\alpha$  simply denotes the outward normal vector. The jump and average terms on an exterior boundary are defined as:

$$[[\varphi_{,n}]] = \varphi_{,\alpha} n_\alpha, \quad (3.15)$$

$$\langle \varphi_{,n} \rangle = \varphi_{,\alpha} n_\alpha. \quad (3.16)$$

### 3.2.2 Weak formulation

Our goal in this section is to construct a weak formulations for the fourth-order problem of thin plates in which  $C^1$  continuity is weakly imposed. To this end, we apply the concept presented in the previous chapter with the note that the normal rotation across element boundaries is now the object of the weakly continuity enforcement. We start from the equilibrium equation (3.1) for an element  $K$ . Both the sides of equation (3.1) are firstly multiplied by test functions  $w^h \in W^h$  and then integrated on each element  $K$ , which results in

$$\int_K w^h m_{\alpha\beta,\beta\alpha} d\Omega = \int_K w^h F d\Omega \quad \forall w^h \in W^h. \quad (3.17)$$

The first term on the LHS is integrated by parts twice and boundary conditions (3.4) and (3.5) are taken into account, which leads to

$$\begin{aligned} & \int_K w_{,\alpha\beta}^h m_{\alpha\beta} d\Omega - \int_{\partial K \cap (\tilde{\Gamma} \cup \Gamma^\theta)} w_{,\alpha}^h \hat{m}_{\alpha\beta} n_\beta d\Gamma + \int_{\partial K \cap \Gamma^u} w^h \hat{m}_{\alpha\beta,\beta} n_\alpha d\Gamma \\ &= \int_K w^h F d\Omega + \int_{\partial K \cap \Gamma^M} w_{,n}^h M d\Gamma + \int_{\partial K \cap \Gamma^Q} w^h Q d\Gamma \quad \forall w^h \in W^h, \end{aligned} \quad (3.18)$$

where  $n_\alpha$  is the outward unit normal vector to the boundary  $\partial K$  of element  $K$ ,  $w_{,n}^h = w_{,\alpha}^h n_\alpha$  on  $\partial K \cap \Gamma^M$  and  $\hat{m}_{\alpha\beta}$  is the numerical moment on the element boundary. Proper choices of  $\hat{m}_{\alpha\beta}$  will be introduced later. Equation (3.18) is the variational form for elements  $K$  in the Kirchhoff plate model.

Insertion of the constitutive equation (3.7) into equation (3.18) and integrating by parts the first term of equation (3.18) twice results in

$$\begin{aligned} & \int_K w_{,\alpha\beta}^h C_{\alpha\beta\gamma\delta} u_{,\gamma\beta}^h d\Omega - \int_{\partial K} w_{,\alpha\beta}^h C_{\alpha\beta\gamma\delta} (u_{,\gamma}^h - \hat{u}_{,\gamma}^h) n_\delta d\Gamma \\ & \quad - \int_{\partial K \cap (\tilde{\Gamma} \cup \Gamma^\theta)} w_{,\alpha}^h \hat{m}_{\alpha\beta}^h n_\beta d\Gamma + \int_{\partial K \cap \Gamma^u} w^h \hat{m}_{\alpha\beta,\beta}^h n_\alpha d\Gamma \\ &= \int_K w^h F d\Omega + \int_{\partial K \cap \Gamma^M} w_{,n}^h M d\Gamma + \int_{\partial K \cap \Gamma^Q} w^h Q d\Gamma \quad \forall w^h \in W^h. \end{aligned} \quad (3.19)$$

The second term of the above equation represents the contribution to the plate potential energy of the difference between the rotation  $u_{,\gamma}^h$  and its numerical flux  $\hat{u}_{,\gamma}^h$  on the element boundary, due to the discontinuity of the finite element space.

Summing over all elements  $K$  and utilising the jump and average operators in equations (3.13) to (3.16) and the identity (2.17),

$$\begin{aligned} & \int_{\tilde{\Omega}} w_{,\alpha\beta}^h C_{\alpha\beta\gamma\delta} u_{,\gamma\beta}^h d\Omega - \int_{\tilde{\Gamma}} \langle w_{,\alpha\beta}^h C_{\alpha\beta\gamma\delta} n_\delta \rangle \llbracket (u_{,\gamma}^h - \hat{u}_{,\gamma}^h) \Xi \rrbracket d\Gamma \\ & \quad - \int_{\tilde{\Gamma}} \llbracket w_{,\alpha\beta}^h C_{\alpha\beta\gamma\delta} n_\delta \rrbracket \langle (u_{,\gamma}^h - \hat{u}_{,\gamma}^h) \Xi \rangle d\Gamma - \int_{\tilde{\Gamma} \cup \Gamma^\theta} \llbracket w_{,\alpha}^h \Xi \rrbracket \langle \hat{m}_{\alpha\beta}^h n_\beta \rangle d\Gamma \\ & \quad - \int_{\tilde{\Gamma}} \langle w_{,\alpha}^h \Xi \rangle \llbracket \hat{m}_{\alpha\beta}^h n_\beta \rrbracket d\Gamma + \int_{\Gamma^u} w^h \hat{m}_{\alpha\beta,\beta}^h n_\alpha d\Gamma \\ &= \int_{\tilde{\Omega}} w^h F d\Omega + \int_{\Gamma^M} w_{,n}^h M d\Gamma + \int_{\Gamma^Q} w^h Q d\Gamma \quad \forall w^h \in W^h. \end{aligned} \quad (3.20)$$

Since  $w^h = 0$  on  $\Gamma^u$ , the sixth term of the LHS of the above equation vanishes. Furthermore, similar to the problem discussed in Remark 3.1.1 for the exterior boundary condition, there is a degree of redundancy on the interior boundaries. As is well known in thin bending analysis, the continuity requirement for the tangent component of the rotation across element boundaries is naturally fulfilled

by means of  $C^0$  finite element constructions (Felippa, 2004). Therefore, we resolve the boundary integral terms into normal and tangential contributions, and consider the normal component only. We may now redefine our problem as: find  $u^h \in V^h$  such that

$$B(w^h, u^h) = L(w^h) \quad \forall w^h \in W^h, \quad (3.21)$$

where  $B(w^h, u^h)$  denotes a bilinear form,

$$\begin{aligned} B(w^h, u^h) = & \int_{\tilde{\Omega}} w^h_{,\alpha\beta} C_{\alpha\beta\gamma\delta} u^h_{,\gamma\beta} d\Omega - \int_{\Gamma} \langle w^h_{,\alpha\beta} C_{\alpha\beta\gamma\delta} n_\gamma n_\delta \rangle \llbracket u^h_{,n} - \hat{u}^h_{,n} \rrbracket d\Gamma \\ & - \int_{\tilde{\Gamma}} \llbracket w^h_{,\alpha\beta} C_{\alpha\beta\gamma\delta} n_\gamma n_\delta \rrbracket \langle u^h_{,n} - \hat{u}^h_{,n} \rangle d\Gamma - \int_{\tilde{\Gamma} \cup \Gamma^\theta} \llbracket w^h_{,n} \rrbracket \langle \hat{m}^h_{nn} \rangle d\Gamma \\ & - \int_{\tilde{\Gamma}} \langle w^h_{,n} \rangle \llbracket \hat{m}^h_{nn} \rrbracket d\Gamma, \end{aligned} \quad (3.22)$$

and  $L(w^h)$  denotes a linear form,

$$L(w^h) = \int_{\tilde{\Omega}} w^h F d\Omega + \int_{\Gamma^M} w^h_{,n} M d\Gamma + \int_{\Gamma^Q} w^h Q d\Gamma. \quad (3.23)$$

Equations (3.21) to (3.23) form a weak formulation for thin plates. Recalling that  $u^h_{,n} = u^h_{,\alpha} n_\alpha \Xi$ . The normal components of the numerical displacement  $\hat{u}^h_{,n}$  and of the numerical moment  $\hat{m}^h_{nn}$  will be defined later in order to generate particular formulations.

### 3.2.3 $C^0$ interior penalty formulation

In a similar fashion to the case of second-order problems presented in Chapter 2, various  $C^0$  Galerkin formulations for Kirchhoff plates can be obtained by imposing proper definitions of numerical fluxes on element boundaries. Within the concept behind the interior penalty method presented in the works of Arnold et al. (2002) for second-order problems and Engel et al. (2002) for Kirchhoff plate, we consider the following choice of the numerical displacement  $\hat{u}^h_{,n}$  and the numerical moment  $\hat{m}^h_{nn}$ :

$$\hat{u}^h_{,n} = \langle u^h_{,n} \rangle \quad \text{on } \tilde{\Gamma} \cup \Gamma^M, \quad (3.24)$$

$$\hat{u}^h_{,n} = g^\theta \quad \text{on } \Gamma^\theta, \quad (3.25)$$

$$\hat{m}^h_{nn} = \langle C_{\alpha\beta\gamma\delta} u^h_{,\gamma\delta} n_\alpha n_\beta \rangle - \frac{\eta d}{h} \llbracket u^h_{,n} \rrbracket \quad \text{on } \tilde{\Gamma}, \quad (3.26)$$

$$\hat{m}^h_{nn} = C_{\alpha\beta\gamma\delta} u^h_{,\gamma\delta} n_\alpha n_\beta - \frac{\eta d}{h} (u^h_{,n} - g^\theta) \quad \text{on } \Gamma^\theta, \quad (3.27)$$

where  $d$  denotes a stiffness on inter-element edges which may be quantified in various ways but in order to obtain the dimensional consistency of the formulation, for instance,  $d = \mathcal{C}_{\alpha\beta\gamma\delta} \delta_{\alpha\beta} \delta_{\gamma\delta}$  or  $d = E t^3$ . Recalling the jump and average operators in equations (3.13) to (3.16), we have

$$\langle u_{,n}^h \rangle = \frac{1}{2} (u_{,\alpha}^{h+} n_{\alpha}^+ - u_{,\alpha}^{h-} n_{\alpha}^-) \quad \text{on } \tilde{\Gamma}, \quad (3.28)$$

$$\langle u_{,n}^h \rangle = u_{,\alpha} n_{\alpha} \quad \text{on } \Gamma^M, \quad (3.29)$$

$$\llbracket u_{,n}^h \rrbracket = u_{,\alpha}^{h+} n_{\alpha}^+ + u_{,\alpha}^{h-} n_{\alpha}^- \quad \text{on } \tilde{\Gamma}, \quad (3.30)$$

$$\llbracket u_{,n}^h \rrbracket = u_{,\alpha} n_{\alpha} \quad \text{on } \Gamma^{\theta}. \quad (3.31)$$

Substitution of numerical flux choices (3.24) to (3.27) into the weak formulation in equations (3.21) to (3.23) leads to a problem which reads: find  $u^h \in V^h$  such that

$$B^{\text{int}}(w^h, u^h) = L^{\text{int}}(w^h) \quad \forall w^h \in W^h, \quad (3.32)$$

where

$$\begin{aligned} B^{\text{int}}(w^h, u^h) = & \int_{\tilde{\Omega}} w_{,\alpha\beta}^h m_{\alpha\beta}^h d\Omega - \int_{\tilde{\Gamma} \cup \Gamma^{\theta}} \langle w_{,\alpha\beta}^h \mathcal{C}_{\alpha\beta\gamma\delta} n_{\gamma} n_{\delta} \rangle \llbracket u_{,n}^h \rrbracket d\Gamma \\ & - \int_{\tilde{\Gamma} \cup \Gamma^{\theta}} \llbracket w_{,n}^h \rrbracket \langle m_{\alpha\beta}^h n_{\alpha} n_{\beta} \rangle d\Gamma + \int_{\tilde{\Gamma} \cup \Gamma^{\theta}} \frac{\eta d}{h} \llbracket w_{,n}^h \rrbracket \llbracket u_{,n}^h \rrbracket d\Gamma, \end{aligned} \quad (3.33)$$

and

$$\begin{aligned} L^{\text{int}}(w^h) = & \int_{\tilde{\Omega}} w^h F d\Omega + \int_{\Gamma^M} w_{,n}^h M d\Gamma + \int_{\Gamma^Q} w^h Q d\Gamma \\ & - \int_{\Gamma^{\theta}} w_{,\alpha\beta}^h \mathcal{C}_{\alpha\beta\gamma\delta} n_{\gamma} n_{\delta} g^{\theta} d\Gamma + \int_{\Gamma^{\theta}} \frac{\eta d}{h} w_{,n}^h g^{\theta} d\Gamma. \end{aligned} \quad (3.34)$$

Equations (3.32) to (3.34) form a  $C^0$  interior penalty formulation for a Kirchhoff plate which has been presented by Engel et al. (2002) and Hughes and Garikipati (2004).

### 3.2.4 Lifting-type $C^0$ formulation

The plate formulation developed here is inspired by the works of Bassi and Rebay (1997) and Brezzi et al. (2000) on second-order problems. Essentially, jumps in normal derivatives across element boundaries will be transformed into additional curvature-like terms on element interiors.

### Lifting operations

Consider the function space  $\mathcal{R}^h$ ,

$$\mathcal{R}^h = \left\{ r_{\alpha\beta} \in L^2(\Omega) : r_{\alpha\beta}(K_i) \in P^l(K_i) \forall K_i \in \mathcal{P}^h, r_{\alpha\beta}(K_i) = r_{\beta\alpha}(K_i) \right\}. \quad (3.35)$$

A condition on the space  $\mathcal{R}^h$  is that it must contain at least all second derivatives of functions in  $V^h$  on element interiors. For triangular elements, this simply implies that  $l \geq k - 2$ , where  $k$  is the polynomial order of the functions in  $V^h$ .

For each element edge  $e \in \tilde{\Gamma}$ , a lifting operation is defined by: given  $\varphi \in H^1(\Omega)$ , find  $r_{\alpha\beta}^e \in \mathcal{R}^h$  such that

$$\int_{K^+ \cup K^-} \tau_{\alpha\beta}^h r_{\alpha\beta}^e(\varphi) d\Omega = - \int_e \langle \tau_{nn}^h \rangle \llbracket \varphi, n \rrbracket d\Gamma \quad \forall \tau_{\alpha\beta}^h \in \mathcal{R}^h, \quad (3.36)$$

where  $\tau_{nn}^h = \tau_{\alpha\beta}^h n_\alpha n_\beta$ ,  $K^+$  and  $K^-$  are elements sharing edge  $e$ . For  $e \in \Gamma^\theta$ , the lifting operation is defined by: given  $\varphi \in H^1(\Omega)$ , find  $r_{\alpha\beta}^e \in \mathcal{R}^h$  such that

$$\int_{K^+} \tau_{\alpha\beta}^h r_{\alpha\beta}^e(\varphi) d\Omega = - \int_e \langle \tau_{nn}^h \rangle \llbracket \varphi, n \rrbracket d\Gamma \quad \forall \tau_{\alpha\beta}^h \in \mathcal{R}^h, \quad (3.37)$$

A similar lifting operation is defined which includes the effect of rotation boundary conditions. For  $e \in \tilde{\Gamma}$ , it is defined by: given  $\varphi \in H^1(\Omega)$ , find  $r_{\alpha\beta}^{e,g} \in \mathcal{R}^h$  such that

$$\int_{K^+ \cup K^-} \tau_{\alpha\beta}^h r_{\alpha\beta}^{e,g}(\varphi) d\Omega = - \int_e \langle \tau_{nn}^h \rangle \llbracket \varphi, n \rrbracket d\Gamma \quad \forall \tau_{\alpha\beta}^h \in \mathcal{R}^h, \quad (3.38)$$

and for  $e \in \Gamma^\theta$  it is defined by: given  $\varphi \in H^1(\Omega)$ , find  $r_{\alpha\beta}^{e,g} \in \mathcal{R}^h$  such that

$$\int_{K^+} \tau_{\alpha\beta}^h r_{\alpha\beta}^{e,g}(\varphi) d\Omega = - \int_e \tau_{nn}^h (\varphi, n - g^\theta) d\Gamma \quad \forall \tau_{\alpha\beta}^h \in \mathcal{R}^h. \quad (3.39)$$

The lifting functions  $r_{\alpha\beta}^e$  and  $r_{\alpha\beta}^{e,g}$  are equal to zero for  $e \in \Gamma^M$ . Here,  $r_{\alpha\beta}^e$  and  $r_{\alpha\beta}^{e,g}$  vanish outside the elements sharing the edge  $e$ . Two functions  $R_{\alpha\beta}$  and  $R_{\alpha\beta}^g$  are now defined as follows:

$$R_{\alpha\beta} = \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} r_{\alpha\beta}^e, \quad (3.40)$$

$$R_{\alpha\beta}^g = \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} r_{\alpha\beta}^{e,g}. \quad (3.41)$$

### Lifting formulation

We consider the following choice of the numerical fluxes:

$$\hat{u}_{,n}^h = \langle u_{,n}^h \rangle \quad \text{on } \tilde{\Gamma} \cup \Gamma^M, \quad (3.42)$$

$$\hat{u}_{,n}^h = g^\theta \quad \text{on } \Gamma^\theta, \quad (3.43)$$

$$\hat{m}_{nn}^h = \langle m_{\alpha\beta}^{\star h} n_\alpha n_\beta \rangle + \eta \langle r_{\alpha\beta}^e(u^h) n_\alpha n_\beta \rangle \quad \text{on } \tilde{\Gamma}, \quad (3.44)$$

$$\hat{m}_{nn}^h = m_{\alpha\beta}^{\star h} n_\alpha n_\beta + \eta r_{\alpha\beta}^{e,g}(u^h) n_\alpha n_\beta \quad \text{on } \Gamma^\theta, \quad (3.45)$$

where

$$m_{\alpha\beta}^{\star h} = \mathcal{C}_{\alpha\beta\gamma\delta} \left\{ u_{,\gamma\delta}^h + R_{\gamma\delta}^g(u^h) \right\}, \quad (3.46)$$

and  $\eta$  is a dimensionless penalty parameter. Substituting equations (3.42) to (3.46) into equations (3.21) to (3.23), we get

$$\begin{aligned} & \int_{\tilde{\Omega}} w_{,\alpha\beta}^h m_{\alpha\beta}^h d\Omega - \int_{\tilde{\Gamma} \cup \Gamma^\theta} \langle w_{,\alpha\beta}^h \mathcal{C}_{\alpha\beta\gamma\delta} n_\gamma n_\delta \rangle \llbracket u_{,n}^h \rrbracket d\Gamma \\ & - \int_{\tilde{\Gamma} \cup \Gamma^\theta} \llbracket w_{,n}^h \rrbracket \langle m_{\alpha\beta}^h n_\alpha n_\beta \rangle d\Gamma - \int_{\tilde{\Gamma} \cup \Gamma^\theta} \llbracket w_{,n}^h \rrbracket \langle \mathcal{C}_{\alpha\beta\gamma\delta} R_{\gamma\delta}^g(u^h) n_\alpha n_\beta \rangle d\Gamma \\ & - \int_{\tilde{\Gamma} \cup \Gamma^\theta} \eta \llbracket w_{,n}^h \rrbracket r_{\gamma\delta}^{e,g}(u^h) d\Gamma = \int_{\tilde{\Omega}} w^h F d\Omega + \int_{\Gamma^M} w_{,n}^h M d\Gamma + \int_{\Gamma^Q} w^h Q d\Gamma \\ & - \int_{\Gamma^\theta} w_{,\alpha\beta}^h \mathcal{C}_{\alpha\beta\gamma\delta} n_\gamma n_\delta g^\theta d\Gamma \quad \forall w^h \in W^h. \end{aligned} \quad (3.47)$$

The lifting operations presented in equations (3.36) to (3.41) are applied in equation (3.47). This results in a variational formulation for Kirchhoff plates which involves: find  $u^h \in V^h$  such that

$$B^{\text{lif}}(w^h, u^h) = L^{\text{lif}}(w^h) \quad \forall w^h \in W^h, \quad (3.48)$$

where

$$\begin{aligned} B^{\text{lif}}(w^h, u^h) &= \int_{\tilde{\Omega}} \left( w_{,\alpha\beta}^h + R_{\alpha\beta}(w^h) \right) m_{\alpha\beta}^{\star h} d\Omega \\ &+ \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} \int_{\tilde{\Omega}} \eta r_{\alpha\beta}^e(w^h) \mathcal{C}_{\alpha\beta\gamma\delta} r_{\gamma\delta}^{e,g}(u^h) d\Omega, \end{aligned} \quad (3.49)$$

and

$$L^{\text{lif}}(w^h) = \int_{\tilde{\Omega}} w^h F d\Omega + \int_{\Gamma^M} w_{,n}^h M d\Gamma + \int_{\Gamma^Q} w^h Q d\Gamma. \quad (3.50)$$

Functions  $R_{\alpha\beta}(w^h)$  and  $R_{\gamma\delta}^g(u^h)$  are test and trial curvature-like functions, respectively. They represent the translation of jumps in the normal derivative

across element boundaries into additional curvature-like terms on element interiors. The last term in the RHS of equation (3.49) is a penalty-like term to enhance stability of the formulation. The lifting-type formulation in equations (3.48) to (3.50) has been presented in the work of Wells and Dung (2007).

The bilinear form in equation (3.49) is analogous to that presented by Bassi and Rebay (1997) for the diffusion equation, which was analysed and suitably stabilised by Brezzi et al. (2000). The lifting-type formulation has several interesting properties. As curvature-like terms on element interiors are employed, the method can straightforwardly be extended to nonlinear problems. As will be shown shortly, stability conditions can be readily defined.

### Modified lifting formulation

A special case of the above formulation can be obtained by adopting the following choice of numerical fluxes in the primal variational form in equation (3.21)

$$\hat{u}_{,n}^h = \langle u_{,n}^h \rangle \quad \text{on } \tilde{\Gamma} \cup \Gamma^M, \quad (3.51)$$

$$\hat{u}_{,n}^h = g^\theta \quad \text{on } \Gamma^\theta, \quad (3.52)$$

$$\hat{m}_{nn}^h = \langle m_{\alpha\beta}^h n_\alpha n_\beta \rangle + \eta \langle r_{\alpha\beta}^e(u^h) n_\alpha n_\beta \rangle \quad \text{on } \tilde{\Gamma}, \quad (3.53)$$

$$\hat{m}_{nn}^h = m_{\alpha\beta}^h n_\alpha n_\beta + \eta r_{\alpha\beta}^{e,g}(u^h) n_\alpha n_\beta \quad \text{on } \Gamma^\theta. \quad (3.54)$$

The bilinear form for this case is given by

$$\begin{aligned} B_{\text{mod}}^{\text{lif}}(w^h, u^h) &= \int_{\tilde{\Omega}} w_{,\alpha\beta}^h C_{\alpha\beta\gamma\delta} u_{,\gamma\delta}^h d\Omega + \int_{\tilde{\Omega}} w_{,\alpha\beta}^h C_{\alpha\beta\gamma\delta} R_{\gamma\delta}^g(u^h) d\Omega \\ &+ \int_{\tilde{\Omega}} R_{\alpha\beta}(w^h) C_{\alpha\beta\gamma\delta} u_{,\gamma\delta}^h d\Omega + \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} \int_{\tilde{\Omega}} \eta r_{\alpha\beta}^e(w^h) C_{\alpha\beta\gamma\delta} r_{\gamma\delta}^{e,g}(u^h) d\Omega, \end{aligned} \quad (3.55)$$

while the linear form is maintained as in equation (3.50). The modified bilinear form is similar to that of Bassi et al. (1997) for the diffusion equation, when  $\eta = 1$ . An attractive feature is that it leads to a sparser stiffness matrix than the formulation in equation (3.49) due to the absence of the term  $R_{\alpha\beta}(w^h) R_{\alpha\beta}^g(u^h)$ . However, an extension of the formulation to nonlinear problems is not obvious.

## 3.3 *A priori* analysis for the lifting formulation

Analyses for the interior penalty formulation can be found in the work of Engel et al. (2002). The analyses show that the dimensionless penalty parameter  $\eta$  must

be chosen suitably large to guarantee stability. The difficulty lies in how large the term must be, as it cannot be suitably quantified *a priori*. Furthermore, the conservative choice of a large penalty parameter to ensure stability can have a negative impact on accuracy of the numerical solution.

Here, we focus on the lifting formulation with *a priori* analyses for consistency, boundedness, stability and convergence.

### 3.3.1 Consistency

Consistency of the formulation can be proven straightforwardly by applying integration by parts to the bilinear form (3.49). This will be performed in terms of the moment tensor  $m_{\alpha\beta}^{*h}$  so that the approach holds for problems in which the moment tensor is a nonlinear function of the curvature. Expanding the first term on the RHS of equation (3.49) and considering the definition of the lifting function in equations (3.36) to (3.41), the Galerkin problem involves: find  $u^h \in V^h$  such that

$$\begin{aligned} \int_{\tilde{\Omega}} w^h_{,\alpha\beta} m_{\alpha\beta}^{*h} d\Omega - \int_{\tilde{\Gamma} \cup \Gamma^\theta} \llbracket w^h_{,n} \rrbracket \langle m_{nn}^{*h} \rangle d\Gamma \\ + \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} \int_{\Omega} \eta r_{\alpha\beta}^e(w^h) C_{\alpha\beta\gamma\delta} r_{\gamma\delta}^{e,g}(u^h) d\Omega \\ = \int_{\tilde{\Omega}} w^h F d\Omega + \int_{\Gamma^M} w^h_{,n} M d\Gamma + \int_{\Gamma^Q} w^h Q d\Gamma \quad \forall w^h \in W^h, \end{aligned} \quad (3.56)$$

where the term  $\int_{\tilde{\Omega}} R_{\alpha\beta}(w^h) m_{\alpha\beta}^{*h} d\Omega$  has been replaced by  $-\int_{\tilde{\Gamma} \cup \Gamma^\theta} \llbracket w^h_{,n} \rrbracket \langle m_{nn}^{*h} \rangle d\Gamma$ . This is possible as it is required that  $\mathcal{R}^h$  contain the second derivatives of all functions in  $V^h$  on element interiors. The reader is reminded that  $w^h_{,n} = w^h_{,\alpha} n_\alpha \Xi$  and  $m_{nn}^{*h} = m_{\alpha\beta}^* n_\alpha n_\beta$ . Replacing now  $u^h$  with the solution  $u$  to the problem defined in Section 3.1, and applying integration by parts twice to the first term in equation (3.56), we obtain

$$\begin{aligned} \int_{\tilde{\Omega}} w^h m_{\alpha\beta,\alpha\beta}^* d\Omega + \sum_{K \in \mathcal{P}^h} \int_{\partial K} w^h_{,\alpha} m_{\alpha\beta}^* n_\beta d\Gamma - \sum_{K \in \mathcal{P}^h} \int_{\partial K} w^h m_{\alpha\beta,\alpha}^* n_\beta d\Gamma \\ - \int_{\tilde{\Gamma} \cup \Gamma^\theta} \llbracket w^h_{,n} \rrbracket \langle m_{nn}^* \rangle d\Gamma + \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} \int_{\Omega} \eta r_{\alpha\beta}^e(w^h) C_{\alpha\beta\gamma\delta} r_{\gamma\delta}^{e,g}(u) d\Omega \\ = \int_{\tilde{\Omega}} w^h F d\Omega + \int_{\Gamma^M} w^h_{,n} M d\Gamma + \int_{\Gamma^Q} w^h Q d\Gamma \quad \forall w^h \in W^h. \end{aligned} \quad (3.57)$$

For a smooth line  $S$ , running from the point  $x^a$  to the point  $x^b$  in the direction  $s_\alpha$  on an element boundary, along which  $w^h$  and  $m_{\alpha\beta}^*$  are continuous, we have

$$\int_S w_{,\alpha}^h m_{\alpha\beta}^* n_\beta d\Gamma = \int_S w_{,\alpha}^h n_\alpha m_{nn}^* d\Gamma - \int_S w^h \left( m_{\alpha\beta}^* n_\beta s_\alpha \right)_{,s} d\Gamma + w^h m_{\alpha\beta}^* n_\beta s_\alpha \big|_{x^a}^{x^b}. \quad (3.58)$$

Recall that  $n_\alpha$  and  $s_\alpha$  are components of the unit outward normal and the unit tangent vectors on element boundaries. With the use of equation (3.58) together the identity (2.17), equation (3.57) can be rewritten as follows:

$$\begin{aligned} & \int_{\tilde{\Omega}} w^h m_{\alpha\beta,\alpha\beta}^* d\Omega + \int_{\Gamma} \llbracket w_{,\alpha}^h n_\alpha \Xi \rrbracket \langle m_{nn}^* \rangle d\Gamma + \int_{\tilde{\Gamma}} \langle w_{,\alpha}^h n_\alpha \Xi \rangle \llbracket m_{nn}^* \rrbracket d\Gamma \\ & - \int_{\Gamma} w^h \llbracket \left( m_{\alpha\beta}^* n_\beta s_\alpha \right)_{,s} \Xi \rrbracket d\Gamma + \sum_e w^h m_{ns}^* \big|_{x_e^a}^{x_e^b} - \int_{\Gamma} w^h \llbracket m_{\alpha\beta,\alpha}^* n_\beta \Xi \rrbracket d\Gamma \\ & - \int_{\tilde{\Gamma} \cup \Gamma^\theta} \llbracket w_{,n}^h \rrbracket \langle m_{nn}^* \rangle d\Gamma + \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} \int_{\Omega} \eta r_{\alpha\beta}^e \left( w^h \right) \mathcal{C}_{\alpha\beta\gamma\delta} r_{\gamma\delta}^{e,g}(u) d\Omega \\ & = \int_{\tilde{\Omega}} w^h F d\Omega + \int_{\Gamma^M} w_{,n}^h M d\Gamma + \int_{\Gamma^Q} w^h Q d\Gamma \quad \forall w^h \in W^h. \end{aligned} \quad (3.59)$$

Taking definitions  $w_{,n}^h = w_{,\alpha}^h n_\alpha \Xi$ ,  $m_{ns} = m_{\alpha\beta}^* n_\beta s_\alpha$  and  $m_{\alpha n,\alpha} = m_{\alpha\beta,\alpha}^* n_\beta \Xi$  into equation (3.57), we obtain

$$\begin{aligned} & \int_{\tilde{\Omega}} w^h m_{\alpha\beta,\alpha\beta}^* d\Omega + \int_{\Gamma} \llbracket w_{,n}^h \rrbracket \langle m_{nn}^* \rangle d\Gamma + \int_{\tilde{\Gamma}} \langle w_{,n}^h \rangle \llbracket m_{nn}^* \rrbracket d\Gamma \\ & - \int_{\Gamma} w^h \llbracket m_{ns,s}^* \rrbracket d\Gamma + \sum_e w^h m_{ns}^* \big|_{x_e^a}^{x_e^b} - \int_{\Gamma} w^h \llbracket m_{\alpha n,\alpha}^* \rrbracket d\Gamma \\ & - \int_{\tilde{\Gamma} \cup \Gamma^\theta} \llbracket w_{,n}^h \rrbracket \langle m_{nn}^* \rangle d\Gamma + \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} \int_{\Omega} \eta r_{\alpha\beta}^e \left( w^h \right) \mathcal{C}_{\alpha\beta\gamma\delta} r_{\gamma\delta}^{e,g}(u) d\Omega \\ & = \int_{\tilde{\Omega}} w^h F d\Omega + \int_{\Gamma^M} w_{,n}^h M d\Gamma + \int_{\Gamma^Q} w^h Q d\Gamma \quad \forall w^h \in W^h, \end{aligned} \quad (3.60)$$

where the term  $\sum_e w^h m_{ns}^* \big|_{x_e^a}^{x_e^b}$  is related to the virtual work of corner forces at element vertices. The end points of each edge are denoted  $x_e^b$  and  $x_e^a$ . Further details on this issue can be found in Hughes and Garikipati (2004). The average and

jump terms in the above equation are expanded as

$$\langle w_{,n} \rangle = \frac{1}{2} (w_{,\alpha}^+ n_{\alpha}^+ - w_{,\alpha}^- n_{\alpha}^-) \quad (3.61)$$

$$[[w_{,n}]] = w_{,\alpha}^+ n_{\alpha}^+ + w_{,\alpha}^- n_{\alpha}^- \quad (3.62)$$

$$\langle m_{nn}^* \rangle = \frac{1}{2} (m_{\alpha\beta}^{*+} n_{\alpha}^+ n_{\beta}^+ + m_{\alpha\beta}^{*-} n_{\alpha}^- n_{\beta}^-) \quad (3.63)$$

$$[[m_{nn}^*]] = m_{\alpha\beta}^{*+} n_{\alpha}^+ n_{\beta}^+ - m_{\alpha\beta}^{*-} n_{\alpha}^- n_{\beta}^- \quad (3.64)$$

$$[[m_{ns,s}^*]] = (m_{\alpha\beta}^{*+} n_{\alpha}^+ s_{\beta}^+)_{,s^+} + (m_{\alpha\beta}^{*-} n_{\alpha}^- s_{\beta}^-)_{,s^-} \quad (3.65)$$

$$[[m_{\alpha n,\alpha}^*]] = m_{\alpha\beta,\alpha}^{*+} n_{\beta}^+ + m_{\alpha\beta,\alpha}^{*-} n_{\beta}^-, \quad (3.66)$$

where the superscripts  $+$  and  $-$  again denote the elements  $K^+$  and  $K^-$  that share an edge. Equation (3.60) can be simplified to

$$\begin{aligned} & \int_{\Omega} w^h (m_{\alpha\beta,\alpha\beta}^* - F) d\Omega - \int_{\Gamma^Q} w^h (m_{\alpha n,\alpha}^* + m_{nn,s}^* + Q) d\Gamma \\ & + \int_{\Gamma^M} w_{,n}^h (m_{nn}^* - M) d\Gamma + \int_{\bar{\Gamma}} \langle w_{,n}^h \rangle [[m_{nn}^*]] d\Gamma - \int_{\bar{\Gamma}} w^h [[m_{\alpha n,\alpha}^* n_{\alpha} + m_{ns,s}^*]] d\Gamma \\ & + \sum_e w^h m_{ns}^* |_{x_e^a}^{x_e^b} + \sum_{e \in \bar{\Gamma} \cup \Gamma^{\theta}} \int_{\Omega} \eta r_{\alpha\beta}^e (w^h) C_{\alpha\beta\gamma\delta} r_{\gamma\delta}^{e,g} (u) d\Omega = 0 \quad \forall w^h \in W^h, \end{aligned} \quad (3.67)$$

in which, due to the continuity of the exact solution  $w$ , all jump terms vanish. The above equation holds for all  $w^h \in W^h$ , hence the formulation is consistent. It is also possible to interpret the Euler-Lagrange equations (3.1) to (3.6) from the above expression. The first integral in equation (3.67) implies the governing equation in (3.1), the second integral implies the modified shear boundary condition (3.5), the third term implies satisfaction of the moment boundary condition, the fourth term implies continuity of the normal moment across inter-element boundaries, the fifth term implies continuity of the modified normal shear across element boundaries and the sixth term implies equilibrium of corner forces. The stabilising term enforces the normal rotation boundary condition and continuity of the normal derivative of the deflection across element boundaries. Enforcement of the normal rotation boundary condition and the continuity of normal rotations across element boundaries implies the  $R_{\alpha\beta}^g = 0$ , hence  $m_{\alpha\beta}^* = m_{\alpha\beta}$ .

A direct consequence of consistency is Galerkin orthogonality. It is trivial to show that if  $u$  is the exact solution and  $u^h$  is the solution to the finite element problem, then

$$B(w^h, e) = 0 \quad \forall w^h \in W^h, \quad (3.68)$$

where  $e = u - u^h$ .

### 3.3.2 Boundedness and stability

Besides consistency, boundedness and stability are important ingredients for convergence analyses of the formulation. These properties can be demonstrated following the analyses of Brezzi et al. (2000) and Arnold et al. (2002) for second-order problems in terms of a mesh-dependent norm. It is assumed that the constitutive tensor  $\mathcal{C}_{\alpha\beta\gamma\delta}$  given by equation (3.7) is positive definite, which implies that there exists a positive number  $C_0$  such that

$$a_{\alpha\beta}\mathcal{C}_{\alpha\beta\gamma\delta}a_{\gamma\delta} \geq C_0 a_{\alpha\beta}a_{\alpha\beta} \quad \forall a_{\alpha\beta} \in \mathcal{R}^h. \quad (3.69)$$

For the sake of convenience, homogeneous Dirichlet problems are considered,  $\Gamma^u = \partial\Omega$ ,  $g^u = 0$  and  $g^\theta = 0$ . We consider the following broken Sobolev space  $V_\star^h$

$$V_\star^h = \left\{ v \in H_0^1(\Omega) : v \in H^2(\tilde{\Omega}) \right\}, \quad (3.70)$$

and we define the mesh dependent energy-like norm  $|||v|||$  as follows

$$|||v|||^2 = |v|_{2,\tilde{\Omega}}^2 + \sum_e \left\| r_{\alpha\beta}^e(v) \right\|_{0,\tilde{\Omega}}^2. \quad (3.71)$$

Equation (3.71) is a norm due to the assumption of homogeneous displacement boundary conditions on the entire boundary.

#### Boundedness

The bilinear form (3.49) is bounded in terms of the norm  $|||\cdot|||$  as there exists a positive number  $C_b$  such that

$$B^{\text{lif}}(w, v) \leq C_b |||w||| |||v||| \quad \forall w, v \in V_\star^h, \quad (3.72)$$

where  $C_b$  depends only on  $\mathcal{C}_{\alpha\beta\gamma\delta}$ ,  $\eta$ , the minimum corner angle of elements in the mesh and the polynomial orders which are used to construct elements (Engel et al., 2002).

Decomposing the bilinear form (3.49) for the case of homogeneous Dirichlet boundary condition, we obtain

$$\begin{aligned} B^{\text{lif}}(w, v) = & (w_{,\alpha\beta}, \mathcal{C}_{\alpha\beta\gamma\delta} v_{,\gamma\delta})_{\tilde{\Omega}} + (R_{\alpha\beta}(w), \mathcal{C}_{\alpha\beta\gamma\delta} v_{,\gamma\delta})_{\tilde{\Omega}} \\ & + (w_{,\alpha\beta}, \mathcal{C}_{\alpha\beta\gamma\delta} R_{\gamma\delta}(v))_{\tilde{\Omega}} + (R_{\alpha\beta}(w), \mathcal{C}_{\alpha\beta\gamma\delta} R_{\gamma\delta}(v))_{\tilde{\Omega}} \\ & + \eta \sum_{e \in \Gamma} \left( r_{\alpha\beta}^e(w), \mathcal{C}_{\alpha\beta\gamma\delta} r_{\gamma\delta}^e(v) \right)_{\tilde{\Omega}}, \end{aligned} \quad (3.73)$$

where  $(\cdot, \cdot)_{\tilde{\Omega}}$  denotes the  $L^2$  inner product on  $\tilde{\Omega}$ . Now, we define a norm of the fourth-order constitutive tensor  $\mathcal{C}_{\alpha\beta\gamma\delta}$  as follows (Engel et al., 2002):

$$\mathcal{C} = \|\mathcal{C}_{\alpha\beta\gamma\delta}\| = \sup_{(\varphi_{\alpha\beta} \in \mathcal{R}^h, \varphi_{\alpha\beta} \neq 0)} \frac{\|\mathcal{C}_{\alpha\beta\gamma\delta} \varphi_{\alpha\beta}\|}{\|\varphi_{\alpha\beta}\|}, \quad (3.74)$$

where  $\|\varphi_{\alpha\beta}\|$  denotes the  $L^2$  norm of the second-order tensor  $\varphi_{\alpha\beta}$ . Application of the Cauchy-Schwarz inequality  $\|(\varphi, \phi)\| \leq (\varphi, \varphi)^{1/2} (\phi, \phi)^{1/2}$ , where  $\varphi$  and  $\phi$  are members of the  $L^2$  space, in (3.73) leads to

$$\begin{aligned} B^{\text{lif}}(w, v) &\leq \mathcal{C} |w|_{2, \tilde{\Omega}} |v|_{2, \tilde{\Omega}} + \mathcal{C} \|R_{\alpha\beta}(v)\|_{0, \tilde{\Omega}} |v|_{2, \tilde{\Omega}} + \mathcal{C} |w|_{2, \tilde{\Omega}} \|R_{\alpha\beta}(v)\|_{0, \tilde{\Omega}} \\ &\quad + \mathcal{C} \|R_{\alpha\beta}(w)\|_{0, \tilde{\Omega}} \|R_{\alpha\beta}(v)\|_{0, \tilde{\Omega}} + \mathcal{C} \eta \sum_{e \in \Gamma} \left( \|r_{\alpha\beta}^e(w)\|_{0, \tilde{\Omega}} \|r_{\alpha\beta}^e(v)\|_{0, \tilde{\Omega}} \right). \end{aligned} \quad (3.75)$$

In addition, from the definition of the lifting function  $R_{\alpha\beta}$  in equation (3.40) and the triangle inequality, it can be shown that

$$\|R_{\alpha\beta}(v)\|_{0, \tilde{\Omega}}^2 \leq N_e \sum_{e \in \Gamma} \|r_{\alpha\beta}^e(v)\|_{0, \tilde{\Omega}}^2, \quad (3.76)$$

where  $N_e$  is the number of edges belonging to an element  $K$ ,  $N_e = 3$  for triangles. With this result together the Cauchy-Schwarz inequality, it is straightforward to demonstrate that

$$B^{\text{lif}}(w, v) \leq C_b \left( |w|_{2, \tilde{\Omega}}^2 + \sum_e \|r_{\alpha\beta}^e(w)\|_{0, \tilde{\Omega}}^2 \right)^{\frac{1}{2}} \left( |v|_{2, \tilde{\Omega}}^2 + \sum_{e \in \Gamma} \|r_{\alpha\beta}^e(v)\|_{0, \tilde{\Omega}}^2 \right)^{\frac{1}{2}}, \quad (3.77)$$

where  $C_b = \mathcal{C} (N_e + \eta)$ , which means that equation (3.72) holds.

A similar procedure can be used to prove the boundedness property of the modified bilinear form (3.55).

### Stability

The presented formulation (3.48) is stable with respect to the norm  $|||\cdot|||$  if there exists a positive number  $C_s$  such that

$$B^{\text{lif}}(v, v) \geq C_s |||v|||^2 \quad \forall v \in V_{\star}^h. \quad (3.78)$$

Setting  $w^h = u^h = v$  in equation (3.49) and the using of the positive definite identity (3.69) lead to

$$B^{\text{lif}}(v, v) \geq C_0 |v|_{2, \tilde{\Omega}}^2 + 2 (R_{\alpha\beta}(v), \mathcal{C}_{\alpha\beta\gamma\delta} v_{,\gamma\delta})_{\tilde{\Omega}} + C_0 \|R_{\alpha\beta}(v)\|_{0, \tilde{\Omega}}^2 + \eta C_0 \sum_{e \in \Gamma} \|r_{\alpha\beta}^e(v)\|_{0, \tilde{\Omega}}^2. \quad (3.79)$$

Application of Young's inequality  $2\varphi\phi \leq \epsilon\varphi^2 + (1/\epsilon)\phi^2$ , where  $\epsilon > 0$ , to the term  $(R_{\alpha\beta}(v), \mathcal{C}_{\alpha\beta\gamma\delta} v_{,\gamma\delta})_{\tilde{\Omega}}$  of the above equation, yields

$$B^{\text{lif}}(v, v) \geq C_0 (1 - \epsilon) |v|_{2, \tilde{\Omega}}^2 + C_0 \left(1 - \frac{1}{\epsilon}\right) \|R_{\alpha\beta}(v)\|_{0, \tilde{\Omega}}^2 + C_0 \eta \sum_{e \in \Gamma} \|r_{\alpha\beta}^e(v)\|_{0, \tilde{\Omega}}^2. \quad (3.80)$$

Considering now equation (3.76) and the case  $\epsilon < 1$ , it follows that

$$B^{\text{lif}}(v, v) \geq C_0 (1 - \epsilon) |v|_{2, \tilde{\Omega}}^2 + C_0 \left(\eta + N_e - \frac{N_e}{\epsilon}\right) \sum_{e \in \Gamma} \|r_{\alpha\beta}^e(v)\|_{0, \tilde{\Omega}}^2. \quad (3.81)$$

This yields  $C_s = \min(1 - \epsilon, \eta + N_e - N_e/\epsilon)$  for equation (3.78). The reader is reminded that  $N_e = 3$  for a mesh of triangles. Since  $C_s$  is a positive number, the requirement for choosing  $\epsilon$  is that  $N_e/(\eta + N_e) < \epsilon < 1$ . This simply implies stability in the sense of equation (3.78) for any  $\eta > 0$ .

Stability analysis for the modified bilinear form in equation (3.55) can be performed analogously, and leads to the stability condition  $\eta > N_e$ .

### 3.3.3 Convergence estimates

Besides the results for proving of consistency, boundedness and stability in the previous sections, a bound on the approximation error is required for *a priori* error estimates. For  $u^I \in V^h$  be the nodal interpolant of the exact solution  $u$ , the following standard approximation bound is used

$$\sum_K |u - u^I|_{m, K}^2 \leq Ch^{2(k+1-m)} |u|_{k+1, \Omega}^2, \quad (3.82)$$

where  $C$  is a positive number,  $k$  and  $m$  are the orders of the approximation polynomial and the considered error norm, respectively.

### Error in the energy-like norm

The consideration of the triangle inequality leads to

$$\left\| u - u^h \right\| = \left\| (u - u^I) + (u^I - u^h) \right\| \leq \left\| u - u^I \right\| + \left\| u^I - u^h \right\|. \quad (3.83)$$

The term  $\left\| u^I - u^h \right\|$  can be bounded by the stability estimate in equation (3.78)

$$\begin{aligned} \left\| u^I - u^h \right\|^2 &\leq C_s^{-1} \left| B^{\text{lif}} \left( u^I - u^h, u^I - u^h \right) \right| \\ &= C_s^{-1} \left| B^{\text{lif}} \left( u^I - u^h, u^I - u \right) + B^{\text{lif}} \left( u^I - u^h, u - u^h \right) \right| \\ &= C_s^{-1} \left| B^{\text{lif}} \left( u^I - u^h, u^I - u \right) \right|, \end{aligned} \quad (3.84)$$

where Galerkin orthogonality (3.68) has been used. Applying the boundedness property of the bilinear form in equation (3.72) to the RHS of the above equation, we get

$$\left\| u^I - u^h \right\|^2 \leq C_s^{-1} C_b \left\| u^I - u \right\| \left\| u^I - u^h \right\|. \quad (3.85)$$

The results in equations (3.83) and (3.85) immediately lead to

$$\left\| u - u^h \right\| \leq \left( 1 + C_s^{-1} C_b \right) \left\| u - u^I \right\|. \quad (3.86)$$

As usual, the error estimation is now an approximation problem, in which the error is determined by the difference between the exact solution and the nodal interpolant, in the defined energy-like norm. The norm in the RHS of equation (3.86) is expressed as

$$\left\| u - u^I \right\|^2 = \left| u - u^I \right|_{2,\tilde{\Omega}}^2 + \sum_{e \in \Gamma} \left\| r_{\alpha\beta}^e \left( u - u^I \right) \right\|_{0,\tilde{\Omega}}^2. \quad (3.87)$$

While the first semi-norm term on the RHS of the above equation can be bounded by the standard interpolation estimate (3.82), the second term requires some extra work. Note that, here, the inter-element jump terms in the norm  $\left\| u^I \right\|$  do not necessarily vanish. For  $v \in V_\star^h$  and for each edge  $e \in \Gamma$ , by recalling the definition of the lifting function  $r_{\alpha\beta}^e(v)$  in equations (3.36) and (3.37) in which  $\tau_{\alpha\beta}^h$  is replaced by  $r_{\alpha\beta}^e(v)$ , we derive the following bound

$$\begin{aligned} \left\| r_{\alpha\beta}^e(v) \right\|_{0,\tilde{\Omega}}^2 &\leq \int_e \left| \langle r_{nn}^e(v) \rangle \llbracket v, n \rrbracket \right| d\Gamma \\ &\leq \left\| \langle r_{nn}^e(v) \rangle \right\|_{0,e} \left\| \llbracket v, n \rrbracket \right\|_{0,e} \\ &\leq \left\| \langle r_{\alpha\beta}^e(v) \rangle \right\|_{0,e} \left\| \llbracket v, n \rrbracket \right\|_{0,e} \\ &\leq Ch_e^{-1/2} \left\| r_{\alpha\beta}^e(v) \right\|_{0,\tilde{\Omega}} \left\| \llbracket v, n \rrbracket \right\|_{0,e}, \end{aligned} \quad (3.88)$$

where  $h_e$  is a mesh-size parameter corresponding to the edge  $e$  and  $C$  is a positive number independent of  $h_e$ . In the last inequality, the scaling argument has been used (Brezzi et al., 2000; Lew et al., 2004). From the above equation, it is clear that

$$\sum_{e \in \Gamma} \left\| r_{\alpha\beta}^e(v) \right\|_{0,\tilde{\Omega}}^2 \leq C \sum_{e \in \Gamma} h_e^{-1} \left\| \llbracket v, n \rrbracket \right\|_{0,e}^2. \quad (3.89)$$

Application of a trace inequality (Arnold, 1982, Eqn 2.5),

$$\|v, n\|_{0,e}^2 \leq C \left( h_e^{-1} |v|_{1,K}^2 + h_e |v|_{2,K}^2 \right) \quad \forall v \in H^2(K), \quad (3.90)$$

leads to

$$\sum_{e \in \Gamma} \left\| r_{\alpha\beta}^e(v) \right\|_{0,\tilde{\Omega}}^2 \leq C \sum_{K \in \mathcal{P}^h} \left( h_e^{-2} |v|_{1,K}^2 + |v|_{2,K}^2 \right). \quad (3.91)$$

With the above result, by taking  $v = u - u^I$ , equation (3.87) follows by

$$\left\| \left\| u - u^I \right\| \right\|^2 \leq \left\| u - u^I \right\|_{2,\tilde{\Omega}}^2 + C \sum_K \left( h_e^{-2} \left\| u - u^I \right\|_{1,K}^2 + \left\| u - u^I \right\|_{2,K}^2 \right). \quad (3.92)$$

Using the standard interpolation estimate (3.82) in the above inequality, we obtain the interpolation estimate

$$\left\| \left\| u - u^I \right\| \right\|^2 \leq C h^{2k-2} |u|_{k+1,\Omega}^2, \quad (3.93)$$

and incorporation into equation (3.86) completes error estimate

$$\left\| \left\| u - u^h \right\| \right\| \leq C_e h^{k-1} |u|_{k+1,\Omega}, \quad (3.94)$$

where  $C_e$  is a positive number. The bound shows that convergence in terms of the norm  $\left\| \left\| \cdot \right\| \right\|$  is optimal for  $k \geq 2$ , subject to  $l \geq k - 2$  which is required to ensure consistency and consequently Galerkin orthogonality. From the above result, the convergence in energy-like norm is predicted to be order one when using quadratic elements and order two when using cubic elements.

### Error in the $L^2$ norm

The convergence rate proof in the  $L^2$  norm follows a standard duality argument. The key point for duality-base error estimates is the use of an auxiliary problem, so-called the dual problem. Consider the dual problem with homogeneous Dirichlet boundary conditions for which the variational formulation reads: find  $\varphi \in H_0^4(\Omega)$  such that

$$B^{\text{lif}}(v, \varphi) = \left( v, u - u^h \right)_\Omega \quad \forall v \in V_\star^h, \quad (3.95)$$

where  $v$  is the test function,  $u$  is the exact solution to the original problem and  $u^h \in V_\star^h$  is the finite element solution to the original problem. It is important to note that the following arguments rely on the regularity assumption  $\varphi \in H_0^4(\Omega)$ . This is a significant assumption, since the solution to a fourth-order problem on polygonal domains does not generally lie in the space  $H_0^4(\Omega)$ . It will however be shown that the resulting error estimate holds for a number of numerical examples. As  $u - u^h$  belongs  $L^2(\Omega)$ , the following standard elliptic regularity estimate holds (Ciarlet, 1978):

$$\|\varphi\|_{4,\Omega} \leq C \|u - u^h\|_{0,\Omega}. \quad (3.96)$$

Setting  $v = u - u^h$  in equation (3.95), we have

$$\begin{aligned} \|u - u^h\|_{0,\Omega}^2 &= B^{\text{lif}}(u - u^h, \varphi) \\ &= B^{\text{lif}}(u - u^h, \varphi - \varphi^h) + B^{\text{lif}}(u - u^h, \varphi^h) \\ &= B^{\text{lif}}(u - u^h, \varphi - \varphi^h) \\ &\leq C_b \|u - u^h\| \|\varphi - \varphi^h\| \quad \forall u^h \in V^h. \end{aligned} \quad (3.97)$$

where  $\varphi^h$  is the finite element solution to the dual problem. We note that  $B^{\text{lif}}(u - u^h, \varphi^h) = 0$  by Galerkin orthogonality. While the bound for  $\|u - u^h\|$  is immediately provided by equation (3.94), the bound for  $\|\varphi - \varphi^h\|$  is given by the incorporation of equations (3.94) and (3.96). It follows that for  $k \geq 3$

$$\begin{aligned} \|\varphi - \varphi^h\| &\leq Ch^2 \|\varphi\|_{4,\Omega} \\ &\leq Ch^2 \|u - u^h\|_{0,\Omega}, \end{aligned} \quad (3.98)$$

and for  $k = 2$

$$\begin{aligned} \|\varphi - \varphi^h\| &\leq Ch \|\varphi\|_{4,\Omega} \\ &\leq Ch \|u - u^h\|_{0,\Omega}. \end{aligned} \quad (3.99)$$

For  $k \geq 3$ , inserting the bound in equation (3.98) into equation (3.97) with the consideration of the energy-like norm bound (3.94),

$$\|u - u^h\|_{0,\Omega}^2 \leq Ch^{k+1} \|u - u^h\|_{0,\Omega} \|u\|_{k+1,\Omega}, \quad (3.100)$$

therefore, for  $k \geq 3$ ,

$$\|u - u^h\|_{0,\Omega} \leq Ch^{k+1} \|u\|_{k+1,\Omega}, \quad (3.101)$$

which is optimal. For example, when cubic ( $k = 3$ ) elements are used the convergence rate is predicted to be order of four. For the case of quadratic elements ( $k = 2$ ), the same steps are followed with the use of equation (3.99) instead equation (3.98) and lead to

$$\|u - u^h\|_{0,\Omega} \leq Ch^k \|u\|_{k+1,\Omega}, \quad (3.102)$$

that indicates convergence of order of two. While this estimate may appear disappointing, the simplicity of the formulation with constant lifting functions still makes it attractive.

Following a similar procedure, error estimates in terms of the  $H^1$  norm can be proven in which the convergence is optimal for  $k \geq 2$ :

$$\|u - u^h\|_{1,\Omega} \leq Ch^k \|u\|_{k+1,\Omega}. \quad (3.103)$$

## 3.4 Implementation aspects

Besides the standard procedure used in continuous Galerkin finite element methods, some additional work is required when implementing discontinuous Galerkin formulations. In this section, key points for implementing the presented formulations are discussed.

### 3.4.1 Finite elements

Although the proposed method is not restricted to a particular type of finite elements, we confine our attention to standard isoparametric triangular elements of order two (quadratic) and order three (cubic).

Within each finite element, the position of a point on the plate relative to the Cartesian coordinate system is given by an isoparametric mapping from a point in a *parent triangle* using area coordinates  $(s_1, s_2, s_3)$ . Note that  $s_1 + s_2 + s_3 = 1$ . Figure 3.3 illustrates the 6-node and 10-node parent triangles in the area coordinates. Let  $A_i$ ,  $i = 1 \rightarrow 3$ , denote the areas of triangles as sketched in Figure 3.3. It is obvious that  $s_i = A_i / (A_1 + A_2 + A_3) = 2 A_i$ .

Since elements are isoparametric, the position coordinates and the displacement at a point are evaluated using the same shape functions. In each element, the representing functions of  $u^h$  and  $w^h$  and their derivatives can be expressed in

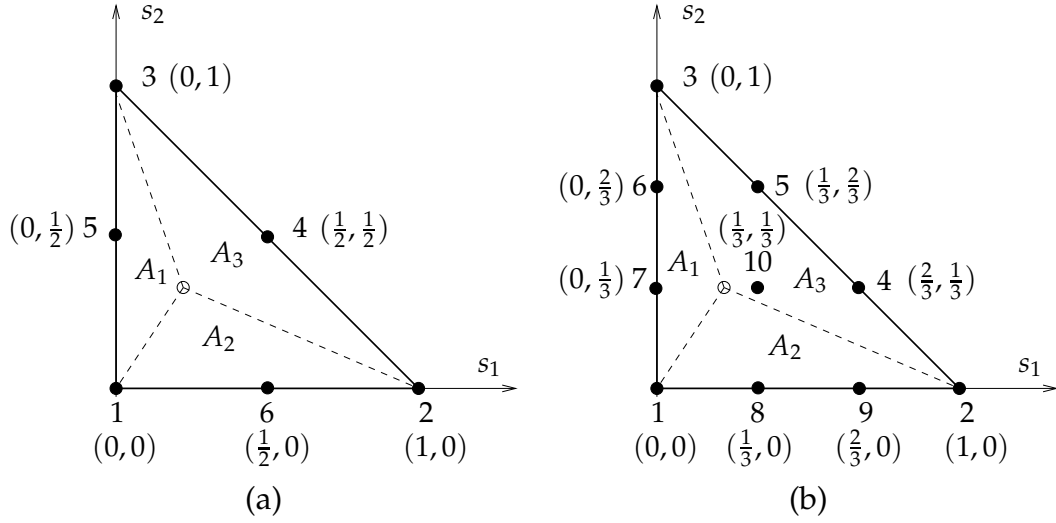


Figure 3.3: Parent triangles using area coordinates: (a) Quadratic six-node elements; (b) Cubic ten-node elements.

term of nodal displacement vectors which are respectively denoted by  $\mathbf{a}$  and  $\mathbf{b}$ . These functions can be expressed by:

$$u^h = \mathbf{N}^k \mathbf{a}, \quad (3.104)$$

$$u^h_{,\alpha} = \mathbf{N}^k_{\alpha} \mathbf{a}, \quad (3.105)$$

$$u^h_{,\alpha\beta} = \mathbf{N}^k_{\alpha\beta} \mathbf{a}, \quad (3.106)$$

and

$$w^h = \mathbf{N}^k \mathbf{b}, \quad (3.107)$$

$$w^h_{,\alpha} = \mathbf{N}^k_{\alpha} \mathbf{b}, \quad (3.108)$$

$$w^h_{,\alpha\beta} = \mathbf{N}^k_{\alpha\beta} \mathbf{b}, \quad (3.109)$$

where  $\mathbf{N}^k$  denotes the matrix of polynomial shape functions  $P^k(K_i)$ , and  $\mathbf{N}^k_{\alpha}$  and  $\mathbf{N}^k_{\alpha\beta}$  are matrices of the first and the second derivatives of the shape functions, respectively. Since  $u^h_{,\alpha\beta}$  and  $w^h_{,\alpha\beta}$  are symmetric second-order tensors, they can be simply rewritten as vectors of three components.

On each element edge  $e$ , let  $\mathbf{a}_e$  and  $\mathbf{b}_e$  denote the trial and weight nodal displacement vectors of elements  $K^+$  and  $K^-$  sharing  $e$ . Jump terms on  $e$  are discretised as

$$\llbracket u^h_{,n} \rrbracket = \llbracket u^h_{,\alpha} n_{\alpha} \Xi \rrbracket = \mathbf{N}^k_j \mathbf{a}_e, \quad (3.110)$$

$$\llbracket w^h_{,n} \rrbracket = \llbracket w^h_{,\alpha} n_{\alpha} \Xi \rrbracket = \mathbf{N}^k_j \mathbf{b}_e, \quad (3.111)$$

and average terms on  $e$  are discretised as

$$\left\langle u_{,\alpha\beta}^h \mathcal{C}_{\alpha\beta\gamma\delta} n_\gamma n_\delta \right\rangle = \mathbf{N}_a^k \mathbf{a}_e, \quad (3.112)$$

$$\left\langle w_{,\alpha\beta}^h \mathcal{C}_{\alpha\beta\gamma\delta} n_\gamma n_\delta \right\rangle = \mathbf{N}_a^k \mathbf{b}_e, \quad (3.113)$$

where  $\mathbf{N}_j^k$  and  $\mathbf{N}_a^k$  denote interpolation matrices for the jump of normal slopes and the average of normal bending moments on  $e$ , respectively.

### 3.4.2 Lifting implementation

When the lifting formulation is used, shape functions are used to represent  $r_{\alpha\beta}^e(u^h)$ ,  $r_{\alpha\beta}^e(w^h)$  and  $\tau_{\alpha\beta}^h \in \mathcal{R}^h$ . These second order tensors are rewritten in vectors of three components. Denoting  $\mathbf{r}_i$ ,  $\bar{\mathbf{r}}_i$ , and  $\boldsymbol{\tau}_i$  respectively be vectors of the trial, the test, and an arbitrary curvature-like function at nodes of element  $K_i$  and  $\boldsymbol{\tau}_e$  be a vector of arbitrary curvature-like functions at nodes of elements  $K_i$  sharing  $e$ , we adopt the following relation:

$$r_{\alpha\beta}^e(u^h) = \mathbf{N}^l \mathbf{r}_i, \quad (3.114)$$

$$r_{\alpha\beta}^e(w^h) = \mathbf{N}^l \bar{\mathbf{r}}_i, \quad (3.115)$$

and

$$\tau_{\alpha\beta}^h = \mathbf{N}^l \boldsymbol{\tau}_i, \quad (3.116)$$

$$\left\langle \tau_{\alpha\beta}^h n_\alpha n_\beta \right\rangle = \mathbf{N}_a^l \boldsymbol{\tau}_e, \quad (3.117)$$

where  $\mathbf{N}^l$  and  $\mathbf{N}_a^l$  are the interpolation matrices containing the shape functions. The lifting operations in equations (3.36) to (3.39) for each sharing edge  $e$  lead to the following expressions:

$$r_{\alpha\beta}^e(u^h) = -\mathbf{N}^l \left( \int_K \mathbf{N}^{lT} \mathbf{N}^l d\Omega \right)^{-1} \int_e \mathbf{N}_a^{lT} \mathbf{N}_j^k d\Gamma \mathbf{a}_e, \quad (3.118)$$

$$r_{\alpha\beta}^e(w^h) = -\mathbf{N}^l \left( \int_K \mathbf{N}^{lT} \mathbf{N}^l d\Omega \right)^{-1} \int_e \mathbf{N}_a^{lT} \mathbf{N}_j^k d\Gamma \mathbf{b}_e, \quad (3.119)$$

$$r_{\alpha\beta}^{e,g}(u^h) = r_{\alpha\beta}^e(u^h) - \mathbf{N}^l \left( \int_K \mathbf{N}^{lT} \mathbf{N}^l d\Omega \right)^{-1} \int_e \mathbf{N}_a^{lT} g^\theta d\Gamma. \quad (3.120)$$

An important point is the choice of lifting shape functions  $P^l(K_i)$ . It is required that  $l \geq k - 2$ . The global stiffness matrices for the interior penalty formulation in equation (3.33) and the modified lifting formulation in equation (3.55)

are sparser than for the lifting formulation in equation (3.49). This derives from the fact that while the additional integrations for the interior penalty formulation and the modified lifting formulation couple neighbouring elements, the term  $R_{\alpha\beta}(w^h) R_{\alpha\beta}^g(u^h)$  for the lifting formulation couples all elements surrounding a given element. Shown in Figure 3.4 are the involved elements when a displacement with respect to a degree of freedom inside an element  $K$  is given. In general, ten elements are involved for the case of the lifting formulation and four elements are involved for the case of the interior penalty formulation as well as the modified lifting formulation (Brezzi et al., 2000).

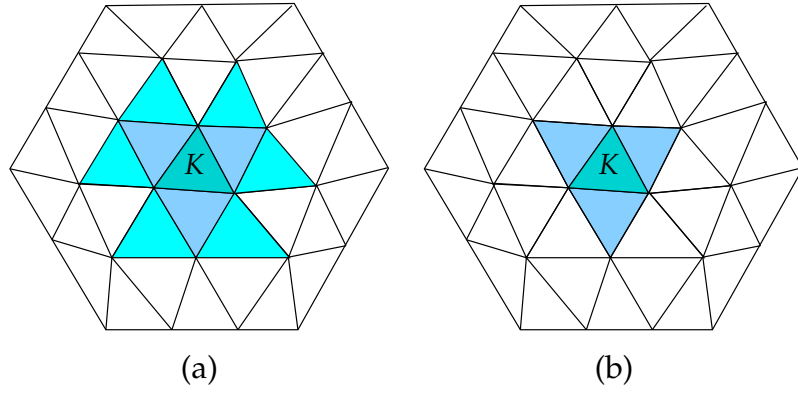


Figure 3.4: Elements coupled in: (a) the lifting formulation; (b) the interior penalty formulation and the modified lifting formulation.

### 3.4.3 Development environment

The code which has been developed for the finite element implementation in this chapter as well as in Chapters 5 and 6 is written in the C++ programming language. Specifically, it is built on DOLFIN, the C++/Python solving interface of FEniCS - an open source software for automated solution of differential equations. For more information about DOLFIN and FEniCS, the reader can refer to works by Hoffman et al. (2007) and Logg (2007).

## 3.5 Numerical examples

In this section, four practical problems of thin plates are addressed. Firstly, thin plate bending under static loading is examined. Next, the performance of the

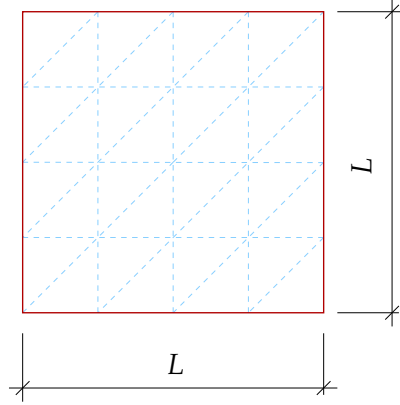


Figure 3.5: Mesh refinement example for square plate, 4 partitions per edge

method for the eigenvalue problems of plate buckling is considered. Then, the free vibration of plate is investigated. Finally, the behaviour of elements with respect to distortion in an unstructured mesh is examined. In these examples, the influence of the penalty parameter on the structural behaviour is of particular interest. In the first three examples, structured meshes of triangular elements are used with an equal number of elements in each spatial direction as shown in Figure 3.5. For all examples, symmetry is not exploited, and the entire structure is modelled.

### 3.5.1 Plate bending

We consider a simply-supported square plate with side length  $L = 2$  m, thickness  $t = 0.01$  m, Young's modulus  $E = 2 \times 10^8$  N/m<sup>2</sup> and Poisson's ratio  $\nu = 0.3$ . As the plate is simply supported,  $\Gamma^u = \Gamma^M = \partial\Omega$ ,  $g^u = 0$  and  $M = 0$ . A uniform distributed force  $F = 10$  N/m<sup>2</sup> is applied. A series reference solution of the displacement for this problem can be found in Timoshenko and Woinowsky-Krieger (1959, pages 197-202). It is used as the reference in computing the error for various finite element solutions. The deformed plate is illustrated in Figure 3.6. We present here results for quadratic ( $k = 2$ ) and cubic ( $k = 3$ ) elements with various values of the lifting function order  $l$  and the penalty parameter  $\eta$ .

#### Quadratic elements

The case of quadratic triangular elements ( $k = 2$ ) with constant lifting functions ( $l = 0$ ) are considered here. According to the *a priori* error analysis presented

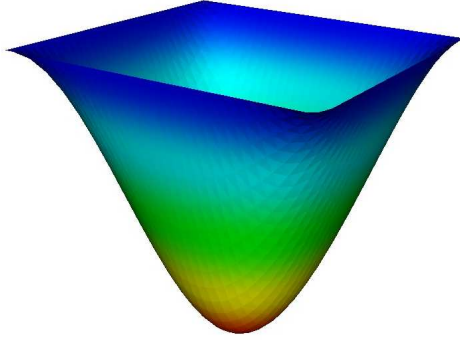


Figure 3.6: Illustration of the deformation of a simply support plate.

in Section 3.3.3, the predicted convergence rate in the  $L^2$  norm and in the  $H^1$  semi-norm is two, and in the  $H^2$  energy-like norm is one.

In Figure 3.7, the  $L^2$  error in the displacement is shown for various values of the penalty parameter. The predicted convergence rate of two is observed for higher penalty values, roughly  $\eta > 1$ , and is approximated for lower values. The computed error in the  $H^1$  semi-norm is shown in Figure 3.8, demonstrating the predicted rate of convergence. The error in the ‘broken’ energy norm  $|u - u^h|_{2,\tilde{\Omega}}$ , which is of practical interest, is presented in Figure 3.9. Note that  $|u - u^h|_{2,\tilde{\Omega}} \leq |||u - u^h|||$ . As with the error in the  $L^2$  norm and the  $H^1$  semi-norm, the convergence behaviour is robust for  $\eta \geq 1$ , with the predicted rate of one being observed.

### Cubic elements

We consider now the case of cubic triangular elements ( $k = 3$ ) with linear lifting functions ( $l = 1$ ). For this case, the predicted convergence rate in the  $L^2$  norm is four, in the  $H^1$  semi-norm it is three and in the  $H^2$  energy-like norm it is two.

The observed convergence behaviour in the  $L^2$  norm is presented in Figure 3.10, again for various values of the penalty parameter. As predicted, in all cases the convergence rate is of order four. Notably, there is less sensitivity to the penalty term for this case compared to the case of quadratic elements. In the  $H^1$  semi-norm, the convergence rate is order three, as shown in Figure 3.11, and is also insensitive to the penalty value. In the broken energy norm, the convergence behaviour is shown in Figure 3.12, and again the predicted rate of two in this case is observed. In all considered norms, the convergence rate is insensitive to the magnitude of the penalty parameter. As for the quadratic case,  $\eta = 1$

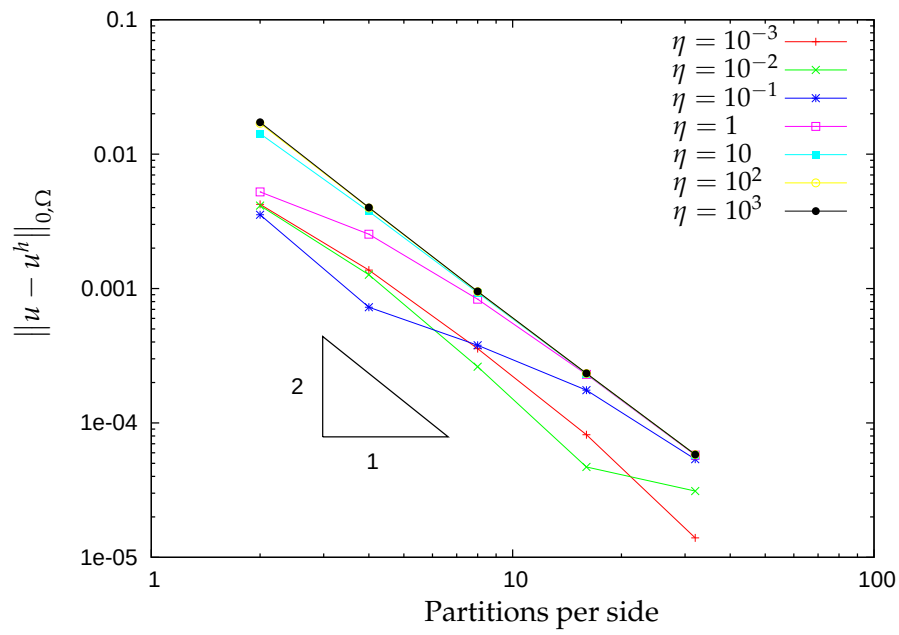


Figure 3.7:  $L^2$  error for the case  $k = 2$  and  $l = 0$  with various penalty values.

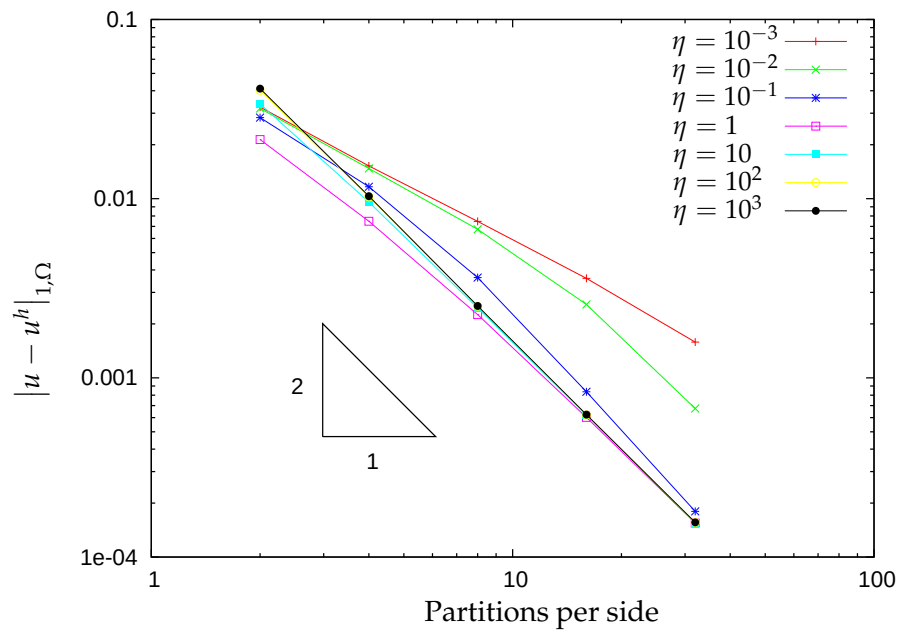


Figure 3.8: Error in the  $H^1$  semi-norm error for the case  $k = 2$  and  $l = 0$  with various penalty values.

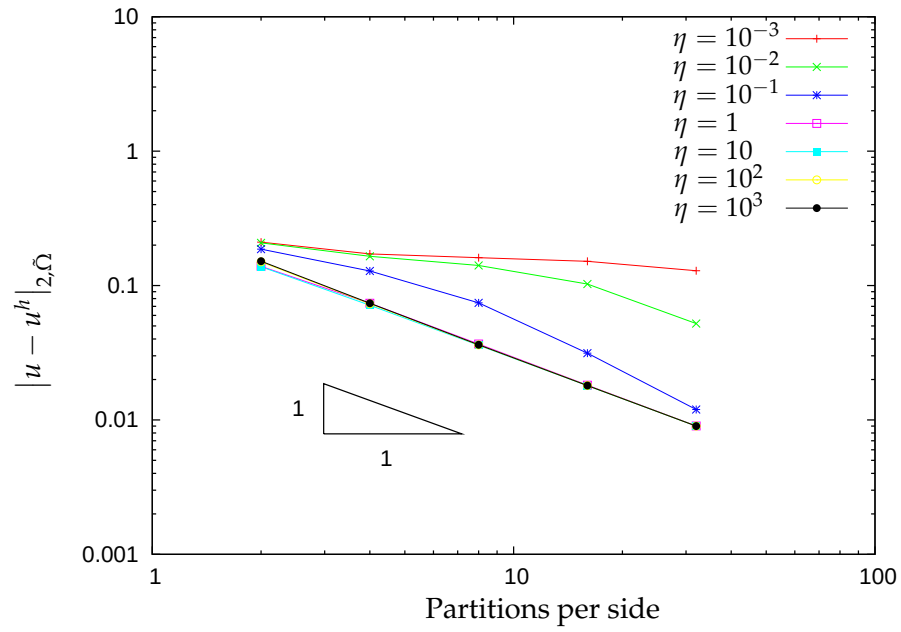


Figure 3.9: Error in the broken energy norm for the case  $k = 2$  and  $l = 0$  with various penalty values.

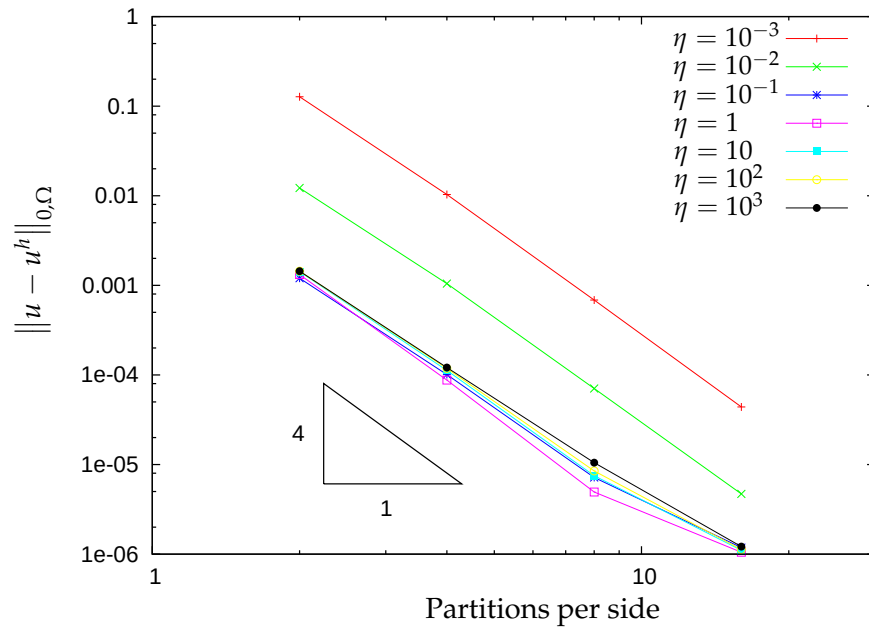


Figure 3.10:  $L^2$  error for the case  $k = 3$  and  $l = 1$  with various penalty values.

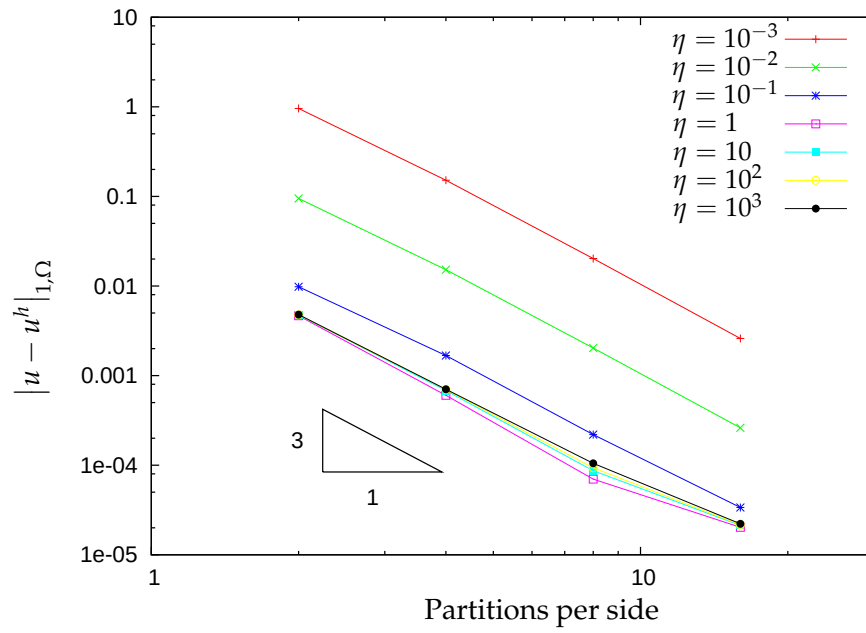


Figure 3.11: Error in the  $H^1$  semi-norm for the case  $k = 3$  and  $l = 1$  with various penalty values.

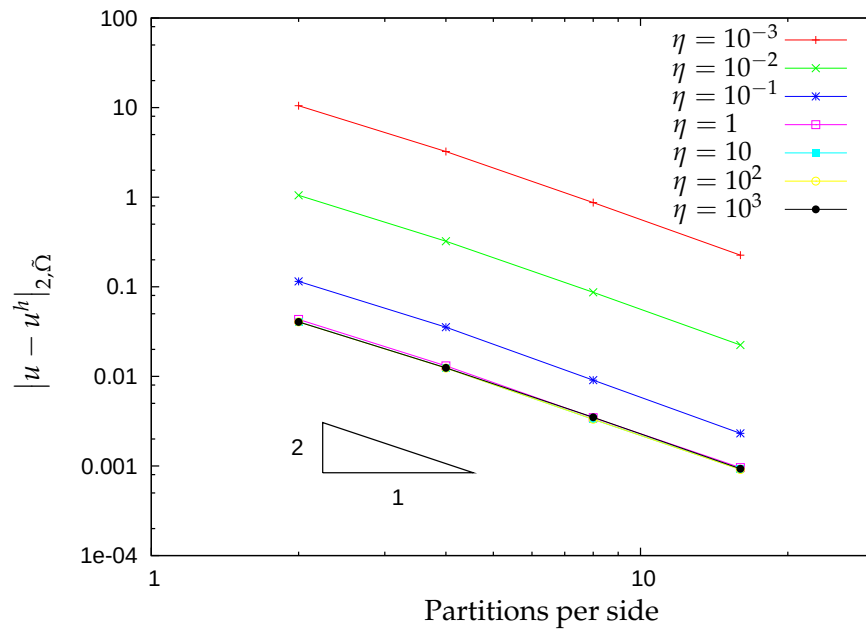


Figure 3.12: Error in the broken energy norm for the case  $k = 3$  and  $l = 1$  with various penalty values.

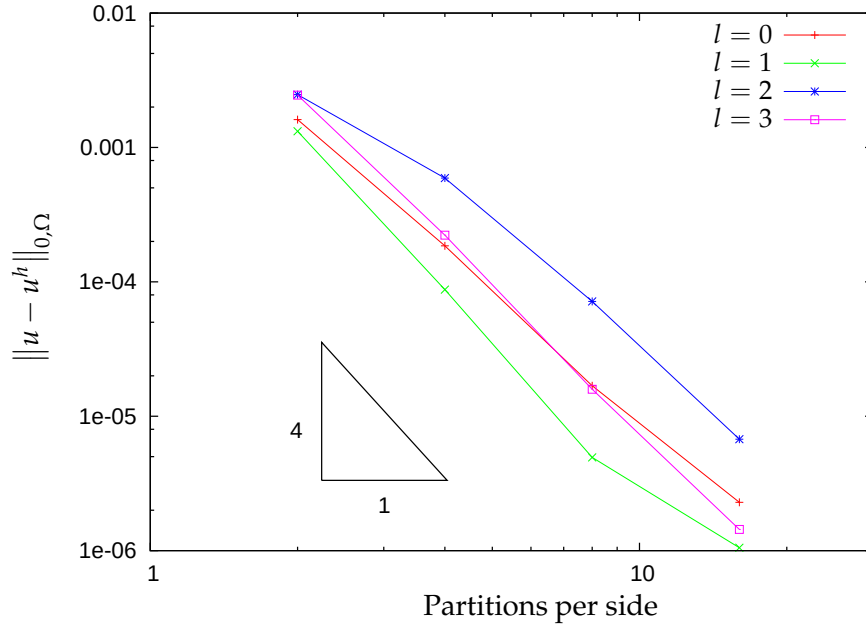


Figure 3.13:  $L^2$  error for the case  $k = 3$  and  $\eta = 1$  with  $l = 0, 1, 2, 3$ .

appears to be a suitable choice.

For all cases, using higher order lifting functions ( $l > k - 2$ ) leads to identical convergence rates. The computed  $L^2$  error for  $k = 3$  and  $\eta = 1$  for lifting functions of various orders is presented in Figure 3.13. For the case  $k = 3$  and  $l = 0$ , the presented analysis in Section 3.3 theoretically is no longer valid since the formulation violates consistency. However, the observed convergence rate in the  $L^2$  norm for the case  $k = 3$  and  $l = 0$  appears from Figure 3.13 to be optimal.

The error in the  $L^2$  norm for the simplified formulation in equation (3.55) for the cases  $k = 2$  and  $k = 3$  with  $l = k - 2$  and  $\eta = 4$  is shown in Figure 3.14. Recall that stability of the modified formulation requires  $\eta > 3$  for triangular elements. The convergence rate in this case appears as predicted for the full lifting formulation. The obtained results, together with the fact that the modified formulation is simpler and results in a stiffness matrix with a smaller bandwidth make this approach attractive for a range of linear problems.

Based on the range of results, it appears that the formulation with  $\eta = 1$  yields reliable results (recalling that stability analysis requires that  $\eta > 0$ ) and for the simplified version  $\eta = 4$  provides reliable results. Cubic elements robustly reproduce the expected convergence behaviour with no observed sensitivity to the penalty term. The  $L^2$  convergence rate of four makes its use attractive. Although

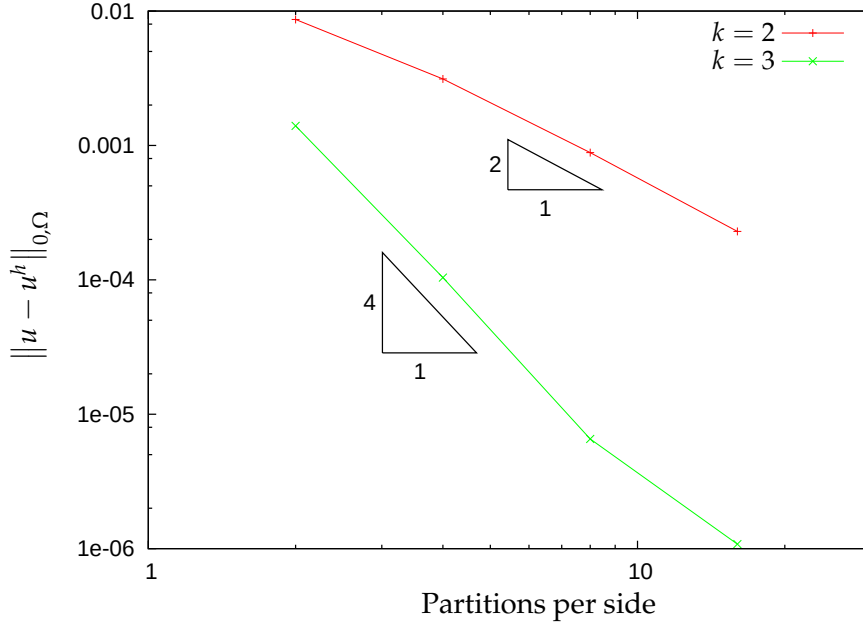


Figure 3.14:  $L^2$  error for the simplified formulation in equation (3.55) for  $k = 2, 3$  with  $l = k - 2$  and  $\eta = 4$ .

the convergence behaviour of the quadratic element is less reliable when using a low value for the penalty parameter and the convergence rate in the  $L^2$  norm is only order two, the simplicity of the formulation means that it is still of practical value and interest.

### 3.5.2 Plate buckling

The performance of the method for a plate buckling problem is considered now. The influence of the penalty parameter on the structural behaviour is of particular interest.

The finite element problem for plate buckling involves: find the eigenpair  $\{u^h, \lambda^h\}$ ,  $u^h \in V^h$ , such that

$$B^{\text{lif}}(w^h, u^h) - \lambda^h \int_{\partial\Omega} w^h_{,\alpha} f_{\alpha\beta} u^h_{,\beta} d\Gamma = 0 \quad \forall w^h \in W^h, \quad (3.121)$$

where  $f_{\alpha\beta}(x, y) = f_{\beta\alpha}(x, y)$ ,  $\lambda^h$  denotes the eigenvalues of the problem and  $f_{\alpha\beta}(x, y)$  is the traction applied at the mid-plane of the plate on the boundary  $\partial\Omega$ . This is illustrated in Figure 3.15. The lowest eigenvalue  $\lambda^h_{\min}$  is of interest as it corresponds to the critical buckling load for a given distribution of forces (determined by  $f_{\alpha\beta}(x, y)$ ).

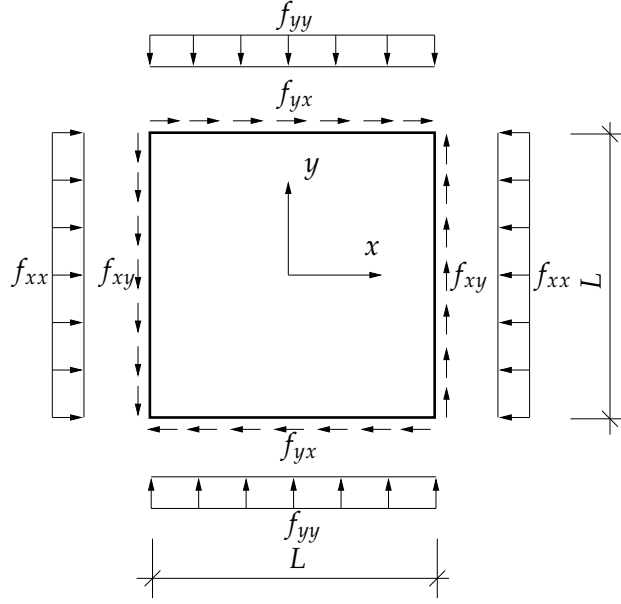


Figure 3.15: Applied boundary forces for the plate buckling problem.

For the simply supported problem in the previous section, with  $f_{xx} = f_{yy} = 1N/m$  and  $f_{xy} = 0$  and in the absence of a distributed load ( $F = 0$ ), the lowest eigenvalue has been computed for the case  $l = k - 2$  and with various values of the penalty parameter on a structured mesh of 128 elements. The computed results, which have been normalised using the exact solution, are shown in Figure 3.16. The exact solution can be found in Timoshenko and Gere (1963, pages 356-360).

Also shown in Figure 3.16 are results for the symmetric  $C^0$  interior penalty method (denoted IP), which is presented in Hughes and Garikipati (2004). In Hughes and Garikipati (2004), the penalty term  $\tau_\alpha$  is used, which we take as  $\tau_\alpha = \eta Et^2/12h_e$ , where  $h_e$  is the distance between the centroids of neighboring elements. Results for the IP method are presented for  $\eta \geq 2$  since a lower value may lead to an unstable result. It is clear that for the lifting formulation, once  $\eta > 0.1$  for  $k = 2$  and  $\eta > 0.01$  for  $k = 3$ , the critical buckling load is insensitive to the magnitude of the penalty term. Notably, for the interior penalty formulation with  $k = 2$ , the critical buckling load increases with increasing  $\eta$ , which is highly undesirable. This is compounded by the fact that the required value of the penalty term to ensure stability is unknown therefore one is inclined to choose a large value to be assured of stability, which leads to a non-conservative estimate of the critical buckling load.

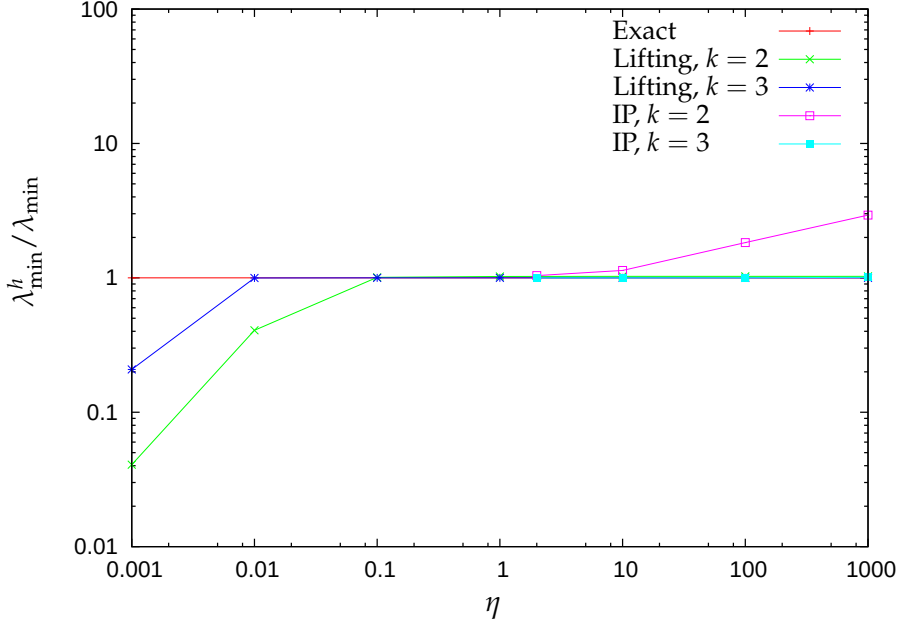


Figure 3.16: Smallest eigenvalue as a function of the penalty parameter for the buckling problem.

### 3.5.3 Plate vibration

A free vibration problem of a thin plate is now considered. For conditionally stable time integration schemes applied for various dynamic problems, the critical time step is usually proportional to the maximum frequency  $\omega_{\max}^h$ . Therefore, the performance of the method corresponding to the maximum frequency is of interest here.

The free vibration of a plate involves the problem: find the eigenpair  $\{u^h, \lambda^h\}$ ,  $u^h \in V^h$ , such that

$$B^{\text{lif}}(w^h, u^h) - \lambda^h \int_{\Omega} \rho w^h u^h d\Omega = 0 \quad \forall w^h \in W^h, \quad (3.122)$$

where  $\rho = \rho_0 t$  ( $\rho_0$  is the mass density and  $t$  is the plate thickness),  $u^h$  is the vibration mode and  $\lambda^h$  is the square of the natural frequency  $\omega^h$ ,  $\lambda^h = (\omega^h)^2$ .

The simply supported problem in the absence of a distributed load is analysed, with  $\rho_0 = 1 \times 10^4 \text{Ns}^2/\text{m}^4$  and the mesh is composed of 128 elements. The maximum eigenvalue for different penalty parameters is shown in Figure 3.17 for both the lifting formulation and the IP method. Not surprisingly, the maximum eigenvalue is proportional to the penalty parameter for larger penalty values (approximately  $\eta > 0.1$  for  $k = 3$  and  $\eta > 1$  for  $k = 2$ ).

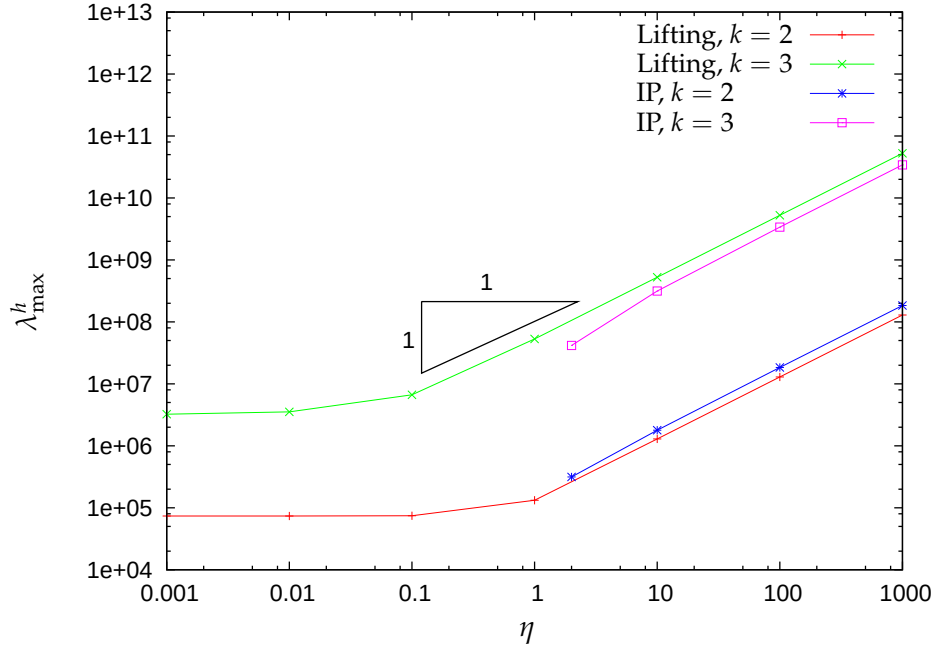


Figure 3.17: Largest eigenvalue as a function of the penalty parameter for the vibration problem.

Relating to a conditionally stable time stepping algorithm, the critical time step  $\Delta t^{\text{crit}}$  decreases with increasing  $\eta$ ,

$$\Delta t^{\text{crit}} \propto \frac{1}{\omega_{\max}} \propto \frac{1}{\sqrt{\eta}}, \quad (3.123)$$

where  $\omega_{\max}$  denotes the maximum natural frequency. In this respect, the lifting formulation has an advantage over the IP formulation in terms of time step selection as a small penalty parameter ( $\eta \approx 1$  for  $k = 2$  and  $\eta \approx 0.1$  for  $k = 3$ ) can be chosen with confidence.

### 3.5.4 Plate bending with distorted meshes

The behaviour of quadrilateral elements with respect to distortion in an unstructured mesh has been investigated in various works, including those of Bathe and Dvorkin (1986) and Sansour and Bednarczyk (1995). In this example, we consider a simply-supported square plate with side length  $L = 100$  mm, thickness  $t = 1$  mm, Young's modulus  $E = 1 \times 10^7$  N/mm<sup>2</sup> and Poisson's ratio  $\nu = 0.3$ . A central force  $F = 4$  N is applied. Meshes of 32 and 128 triangular elements

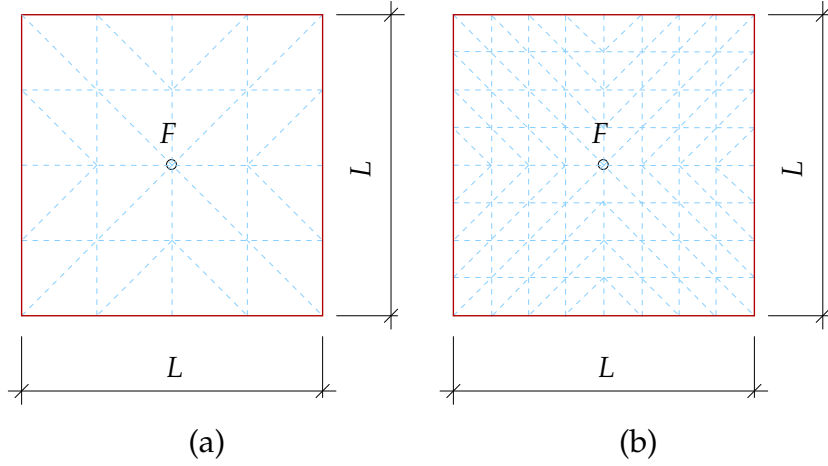


Figure 3.18: Structured mesh of: (a) 32 elements; (b) 128 elements.

are considered. The structured meshes (Figures 3.18) are used to compute the reference result which is the deflection at the loaded point. The deflection at the loaded point in distorted meshes (Figures 3.19) is computed and then normalised by the reference. Shown in Table 3.1 are results for quadratic ( $k = 2$ ) and cubic ( $k = 3$ ) elements with the lifting function order  $l = k - 2$ , the penalty parameter  $\eta = 1$  and various values of the length  $a$ .

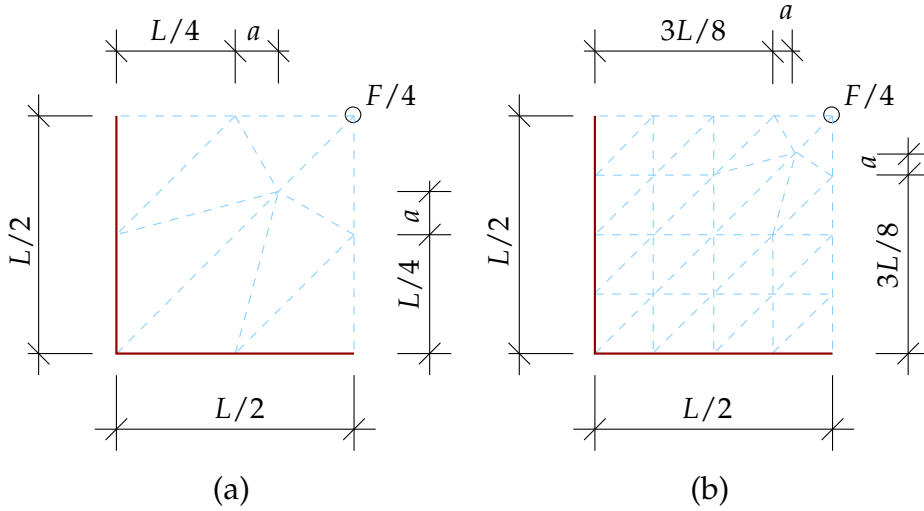


Figure 3.19: A quarter of non-structured mesh of: (a) 32 elements; (b) 128 elements.

The results show that both quadratic and cubic elements in the distorted meshes perform well. In particular, there is less sensitivity to the distortion for the case of cubic elements compared to the case of quadratic elements. For a

| a<br>(mm) | 32 elements |          | 128 elements |          |
|-----------|-------------|----------|--------------|----------|
|           | Quadratic   | Cubic    | Quadratic    | Cubic    |
| 0         | 1.000000    | 1.000000 | 1.000000     | 1.000000 |
| 2.5       | 0.974533    | 0.996146 | 0.984645     | 0.998039 |
| 5         | 0.954160    | 0.994220 | 0.975048     | 0.996078 |
| 7.5       | 0.938879    | 0.992293 | 0.969290     | 0.996078 |
| 10        | 0.925297    | 0.988439 | 0.992322     | 0.994118 |

Table 3.1: Normalised displacement at the loaded point of distorted meshes

finer mesh, sensitivity to the distortion is smaller, for both quadratic and cubic elements.



# Chapter 4

## Thin shell problems

‘Thin shell’ describes a thin structure with a curved mid-surface. For the simulation of a thin shell problem, a three-dimensional model can be applied directly. However, expense is a barrier to this approach. Shell theories are proposed for simulating the behaviour of thin shells using two-dimensional models. In general, there exist two shell theory families, namely the Koiter and Naghdi families. The former ignores transverse shear effects by applying the Kirchhoff-Love kinematic assumptions, and the latter is based on the Reissner-Mindlin kinematic assumptions and takes transverse shear effects into account. The reader can refer to numerous textbooks for more details, including the works of Mindlin and Cooper (1960), Koiter and Simmonds (1972), Pietraszkiewicz (1983), Makowski and Pietraszkiewicz (1989), Bernadou (1996), Chapelle and Bathe (2003) and Ciarlet (2005).

A number of shell finite elements built upon one of two above-mentioned families have been proposed and developed over the years. Conventionally, elements in the Koiter type model requires the use of  $C^1$  basis functions, which are difficult to construct, while the model built on Naghdi-type demands the use of  $C^0$  basis functions only. Because of its simple implementation, the Naghdi-type model is relatively attractive. However, it poses some other difficulties, in particular shear locking when shell thickness becomes small (Brezzi et al., 1989).

This chapter serves the preliminary concept of thin shell descriptions and introduces governing equations for a geometrically nonlinear and linear Koiter shell model. The following assumptions are made:

- The shell thickness is small in comparison with the least radius of curvature of the shell mid-surface;

- Straight lines that initially are normal to the middle surface remain so during deformation;
- The St. Venant-Kirchhoff elasticity assumption is adopted and the strain-stress relation is linear; and
- The state of stress inside the shell is planar and parallel to the mid-surface.

## 4.1 Shell description

In describing a shell, we adopt largely the presentation of Bernadou (1996) and Chapelle and Bathe (2003).

Consider a shell with a body  $\mathcal{V}$ , surface boundary  $\mathcal{B}$ , mid-surface  $\Omega$  and thickness  $t$  in the undeformed (reference) configuration, see Figure 4.1. The shell lies in the Euclidean space  $\mathbb{E}^3$  with a fixed orthonormal coordinate system  $(E^1, E^2, E^3)$ . The surface boundary  $\mathcal{B}$  of the shell is partitioned into the top surface  $\mathcal{B}_t$ , bottom surface  $\mathcal{B}_b$  and side surface  $\mathcal{B}_s$ . The boundary of the mid-surface  $\Omega$  is denoted by  $\partial\Omega$  and the closure of  $\Omega$  is denoted by  $\bar{\Omega}$ ,  $\bar{\Omega} = \Omega \cup \partial\Omega$ . Note that  $\partial\Omega \in \mathcal{B}_s$ .

### 4.1.1 Shell mid-surface

A geometric mapping will be used to define the shell mid-surface. Let  $\mathbb{R}^d$  denote a parametric space of dimension  $d$ . This space is defined as the set of all collections of  $d$  real numbers  $(\xi^i)$ ,  $i = 1 \rightarrow d$ . Let  $\omega$  denote a two-dimensional parametric domain,  $\omega \subset \mathbb{R}^2$ . Denote the boundary of  $\omega$  by  $\partial\omega$  and the closure of  $\omega$  by  $\bar{\omega}$ ,  $\bar{\omega} = \omega \cup \partial\omega$ . The mid-surface and its boundary of the given shell are defined by an injective geometric mapping  $\Phi$  from the domain  $\bar{\omega} \subset \mathbb{R}^2$  into the Euclidean space  $\mathbb{E}^3$ ,  $\Phi : \mathbb{R}^2 \rightarrow \mathbb{E}^3$ . Hence,  $\bar{\Omega}$  is the image of  $\bar{\omega}$  in the space  $\mathbb{E}^3$  via the mapping  $\Phi$ . The position vector  $X$  of a point  $p_\Omega$  on the mid-surface is now given by  $X = \Phi(\xi^1, \xi^2)$ , where  $(\xi^1, \xi^2)$  is a point on the two-dimensional domain  $\omega$ .

A local coordinate system on the mid-surface will be used frequently from now on. Since its basis vectors are generally not orthogonal, the concepts of covariant and contravariant basis vectors are adopted. Covariant basis vectors are defined as:

$$A_\alpha = \frac{\partial \Phi(\xi^1, \xi^2)}{\partial \xi^\alpha} \quad (4.1)$$

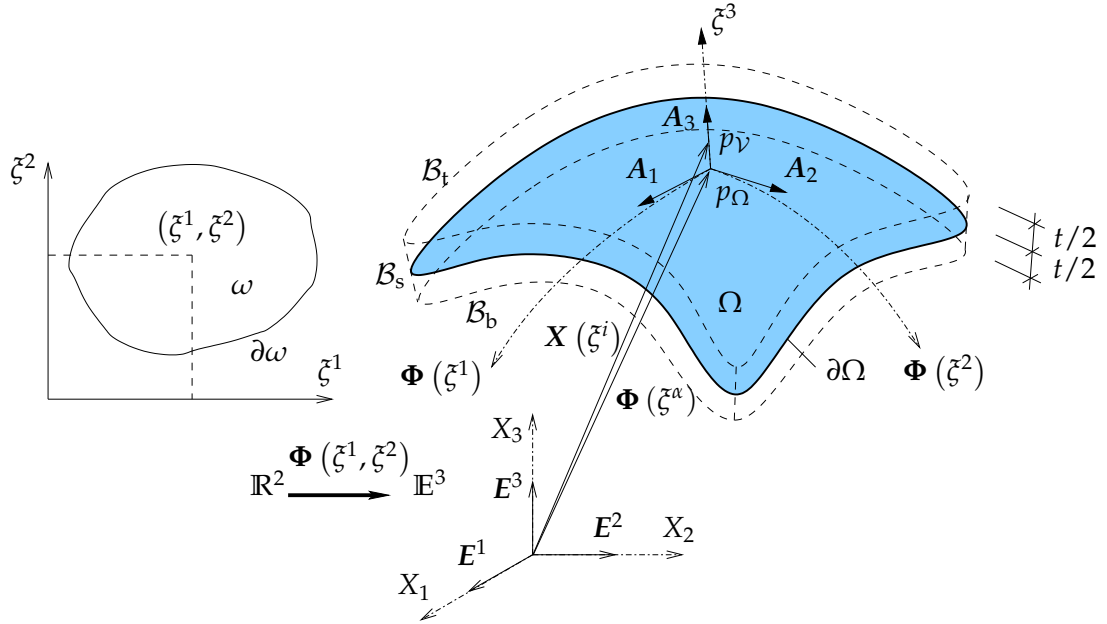


Figure 4.1: Definition of shell geometry.

and

$$A_3 = \frac{A_1 \times A_2}{\|A_1 \times A_2\|}, \quad (4.2)$$

where  $\alpha = 1, 2$  are indices denoting two directions of the surface tangent plane at a point on the mid-surface,  $A_\alpha$  are tangent vectors to the mid-surface and  $A_3$  is the unit normal vector to the mid-surface. The surface contravariant basis vectors  $A^\alpha$  are defined by

$$A_\beta \cdot A^\alpha = \delta_\beta^\alpha, \quad (4.3)$$

where  $\delta_\beta^\alpha$  is the Kronecker delta. Hereinafter, lowered indices represent components of covariant vectors, while raised indices represent components of contravariant vectors. Note that there is no distinction between the contravariant basis vector  $A^3$  and the covariant basis vector  $A_3$  since they are equal.

Covariant components  $A_{\alpha\beta}$  of the metric tensor are defined by

$$A_{\alpha\beta} = A_\alpha \cdot A_\beta, \quad (4.4)$$

and contravariant components  $A^{\alpha\beta}$  of the metric are defined by

$$A^{\alpha\beta} = A^\alpha \cdot A^\beta. \quad (4.5)$$

This metric tensor is also called the first fundamental form of the mid-surface. Covariant components of the second fundamental form, containing curvature information for the mid-surface, are defined as

$$B_{\gamma\alpha} = -A_{3,\alpha} \cdot A_\gamma, \quad (4.6)$$

and the covariant-contravariant components of the second fundamental form are defined as

$$B_\alpha^\gamma = -A_{3,\alpha} \cdot A^\gamma. \quad (4.7)$$

Some other geometric definitions of the mid-surface will be used later including

$$\Gamma_{\gamma\alpha}^\delta = -A_{,\alpha}^\delta \cdot A_\gamma, \quad (4.8)$$

which is called the surface *Christoffel symbol*, and

$$A = \det(A_{\alpha\beta}), \quad (4.9)$$

which relates a differential area  $d\bar{\Omega}$  of the mid-surface and a differential area  $d\bar{\omega}$  of the parametric domain,  $d\bar{\Omega} = \sqrt{A} d\bar{\omega}$ .

#### 4.1.2 Shell domain

Consider now the set of three real parameters  $(\xi^1, \xi^2, \xi^3) \in \mathbb{R}^3$  such that  $(\xi^1, \xi^2)$  is a point on  $\omega$  and  $-t/2 \leq \xi^3 \leq t/2$ . Let  $X$  denote the position vector of a material point  $p_V$  in the shell body in the reference configuration  $\mathcal{V}$ . This vector is determined as a function of three real parameters  $(\xi^1, \xi^2, \xi^3) \in \mathbb{R}^3$  in

$$X(\xi^1, \xi^2, \xi^3) = \Phi(\xi^1, \xi^2) + \xi^3 A_3(\xi^1, \xi^2). \quad (4.10)$$

At each material point, similar to the presentation in the above, local covariant basis vectors  $G_i$  on the reference configuration are defined by

$$G_\alpha = \frac{\partial X}{\partial \xi^\alpha} = \frac{\partial \Phi}{\partial \xi^\alpha} + \xi^3 \frac{\partial A_3}{\partial \xi^\alpha} = A_\alpha + \xi^3 \frac{\partial A_3}{\partial \xi^\alpha}, \quad (4.11)$$

$$G_3 = \frac{\partial \Phi}{\partial \xi^3} = A_3. \quad (4.12)$$

Contravariant basis vectors  $G^i$  are defined such that

$$G_i \cdot G^j = \delta_i^j, \quad (4.13)$$

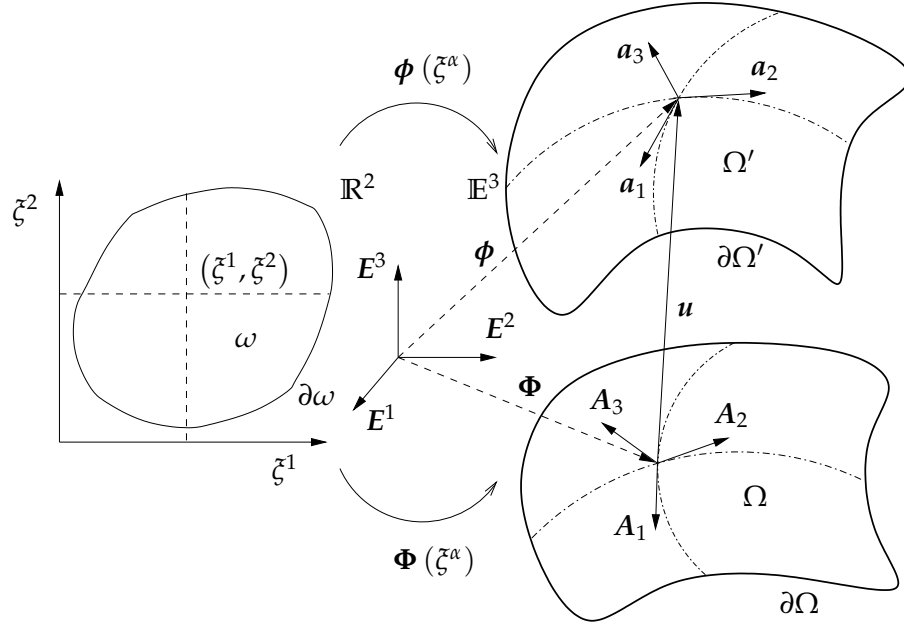


Figure 4.2: Shell kinematic description.

where  $\delta_i^j$  is the Kronecker-delta. Metric coefficients  $G_{ij}$  read

$$G_{ij} = \mathbf{G}_i \cdot \mathbf{G}_j, \quad (4.14)$$

and a scalar  $G$  used in the volume measure reads

$$G = \det(G_{ij}). \quad (4.15)$$

## 4.2 Shell kinematics

Under external loading, the structure deforms. Let  $\mathbf{x}$  denote the position vector of a material point in the deformed shell media. We now adopt the Kirchhoff-Love hypothesis to the expression of the position vector  $\mathbf{x}$ . The hypothesis requires that a material fiber which is initially normal to the mid-surface of the shell remains normal to the mid-surface. With this assumption, the position vector  $\mathbf{x}$  can be given by

$$\mathbf{x} = \boldsymbol{\phi}(\xi^1, \xi^2) + \xi^3 \kappa(\xi^1, \xi^2) \mathbf{a}_3(\xi^1, \xi^2), \quad (4.16)$$

where  $\boldsymbol{\phi} : \mathbb{R}^2 \rightarrow \mathbb{E}^3$  is an injective geometric mapping from the domain  $\bar{\omega}$  into the domain  $\bar{\Omega}'$ , the closure of the deformed mid-surface;  $\kappa$  represents the change

in the shell thickness,

$$\kappa = \frac{t'}{t}, \quad (4.17)$$

where  $t'$  is the thickness of the deformed shell;  $\mathbf{a}_3$  is the unit basis vector which is normal to the deformed mid-surface at the considered point.

#### 4.2.1 Basis vectors in deformed configuration

The covariant basis vectors  $\mathbf{a}_\alpha$  which is tangent to deformed mid-surface in the deformed configuration are defined by

$$\mathbf{a}_\alpha = \frac{\partial \boldsymbol{\phi}(\xi^1, \xi^2)}{\partial \xi^\alpha}, \quad (4.18)$$

and the normal vector  $\mathbf{a}_3$  to the mid-surface in the deformed configuration is defined by

$$\mathbf{a}_3 = \frac{\mathbf{a}_1 \times \mathbf{a}_2}{\|\mathbf{a}_1 \times \mathbf{a}_2\|}. \quad (4.19)$$

Recall that  $\boldsymbol{\phi}(\xi^1, \xi^2)$  is the position vector of a point on the deformed mid-surface  $\Omega'$ , see Figure 4.2. Assuming that the change in shell thickness is negligible, we have  $\kappa \simeq 1$ . With the above results, the displacement vector  $\mathbf{u}$  of a material point in the shell body reads

$$\mathbf{u} = \mathbf{x} - \mathbf{X} = \boldsymbol{\phi} - \boldsymbol{\Phi} + \xi^3 (\mathbf{a}_3 - \mathbf{A}_3). \quad (4.20)$$

For a material point on the shell mid-surface, the displacement vector is simplified by

$$\mathbf{u} = \boldsymbol{\phi} - \boldsymbol{\Phi}. \quad (4.21)$$

For each material point in the shell body on the deformed configuration, local covariant basis vectors  $\mathbf{g}_i$  are defined by

$$\mathbf{g}_\alpha = \frac{\partial \mathbf{x}}{\partial \xi^\alpha} = \frac{\partial \boldsymbol{\phi}}{\partial \xi^\alpha} + \xi^3 \frac{\partial (\mathbf{a}_3)}{\partial \xi^\alpha} = \mathbf{a}_\alpha + \xi^3 \frac{\partial (\mathbf{a}_3)}{\partial \xi^\alpha}, \quad (4.22)$$

$$\mathbf{g}_3 = \frac{\partial \boldsymbol{\phi}}{\partial \xi^3} = \mathbf{a}_3, \quad (4.23)$$

metric coefficients  $g_{ij}$  are given by

$$g_{ij} = \mathbf{g}_i \cdot \mathbf{g}_j, \quad (4.24)$$

and a scalar  $g$  is given by

$$g = \det(g_{ij}). \quad (4.25)$$

### 4.2.2 Green-Lagrange strain tensor

The deformation of the shell body is represented in a Green-Lagrange strain tensor field. In terms of the metric coefficients and basis vectors, the Green-Lagrange strain tensor defined for each material point in the shell media reads

$$\mathbf{E} = \frac{1}{2} (g_{ij} - G_{ij}) \mathbf{G}^i \otimes \mathbf{G}^j. \quad (4.26)$$

As the shell is thin,  $(\xi^3)^2 \approx 0$ , the covariant components of the Green-Lagrange strain tensor can be expressed as

$$E_{ij} = \alpha_{ij} + \xi^3 \beta_{ij}, \quad (4.27)$$

where

$$\alpha_{ij} = \frac{1}{2} (\mathbf{a}_i \cdot \mathbf{a}_j - \mathbf{A}_i \cdot \mathbf{A}_j), \quad (4.28)$$

$$\beta_{ij} = \mathbf{a}_i \cdot \mathbf{a}_{3,j} - \mathbf{A}_i \cdot \mathbf{A}_{3,j}. \quad (4.29)$$

Since  $\mathbf{A}_3 \cdot \mathbf{A}_3 = \mathbf{a}_3 \cdot \mathbf{a}_3 = 1$ , it simply follows that

$$\alpha_{33} = \mathbf{a}_3 \cdot \mathbf{a}_3 - \mathbf{A}_3 \cdot \mathbf{A}_3 = 0, \quad (4.30)$$

$$\beta_{33} = \mathbf{a}_3 \cdot \mathbf{a}_{3,3} - \mathbf{A}_3 \cdot \mathbf{A}_{3,3} = 0. \quad (4.31)$$

The Kirchhoff-Love assumption implies  $\mathbf{A}_\alpha \cdot \mathbf{A}_3 = \mathbf{a}_\alpha \cdot \mathbf{a}_3 = 0$ , that leads to

$$\alpha_{\alpha 3} = \mathbf{a}_\alpha \cdot \mathbf{a}_3 - \mathbf{A}_\alpha \cdot \mathbf{A}_3 = 0, \quad (4.32)$$

$$\beta_{\alpha 3} = \mathbf{a}_\alpha \cdot \mathbf{a}_{3,3} - \mathbf{A}_\alpha \cdot \mathbf{A}_{3,3} = 0, \quad (4.33)$$

and

$$\mathbf{A}_\alpha \cdot \mathbf{A}_{3,\beta} = -\mathbf{A}_{\alpha,\beta} \cdot \mathbf{A}_3, \quad (4.34)$$

$$\mathbf{a}_\alpha \cdot \mathbf{a}_{3,\beta} = -\mathbf{a}_{\alpha,\beta} \cdot \mathbf{a}_3. \quad (4.35)$$

Therefore, the nonzero components are  $\alpha_{\alpha\beta}$  and  $\beta_{\alpha\beta}$  which can be rewritten in

$$\alpha_{\alpha\beta} = \frac{1}{2} (\mathbf{a}_\alpha \cdot \mathbf{a}_\beta - \mathbf{A}_\alpha \cdot \mathbf{A}_\beta), \quad (4.36)$$

$$\beta_{\alpha\beta} = \mathbf{A}_{\alpha,\beta} \cdot \mathbf{A}_3 - \mathbf{a}_{\alpha,\beta} \cdot \mathbf{a}_3. \quad (4.37)$$

Essentially,  $\alpha_{\alpha\beta}$  corresponds to membrane strains and  $\beta_{\alpha\beta}$  corresponds to the bending curvature. The deformation state of the shell body is completely defined by the metric tensors on the mid-surface.

With the aid of equation (4.21), components of the Green-Lagrange strain tensor can be expressed in terms of basis vectors in the undeformed configuration, the displacement vector  $\mathbf{u}$  and the normal vector  $\mathbf{a}_3$  to the deformed mid-surface. The membrane strain components are therefore given by

$$\alpha_{\alpha\beta} = \frac{1}{2} (\mathbf{A}_\alpha \cdot \mathbf{u}_{,\beta} + \mathbf{u}_{,\alpha} \cdot \mathbf{A}_\beta + \mathbf{u}_{,\alpha} \cdot \mathbf{u}_{,\beta}) \quad (4.38)$$

and the bending curvature components are given by

$$\beta_{\alpha\beta} = \mathbf{A}_{\alpha,\beta} \cdot \mathbf{A}_3 - \mathbf{A}_{\alpha,\beta} \cdot \mathbf{a}_3 - \mathbf{u}_{,\alpha\beta} \cdot \mathbf{a}_3. \quad (4.39)$$

### 4.3 Constitutive model

The Green-Lagrange strain  $\mathbf{E}$  presented in the above section is energy conjugate to the second Piola-Kirchhoff stress tensor which is denoted by  $\boldsymbol{\Sigma}$ . This stress relates forces in the reference configuration to areas in the reference configuration. For the sake of simplicity, we consider thin shell problems in the physically linear regime of deformation and we assume that the components of the second Piola-Kirchhoff stress are related to components of the Green-Lagrange strain via

$$\Sigma^{ij} = c^{ijkl} E_{kl}, \quad (4.40)$$

where  $c^{ijkl}$  is a fourth-order constitutive tensor. We assume that  $c^{ijkl}$  does not vary with  $\xi^3$ , therefore it will be represented by the elasticity tensor  $c^{\alpha\beta\gamma\delta}$  on the shell mid-surface. The components  $c^{\alpha\beta\gamma\delta}$  read

$$c^{\alpha\beta\gamma\delta} = \frac{2\lambda\mu}{\lambda + 2\mu} A^{\alpha\beta} A^{\gamma\delta} + \mu (A^{\alpha\delta} a^{\beta\gamma} + A^{\alpha\gamma} a^{\beta\delta}), \quad (4.41)$$

where  $\lambda$  and  $\mu$  are Lamé parameters and the metric coefficients  $A^{\alpha\beta}$  are defined in the reference configuration, see equation (4.5).

### 4.4 Variational problem

Let  $\delta\mathbf{u}$  denote the variation of the displacement  $\mathbf{u}$  about the current configuration. The variation  $\delta\mathbf{u}$  leads to variations of other quantities representing the current configuration. From now on, we denote  $\delta\varphi$  as the variation of the function  $\varphi$  about the current configuration due to  $\delta\mathbf{u}$ .

#### 4.4.1 Variations of strain components

Within the context of directional derivatives (see Bonet and Wood (1997) for more details), the variation of a function can be derived. Straightforwardly, the variation of basis vector  $\mathbf{a}_\alpha$  is given by

$$\delta \mathbf{a}_\alpha = \delta \mathbf{u}_{,\alpha}, \quad (4.42)$$

and the variation of basis vector  $\mathbf{a}_3$  is given by

$$\begin{aligned} \delta \mathbf{a}_3 = \frac{1}{\sqrt{a}} (\delta \mathbf{a}_1 \times \mathbf{a}_2 + \mathbf{a}_1 \times \delta \mathbf{a}_2) \\ - \frac{a_3}{\sqrt{a}} (\delta \mathbf{a}_1 \cdot (\mathbf{a}_2 \times \mathbf{a}_3) + \delta \mathbf{a}_2 \cdot (\mathbf{a}_3 \times \mathbf{a}_1)), \end{aligned} \quad (4.43)$$

or

$$\begin{aligned} \delta \mathbf{a}_3 = \frac{1}{\sqrt{a}} (\delta \mathbf{u}_{,1} \times \mathbf{a}_2 + \mathbf{a}_1 \times \delta \mathbf{u}_{,2}) \\ - \frac{a_3}{\sqrt{a}} (\delta \mathbf{u}_{,1} \cdot (\mathbf{a}_2 \times \mathbf{a}_3) + \delta \mathbf{u}_{,2} \cdot (\mathbf{a}_3 \times \mathbf{a}_1)). \end{aligned} \quad (4.44)$$

Based on equations (4.36) to (4.39), the variation of the membrane strain component reads

$$\begin{aligned} \delta \alpha_{\alpha\beta} &= \frac{1}{2} (\delta \mathbf{a}_\alpha \cdot \mathbf{a}_\beta + \mathbf{a}_\alpha \cdot \delta \mathbf{a}_\beta), \\ &= \frac{1}{2} (\mathbf{A}_\alpha \cdot \delta \mathbf{u}_{,\beta} + \delta \mathbf{u}_{,\alpha} \cdot \mathbf{A}_\beta + \delta \mathbf{u}_{,\alpha} \cdot \mathbf{u}_{,\beta} + \mathbf{u}_{,\alpha} \cdot \delta \mathbf{u}_{,\beta}), \end{aligned} \quad (4.45)$$

and the variation of the bending strain component reads

$$\begin{aligned} \delta \beta_{\alpha\beta} &= -\mathbf{a}_{\alpha,\beta} \cdot \delta \mathbf{a}_3 - \delta \mathbf{a}_{\alpha,\beta} \cdot \mathbf{a}_3, \\ &= -\mathbf{A}_{\alpha,\beta} \cdot \delta \mathbf{a}_3 - \delta \mathbf{u}_{,\alpha\beta} \cdot \mathbf{a}_3 - \mathbf{u}_{,\alpha\beta} \cdot \delta \mathbf{a}_3. \end{aligned} \quad (4.46)$$

#### 4.4.2 Variational formulation

The variational formulation for the considered problem can be derived from the principle of virtual work. In the absence of inertia effects, balance of linear momentum requires that

$$\int_{\mathcal{V}} \delta \mathbf{E} : \boldsymbol{\Sigma} d\mathcal{V} = \int_{\mathcal{V}} \delta \mathbf{u} \cdot \mathbf{f} d\mathcal{V} + \int_{\mathcal{B}_h} \delta \mathbf{u} \cdot \mathbf{h} d\mathcal{B} \quad \forall \delta \mathbf{u}, \quad (4.47)$$

where  $\delta$  indicates a variation about the current configuration,  $\mathbf{f}$  is the body force in the reference configuration and  $\mathbf{h}$  is the surface force on the Neumann boundary acting on a region  $\mathcal{B}_h$  of the side surface  $\mathcal{B}_s$ ,  $\mathcal{B}_h \subset \mathcal{B}_s$ .

Inserting the expression for the Green-Lagrange strain in equation (4.27) into the LHS of equation (4.47), we obtain

$$\int_{\mathcal{V}} \delta \mathbf{E} : \boldsymbol{\Sigma} d\mathcal{V} = \int_{\mathcal{V}} \left( \delta \alpha_{\alpha\beta} + \xi^3 \delta \beta_{\alpha\beta} \right) \Sigma^{\alpha\beta} d\mathcal{V}. \quad (4.48)$$

Taking the constitutive relation (4.40) into account, we have

$$\int_{\mathcal{V}} \delta \mathbf{E} : \boldsymbol{\Sigma} d\mathcal{V} = \int_{\mathcal{V}} \left( \delta \alpha_{\alpha\beta} + \xi^3 \delta \beta_{\alpha\beta} \right) c^{\alpha\beta\gamma\delta} \left( \alpha_{\gamma\delta} + \xi^3 \beta_{\gamma\delta} \right) d\mathcal{V}. \quad (4.49)$$

For measuring the volume  $d\mathcal{V}$ , assuming  $g \simeq a$ , hence  $d\mathcal{V} \simeq t d\Omega = t \sqrt{a} d\omega$ . Integration of the above expression through the thickness leads to

$$\int_{\mathcal{V}} \delta \mathbf{E} : \boldsymbol{\Sigma} d\mathcal{V} = \int_{\Omega} \left( t \delta \alpha_{\alpha\beta} c^{\alpha\beta\gamma\delta} \alpha_{\gamma\delta} + \frac{t^3}{12} \delta \beta_{\alpha\beta} c^{\alpha\beta\gamma\delta} \beta_{\gamma\delta} \right) d\Omega. \quad (4.50)$$

For the case of the St Venant-Kirchhoff elasticity and plane stress conditions, equation (4.50) is Koiter's nonlinear shell model (Koiter and Simmonds, 1972; Ciarlet, 2005).

The terms on the RHS of equation (4.47) can also be integrated through the thickness and simplified, see Hughes (2000) for more elaboration, leading to

$$\begin{aligned} \int_{\mathcal{V}} \delta \mathbf{u} \cdot \mathbf{f} d\Omega + \int_{\mathcal{B}_h} \delta \mathbf{u} \cdot \mathbf{h} dB \\ = \int_{\Omega} \delta \mathbf{u} \cdot \mathbf{F} d\Omega + \int_{\Gamma^H} \delta \mathbf{u} \cdot \mathbf{H} d\Gamma + \int_{\Gamma^M} (\delta \mathbf{a}_3 \cdot \mathbf{n}) M d\Gamma, \end{aligned} \quad (4.51)$$

where  $\mathbf{F}$  is the resultant force acting on the mid-surface,  $\mathbf{H}$  is the resultant force applied on the force boundary  $\Gamma^H \subset \partial\Omega$  and  $M$  is the normal component of the moment vector to the moment boundary  $\Gamma^M \subset \partial\Omega$ . These boundary values are defined on the boundary of the reference mid-surface.

The variational problem which we wish to solve therefore reads: find the displacement field  $\mathbf{u}$  such that

$$\begin{aligned} \int_{\Omega} \left( t \delta \alpha_{\alpha\beta} c^{\alpha\beta\gamma\delta} \alpha_{\gamma\delta} + \frac{t^3}{12} \delta \beta_{\alpha\beta} c^{\alpha\beta\gamma\delta} \beta_{\gamma\delta} \right) d\Omega \\ = \int_{\Omega} \delta \mathbf{u} \cdot \mathbf{F} d\Omega + \int_{\Gamma^H} \delta \mathbf{u} \cdot \mathbf{H} d\Gamma + \int_{\Gamma^M} (\delta \mathbf{a}_3 \cdot \mathbf{n}) M d\Gamma \quad \forall \delta \mathbf{u}, \end{aligned} \quad (4.52)$$

subject to the Dirichlet boundary conditions

$$\mathbf{u} = \mathbf{g}^u \quad \text{on } \Gamma^u, \quad (4.53)$$

$$\mathbf{u}_{,\gamma} n^\gamma \cdot \mathbf{a}_3 = g^\theta \quad \text{on } \Gamma^\theta, \quad (4.54)$$

where  $\delta \mathbf{u}$  is an ‘admissible’ variation which will be defined later,  $n^\gamma$  are the contravariant components of the outward unit normal vector to the undeformed boundary  $\partial\Omega$  and orthogonal to  $\mathbf{A}_3$ . Here, it has been assumed that  $\partial\Omega$  is partitioned such that  $\overline{\Gamma^u \cup \Gamma^H} = \overline{\Gamma^\theta \cup \Gamma^M} = \partial\Omega$  and  $\Gamma^u \cap \Gamma^H = \Gamma^\theta \cap \Gamma^M = \emptyset$  in order to avoid the under- or over-prescription of boundary conditions. More complex combinations of boundary conditions which distinguish between membrane and bending parts of the problem are possible. The rotation boundary condition (4.54) is nonlinear in the sense that it depends on  $\mathbf{a}_3$ .

## 4.5 Linear thin shell problems

We now consider the case of linear thin shell problems in which deformation of the shells is small and geometrically linear kinematic equations can be adopted. Linearisation of the membrane strain (4.38) yields

$$\alpha_{\alpha\beta} = \frac{1}{2} (\mathbf{u}_{,\alpha} \cdot \mathbf{A}_\beta + \mathbf{u}_{,\beta} \cdot \mathbf{A}_\alpha), \quad (4.55)$$

where the second-order infinitesimal term  $\mathbf{u}_{,\alpha} \cdot \mathbf{u}_{,\beta}$  has been ignored. The variation of the membrane strain reads

$$\delta \alpha_{\alpha\beta} = \frac{1}{2} (\delta \mathbf{u}_{,\alpha} \cdot \mathbf{A}_\beta + \delta \mathbf{u}_{,\beta} \cdot \mathbf{A}_\alpha). \quad (4.56)$$

Linearisation of the curvature (4.39) yields

$$\begin{aligned} \beta_{\alpha\beta} = & -\mathbf{u}_{,\alpha\beta} \cdot \mathbf{A}_3 - \frac{1}{\sqrt{A}} (\mathbf{u}_{,1} \cdot (\mathbf{A}_2 \times \mathbf{A}_{\alpha,\beta}) + \mathbf{u}_{,2} \cdot (\mathbf{A}_{\alpha,\beta} \times \mathbf{A}_1)) \\ & + \frac{\mathbf{A}_{\alpha,\beta} \cdot \mathbf{A}_3}{\sqrt{A}} (\mathbf{u}_{,1} \cdot (\mathbf{A}_2 \times \mathbf{A}_3) + \mathbf{u}_{,2} \cdot (\mathbf{A}_3 \times \mathbf{A}_1)), \end{aligned} \quad (4.57)$$

and a variation of the curvature reads

$$\begin{aligned} \delta \beta_{\alpha\beta} = & -\delta \mathbf{u}_{,\alpha\beta} \cdot \mathbf{A}_3 - \frac{1}{\sqrt{A}} (\delta \mathbf{u}_{,1} \cdot (\mathbf{A}_2 \times \mathbf{A}_{\alpha,\beta}) + \delta \mathbf{u}_{,2} \cdot (\mathbf{A}_{\alpha,\beta} \times \mathbf{A}_1)) \\ & + \frac{\mathbf{A}_{\alpha,\beta} \cdot \mathbf{A}_3}{\sqrt{A}} (\delta \mathbf{u}_{,1} \cdot (\mathbf{A}_2 \times \mathbf{A}_3) + \delta \mathbf{u}_{,2} \cdot (\mathbf{A}_3 \times \mathbf{A}_1)). \end{aligned} \quad (4.58)$$

Note that as the shell deformation is small, basis vectors in the above equations are all in the reference configuration. The expressions for the variations  $\delta\alpha_{\alpha\beta}$  and  $\delta\beta_{\alpha\beta}$  are similar to those for  $\alpha_{\alpha\beta}$  and  $\beta_{\alpha\beta}$ , respectively, because the linear regime of the shell deformation has been considered. See Cirak et al. (2000) and Ciarlet (2005) for more elaborations.

Until now, no assumption has been made to the basis in which the components of the displacement vector as well as strain and stress tensors are defined. Both the fixed orthonormal basis  $(E^1, E^2, E^3)$  and the local curvilinear basis  $(A^1, A^2, A^3)$  can be used to express these quantities.

#### 4.5.1 Strain components for a fixed orthonormal basis

Using the fixed orthonormal basis  $(E^1, E^2, E^3)$ , we can express the displacement vector  $\mathbf{u}$  as

$$\mathbf{u} = u_i \mathbf{E}^i, \quad (4.59)$$

where  $u_i$  are displacement components corresponding to the  $\mathbf{E}^i$  directions. Since basis vectors  $\mathbf{E}^i$  are fixed,

$$\mathbf{u}_{,\alpha} = u_{i,\alpha} \mathbf{E}^i. \quad (4.60)$$

Expressions for the membrane components  $\alpha_{\alpha\beta}(\mathbf{u})$  and the bending components  $\beta_{\alpha\beta}(\mathbf{u})$  of the strain tensor in equations (4.55) and (4.57) can be rewritten as

$$\alpha_{\alpha\beta} = \frac{1}{2} \left( u_{i,\alpha} \mathbf{E}^i \cdot \mathbf{A}_\beta + u_{i,\beta} \mathbf{E}^i \cdot \mathbf{A}_\alpha \right), \quad (4.61)$$

and

$$\begin{aligned} \beta_{\alpha\beta} = & -u_{i,\alpha\beta} \mathbf{E}^i \cdot \mathbf{A}_3 - \frac{1}{\sqrt{A}} \left( u_{i,1} \mathbf{E}^i \cdot (\mathbf{A}_2 \times \mathbf{A}_{\alpha,\beta}) + u_{i,2} \mathbf{E}^i \cdot (\mathbf{A}_{\alpha,\beta} \times \mathbf{A}_1) \right) \\ & + \frac{\mathbf{A}_{\alpha,\beta} \cdot \mathbf{A}_3}{\sqrt{A}} \left( u_{i,1} \mathbf{E}^i \cdot (\mathbf{A}_2 \times \mathbf{A}_3) + u_{i,2} \mathbf{E}^i \cdot (\mathbf{A}_3 \times \mathbf{A}_1) \right). \end{aligned} \quad (4.62)$$

Expressions for  $\delta\alpha_{\alpha\beta}(\mathbf{w})$  and  $\delta\beta_{\alpha\beta}(\mathbf{w})$  can be obtained similarly.

The use of the fixed orthonormal basis avoids the complexity related to differentiation of basis vectors. This is particularly attractive when dealing with geometrically nonlinear thin shell problems.

#### 4.5.2 Strain components for a curvilinear basis

In a curvilinear basis system  $(G^1, G^2, G^3)$  in a thin shell, basis vectors  $\mathbf{G}^i$  are defined locally. They may change from point to point in the shell domain. For

differentiating a quantity defined in the shell domain, covariant differentiation is often used.

### Covariant differentiation on a curvilinear basis

For the case of a scalar quantity, denoted by  $\varphi$ , there is no distinction between a covariant derivative and an ordinary derivative (Chapelle and Bathe, 2003). The gradient of  $\varphi$ , which is a vector quantity, can therefore be expressed by

$$\nabla \varphi = \varphi_{,i} \mathbf{G}^i, \quad (4.63)$$

where  $\varphi_{,i} = \partial \varphi / \partial \xi^i$ ,  $\xi^i$  are parametric coordinates,  $i = 1 \rightarrow d$ .

For a vector quantity  $\boldsymbol{\varphi}$ ,  $\boldsymbol{\varphi} = \varphi_i \cdot \mathbf{G}^i$ , the gradient is a second-order tensor and its covariant components are defined by

$$\varphi_{i|j} = \varphi_{i,j} - \Gamma_{ij}^k \varphi_k, \quad (4.64)$$

where  $\Gamma_{ij}^k$  denotes the Christoffel symbol,

$$\Gamma_{ij}^k = \mathbf{G}_{i,j} \cdot \mathbf{G}^k. \quad (4.65)$$

The gradient of  $\boldsymbol{\varphi}$  is then expressed by

$$\nabla \boldsymbol{\varphi} = \varphi_{i|j} \mathbf{G}^i \otimes \mathbf{G}^j. \quad (4.66)$$

For more details about the concept of covariant derivatives, the reader can refer to various textbooks, including those by Pressley (2001), Chapelle and Bathe (2003) and Ciarlet (2005).

### Differentiations of terms on the shell mid-surface

On the shell mid-surface, the displacement vector  $\mathbf{u}$  is expressed by

$$\mathbf{u} = u_i \mathbf{A}^i. \quad (4.67)$$

With the aid of covariant derivatives, membrane and bending strain components on the mid-surface can be expressed in terms of the displacement components  $u_i$  and their covariant derivatives. Based on equations (4.55), the membrane strain components  $\alpha_{\alpha\beta}$  read

$$\alpha_{\alpha\beta} = \frac{1}{2} \left( u_{\alpha|\beta} + u_{\beta|\alpha} \right) - B_{\alpha\beta} u_3, \quad (4.68)$$

where  $u_{\alpha|\beta}$  is the covariant derivative of  $u_\alpha$ , which is expressed as

$$u_{\alpha|\beta} = u_{\alpha,\beta} - \Gamma_{\alpha\beta}^\gamma u_\gamma, \quad (4.69)$$

recalling that  $\alpha, \beta = 1 \rightarrow 2$ . The curvature components in equation (4.57) read

$$\beta_{\alpha\beta} = u_{3|\alpha\beta} + B_{\alpha|\beta}^\gamma u_\gamma + B_\alpha^\gamma u_{\gamma|\beta} + B_\beta^\gamma u_{\gamma|\alpha} - B_\alpha^\gamma B_{\gamma\beta} u_3, \quad (4.70)$$

where the covariant derivatives  $u_{3|\alpha\beta}$  and  $B_{\alpha|\beta}^\gamma$  are defined as

$$u_{3|\alpha\beta} = u_{3,\alpha\beta} - \Gamma_{\alpha\beta}^\gamma u_{3,\gamma}, \quad (4.71)$$

$$B_{\alpha|\beta}^\gamma = B_{\alpha,\beta}^\gamma + \Gamma_{\beta\delta}^\gamma B_\alpha^\delta - \Gamma_{\alpha\beta}^\delta B_\delta^\gamma. \quad (4.72)$$

Expressions of  $B_{\alpha\beta}$  and  $B_\alpha^\beta$  were given in equations (4.6) and (4.7). The above results show that, the in-plane components  $u_\alpha$  of the displacement belong to the space  $H^1(\Omega)$  and the out-plane component  $u_3$  of the displacement belongs to the space  $H^2(\Omega)$ .

This approach is attractive when there exists a simple exact mapping from a two-dimension parametric domain to the shell mid-surface. In this case, fundamental forms of the surface metric tensors can be evaluated simply. Expressions of interests can therefore be built on the two-dimensional parametric domain. Thus, formulations for the considered thin shell problem become similar to those of a plate bending problem. However, the presence of covariant differentiations often leads to a complicated implementation, especially for geometrically non-linear problems.

## Chapter 5

### $C^0$ formulation for linear thin shells\*

As seen in Section 4.2.2, there exist second-order derivatives with respect to the displacement involving to the bending curvature of thin shells. The classical regularity requirement for thin shells therefore demands that the transverse component of the displacement comes from the space  $H^2$ . Motivated largely by the difficulty in constructing suitable  $H^2$  finite element bases, a number of approaches have been presented for simulation of thin shells. A review of various approaches can be found in the books by Bernadou (1996) and Chapelle and Bathe (2003).

A  $C^0$  finite element formulation for linear thin shells is introduced here. Unlike some recent rotation-free approaches for thin shells which are built on novel  $H^2$ -conforming bases (Krysl and Belytschko, 1996; Cirak et al., 2000; Hughes et al., 2005), the considered approach utilises standard finite element basis functions. It bases upon recent developments in discontinuous Galerkin methods for second-order (Bassi and Rebay, 1997; Arnold et al., 2002; Brezzi et al., 2000) and fourth-order (Engel et al., 2002) elliptic problems. The method is a generalisation of the approach in Chapter 3 for thin plates. Here,  $H^1$  functions are used for all displacement components and continuity of the normal rotation across element boundaries is imposed in a weak sense. Unlike the case of thin plates, curved mid-surfaces are considered and the membrane strain component on the shell mid-surface is present. The formulation developed in this work posses several attractive features, including the use of simple basis functions, the absence of shear locking effects and the simplicity with which  $C^0$  geometric features can be incorporated.

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\*This chapter is based on Dung and Wells (2007) and Dung and Wells (2008).

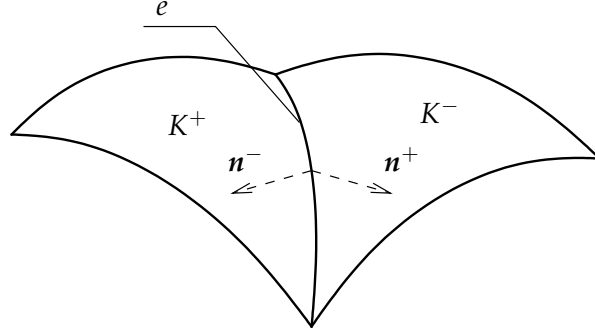


Figure 5.1: Outward normal vectors at an inter-element edge for a  $C^0$  mid-surface.

## 5.1 Finite element discretisation

The linear thin shell problem with the shell description in equations (4.1) to (4.15) and the kinematic equations (4.55) to (4.72) is now considered. Adopt a partition  $\mathcal{P}^h$  of the reference mid-surface  $\Omega$  into  $N$  elements  $K_i$ ,  $i = 1 \rightarrow N$ , such that  $\bigcup_{i=1}^N \bar{K}_i = \bar{\Omega}$  and  $\bigcup_{i=1}^N K_i = \tilde{\Omega}$ . The union of all element edges is denoted by  $\Gamma$ ,  $\Gamma = \bigcup_{i=1}^N \partial K_i$ , and the union of all internal element edges is denoted by  $\tilde{\Gamma}$ ,  $\tilde{\Gamma} = \Gamma \setminus \partial\Omega$ .

The trial and test function spaces on the mid-surface are defined as follows

$$V^h = \left\{ u_i^h \in H^1(\Omega) : u_i^h \in P^k(K_j), \forall K_j \in \mathcal{P}^h; u_i^h|_{\Gamma^u} = g_i^u \right\}, \quad (5.1)$$

$$W^h = \left\{ w_i^h \in H^1(\Omega) : w_i^h \in P^k(K_j), \forall K_j \in \mathcal{P}^h; w_i^h|_{\Gamma^u} = 0 \right\}, \quad (5.2)$$

where  $u_i^h$  and  $w_i^h$ , respectively, are components of vectors  $\mathbf{u}^h$  and  $\mathbf{w}^h$ , relative to either the fixed orthonormal basis  $\mathbf{E}^i$  or the curvilinear basis  $\mathbf{A}^i$ , and  $P^k(K_j)$  are standard Lagrange finite element basis functions of polynomial order  $k$  on elements  $K_j$ .

In the following sections, operators will again be defined at elements boundaries in terms of values on '+' and '-' sides of an element edge with the aid of the trace definition of a function in equation (2.12). Recall that  $\mathbf{n}^+$  is the outward normal vector to the '+' element and  $\mathbf{n}^-$  is the outward normal vector to the '-' element, see Figure 5.1 for an illustration. Note that as we allow for a  $C^0$  representation of the shell mid-surface,  $\mathbf{n}^+ \neq -\mathbf{n}^-$ .

## 5.2 $C^0$ finite element formulation based on a lifting operator

### 5.2.1 Lifting approach for the rotation term

As discussed in the previous chapter, a conforming  $C^1$  continuity for the bending strain is naturally required. Shell finite elements are therefore difficult to construct. The problematic part lies on the first term on the RHS in equation (4.57) for the curvature tensor which involves second derivatives of the displacement. This term is extracted here and reads

$$\beta'_{\alpha\beta} = -\mathbf{u}_{,\alpha\beta}^h \cdot \mathbf{A}_3. \quad (5.3)$$

To deal with this part, we first define a function space  $\mathcal{R}^h$ ,

$$\mathcal{R}^h = \left\{ \tau^{\alpha\beta} \in L^2(\Omega) : \tau^{\alpha\beta}(K_i) \in P^l(K_i) \ \forall K_i \in \mathcal{P}^h, \tau^{\alpha\beta}(K_i) = \tau^{\beta\alpha}(K_i) \right\}. \quad (5.4)$$

A condition on the definition of  $\mathcal{R}^h$  is that it has to contain all second derivatives of a function  $v_i \in V^h$  on an element interior,  $v_{i,\alpha\beta} \in \mathcal{R}^h$  on  $\tilde{\Omega}$ . For triangular elements, this simply implies that  $l \geq k - 2$ .

We now consider a variational form of equation (5.3). It reads: find  $\beta'_{\alpha\beta} \in \mathcal{R}^h$  such that

$$\int_{\tilde{\Omega}} \tau^{\alpha\beta} \beta'_{\alpha\beta} d\Omega = - \int_{\tilde{\Omega}} \tau^{\alpha\beta} \mathbf{u}_{,\alpha\beta}^h \cdot \mathbf{A}_3 d\Omega \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h. \quad (5.5)$$

Applying integration by parts to the RHS of equation (5.5), we have

$$\int_{\tilde{\Omega}} \tau^{\alpha\beta} \beta'_{\alpha\beta} d\Omega = \int_{\tilde{\Omega}} \left( \tau^{\alpha\beta} \mathbf{A}_3 \right)_{,\beta} \cdot \mathbf{u}_{,\alpha}^h d\Omega - \sum_K \int_{\partial K} \tau^{\alpha\beta} n_\beta \hat{\theta}_\alpha d\Gamma, \quad (5.6)$$

where  $\hat{\theta}_\alpha$  is known as the rotation numerical flux, or simply, the numerical rotation. Applying integration by parts to the first term on the RHS of equation (5.6) gives

$$\int_{\tilde{\Omega}} \tau^{\alpha\beta} \beta'_{\alpha\beta} d\Omega = - \int_{\tilde{\Omega}} \tau^{\alpha\beta} \mathbf{u}_{,\alpha\beta}^h \cdot \mathbf{A}_3 d\Omega + \sum_K \int_{\partial K} \tau^{\alpha\beta} n_\beta \left( \mathbf{u}_{,\alpha}^h \cdot \mathbf{A}_3 - \hat{\theta}_\alpha \right) d\Gamma. \quad (5.7)$$

Similar to the approach presented in Chapter 3 for the thin bending analysis, the role of this expression will be to maintain consistency and enforce weak continuity of the normal component of the rotation, projected onto  $\mathbf{A}_3$ . Since there exists a degree of redundancy in the boundary integral (see Remark 3.1.1), we

resolve the surface integral term into normal and tangential contributions, and considering the normal component only,

$$\begin{aligned} \int_{\tilde{\Omega}} \tau^{\alpha\beta} \widehat{\beta}_{\alpha\beta} d\Omega &= - \int_{\tilde{\Omega}} \tau^{\alpha\beta} \mathbf{u}_{,\alpha\beta}^h \cdot \mathbf{A}_3 d\Omega \\ &\quad + \sum_K \int_{\partial K} \tau^{\alpha\beta} n_\alpha n_\beta \left( \mathbf{u}_{,\alpha}^h n^\alpha \cdot \mathbf{A}_3 - \widehat{\theta}_\alpha n^\alpha \right) d\Gamma. \end{aligned} \quad (5.8)$$

The above equation can be expressed in terms of jump and average operators at element edges with the aid of equation (2.17),

$$\begin{aligned} \int_{\tilde{\Omega}} \tau^{\alpha\beta} \widehat{\beta}_{\alpha\beta} d\Omega &= - \int_{\tilde{\Omega}} \tau^{\alpha\beta} \mathbf{u}_{,\alpha\beta}^h \cdot \mathbf{A}_3 d\Omega + \int_{\Gamma} \langle \tau^{\alpha\beta} n_\alpha n_\beta \rangle \left[ \left[ \mathbf{u}_{,\alpha}^h n^\alpha \cdot \mathbf{A}_3 \Xi - \widehat{\theta}_\alpha n^\alpha \Xi \right] \right] d\Gamma \\ &\quad + \int_{\tilde{\Gamma}} \left[ \tau^{\alpha\beta} n_\alpha n_\beta \right] \left\langle \mathbf{u}_{,\alpha}^h n^\alpha \cdot \mathbf{A}_3 \Xi - \widehat{\theta}_\alpha n^\alpha \Xi \right\rangle d\Gamma, \end{aligned} \quad (5.9)$$

where  $\Xi$  is the sign function defined in equation (2.16). By denoting

$$\tau_{nn} = \tau^{\alpha\beta} n_\alpha n_\beta, \quad (5.10)$$

$$\mathbf{u}_{,n}^h = \mathbf{u}_{,\alpha}^h n^\alpha \Xi, \quad (5.11)$$

$$\widehat{\theta}_n = \widehat{\theta}_\alpha n^\alpha \Xi, \quad (5.12)$$

and considering the choice of  $\widehat{\theta}_n$  as follows

$$\widehat{\theta}_n = \begin{cases} \langle \mathbf{u}_{,n}^h \cdot \mathbf{A}_3 \rangle & e \in \tilde{\Gamma} \cup \Gamma^M, \\ g^\theta & e \in \Gamma^\theta, \end{cases} \quad (5.13)$$

equation (5.9) reduces to

$$\begin{aligned} \int_{\tilde{\Omega}} \tau^{\alpha\beta} \widehat{\beta}_{\alpha\beta} d\Omega &= - \int_{\tilde{\Omega}} \tau^{\alpha\beta} \mathbf{u}_{,\alpha\beta}^h \cdot \mathbf{A}_3 d\Omega + \int_{\tilde{\Gamma} \cup \Gamma^\theta} \langle \tau_{nn} \rangle \left[ \left[ \mathbf{u}_{,n}^h \cdot \mathbf{A}_3 \right] \right] d\Gamma \\ &\quad - \int_{\Gamma^\theta} \tau_{nn} g^\theta d\Gamma. \end{aligned} \quad (5.14)$$

The average and jump terms in the above equation are explicitly given as

$$\langle \tau_{nn} \rangle = \begin{cases} \frac{1}{2} \left( \tau^{\alpha\beta+} n_\alpha^+ n_\beta^+ + \tau^{\alpha\beta-} n_\alpha^- n_\beta^- \right) & e \in \tilde{\Gamma}, \\ \tau^{\alpha\beta} n_\alpha n_\beta & e \in \Gamma^\theta, \end{cases} \quad (5.15)$$

$$\left[ \left[ \mathbf{u}_{,n}^h \cdot \mathbf{A}_3 \right] \right] = \begin{cases} \mathbf{u}_{,\alpha}^{h+} n^{\alpha+} \cdot \mathbf{A}_3^+ + \mathbf{u}_{,\alpha}^{h-} n^{\alpha-} \cdot \mathbf{A}_3^- & e \in \tilde{\Gamma}, \\ \mathbf{u}_{,\alpha}^h n^\alpha \cdot \mathbf{A}_3 & e \in \Gamma^\theta. \end{cases} \quad (5.16)$$

### Lifting operations

Within the concept behind the lifting operations presented by Bassi and Rebay (1997) for elliptic problems, a lifting operation is defined here involving to the two last terms of equation (5.14). The lifting operation is defined by: find  $R_{\alpha\beta}^g \in \mathcal{R}^h$  such that

$$\int_{\Omega} \tau^{\alpha\beta} R_{\alpha\beta}^g d\Omega = \int_{\tilde{\Gamma} \cup \Gamma^\theta} \langle \tau_{nn} \rangle \left[ \left[ \mathbf{u}_{,n}^h \cdot \mathbf{A}_3 \right] \right] d\Gamma - \int_{\Gamma^\theta} \tau_{nn} g^\theta d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h. \quad (5.17)$$

Here, jumps in the normal derivative on  $\tilde{\Gamma} \cup \Gamma^\theta$ , projected onto  $\mathbf{A}_3$ , are transformed into a curvature-like term  $R_{\alpha\beta}^g$  on element interiors. A version of the lifting operation corresponding to the test function  $w_i^h \in W^h$  is defined by: find  $R_{\alpha\beta} \in \mathcal{R}^h$  such that

$$\int_{\Omega} \tau^{\alpha\beta} R_{\alpha\beta} d\Omega = \int_{\tilde{\Gamma} \cup \Gamma^\theta} \langle \tau_{nn} \rangle \left[ \left[ \mathbf{w}_{,n}^h \cdot \mathbf{A}_3 \right] \right] d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h, \quad (5.18)$$

where  $\mathbf{w}_{,n}^h = \mathbf{w}_{,\alpha}^h n^\alpha \Xi$ , in which  $\Xi$  is the sign function. Taking the lifting procedure into account, the total curvature, denoted by  $\beta_{\alpha\beta}^*$ , on  $\tilde{\Omega}$  is given by

$$\beta_{\alpha\beta}^* \left( \mathbf{u}^h \right) = \beta_{\alpha\beta} \left( \mathbf{u}^h \right) + R_{\alpha\beta}^g, \quad (5.19)$$

and the total curvature corresponding to the test field  $\mathbf{w}^h$  is given by

$$\beta_{\alpha\beta}^* \left( \mathbf{w}^h \right) = \beta_{\alpha\beta} \left( \mathbf{w}^h \right) + R_{\alpha\beta}, \quad (5.20)$$

where the expression of the standard curvature components  $\beta_{\alpha\beta}$  has been given in equation (4.57). Note that if  $\mathbf{u}^h$  and  $\mathbf{w}^h$  are constructed such that the normal rotation components, projected onto  $\mathbf{A}_3$ , are continuous,  $R_{\alpha\beta}^g$  and  $R_{\alpha\beta}$  vanish. In practice, the terms  $R_{\alpha\beta}^g$  and  $R_{\alpha\beta}$  are computed edge-wise. For  $e \in \tilde{\Gamma}$ ,

$$\int_{K^+ \cup K^-} \tau^{\alpha\beta} r_{\alpha\beta}^{g,e} d\Omega = \int_e \langle \tau_{nn} \rangle \left[ \left[ \mathbf{u}_{,n}^h \cdot \mathbf{A}_3 \right] \right] d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h, \quad (5.21)$$

$$\int_{K^+ \cup K^-} \tau^{\alpha\beta} r_{\alpha\beta}^e d\Omega = \int_e \langle \tau_{nn} \rangle \left[ \left[ \mathbf{w}_{,n}^h \cdot \mathbf{A}_3 \right] \right] d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h, \quad (5.22)$$

and for  $e \in \Gamma^\theta$ ,

$$\int_K \tau^{\alpha\beta} r_{\alpha\beta}^{g,e} d\Omega = \int_e \tau_{nn} \left( \mathbf{u}_{,n}^h \cdot \mathbf{A}_3 - g^\theta \right) d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h, \quad (5.23)$$

$$\int_K \tau^{\alpha\beta} r_{\alpha\beta}^e d\Omega = \int_e \tau_{nn} \mathbf{w}_{,n}^h \cdot \mathbf{A}_3 d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h, \quad (5.24)$$

where  $K^+$  and  $K^-$  denote two elements sharing the edge  $e$ . The lifting terms  $R_{\alpha\beta}^g$  and  $R_{\alpha\beta}$  then read

$$R_{\alpha\beta}^g = \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} r_{\alpha\beta}^{g,e}, \quad (5.25)$$

$$R_{\alpha\beta} = \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} r_{\alpha\beta}^e. \quad (5.26)$$

Note that the functions  $r_{\alpha\beta}^{g,e}$  and  $r_{\alpha\beta}^e$  vanish outside the elements sharing the edge  $e$  and they may be discontinuous across  $e$ .

### Expression of jump terms in different bases

When the displacement vector  $\mathbf{u}^h$  is expressed on the fixed orthonormal basis  $(E^1, E^2, E^3)$ , we have  $\mathbf{u}_{,n}^h = u_{i,\alpha}^h n^\alpha \Xi E^i$ . Therefore the jump term in equation (5.21) can be expressed in terms of displacement components as

$$\llbracket \mathbf{u}_{,n}^h \cdot \mathbf{A}_3 \rrbracket = u_{i,\alpha}^{h+} n^{\alpha+} E^i \cdot \mathbf{A}_3^+ + u_{i,\alpha}^{h-} n^{\alpha-} E^i \cdot \mathbf{A}_3^-. \quad (5.27)$$

When the local curvilinear basis  $(A^1, A^2, A^3)$  is used, we have

$$\begin{aligned} \mathbf{u}_{,n}^h \cdot \mathbf{A}_3 &= u_{i|\alpha}^h n^\alpha A^i \cdot \mathbf{A}_3 \Xi \\ &= u_{3|\alpha}^h n^\alpha \Xi \\ &= u_{3,\alpha}^h n^\alpha \Xi. \end{aligned} \quad (5.28)$$

Thus, the jump term in equation (5.21) is expressed in the local curvilinear basis as

$$\llbracket \mathbf{u}_{,n}^h \cdot \mathbf{A}_3 \rrbracket = u_{3,\alpha}^{h+} n^{\alpha+} + u_{3,\alpha}^{h-} n^{\alpha-}. \quad (5.29)$$

In equation (5.29), only the first-order derivatives of the normal displacement component to the mid-surface appears.

Similar expressions are adopted for jump terms relating to  $\mathbf{w}^h$ .

### 5.2.2 $C^0$ Galerkin formulation

Following the lifting-type discontinuous Galerkin approach presented by Bassi and Rebay (1997) and Brezzi et al. (2000) for second-order problems, we present here a lifting-type  $C^0$  formulation for linear thin shells. The formulation is

obtained by taking the expressions for the total curvatures in equations (5.19) and (5.20) into equation (4.52) and adding a penalty-like term to enhance stability. Then, the Galerkin problem reads: find  $\mathbf{u}^h \in V^h$  such that

$$B^{\text{lif}}(\mathbf{w}^h, \mathbf{u}^h) = L(\mathbf{w}^h) \quad \forall \mathbf{w}^h \in W^h, \quad (5.30)$$

where

$$\begin{aligned} B^{\text{lif}}(\mathbf{w}^h, \mathbf{u}^h) = & \int_{\Omega} \alpha_{\alpha\beta}(\mathbf{w}^h) N^{\alpha\beta}(\mathbf{u}^h) d\Omega + \int_{\bar{\Omega}} \beta_{\alpha\beta}^*(\mathbf{w}^h) M^{\star\alpha\beta}(\mathbf{u}^h) d\Omega \\ & + \sum_{e \in \bar{\Gamma} \cup \Gamma^\theta} \int_{\bar{\Omega}} \frac{\eta t^3}{12} r_{\alpha\beta}^e(\mathbf{w}^h) c^{\alpha\beta\gamma\delta} r_{\gamma\delta}^{g,e}(\mathbf{u}^h) d\Omega, \end{aligned} \quad (5.31)$$

and

$$L(\mathbf{w}) = \int_{\Omega} \mathbf{w} \cdot \mathbf{F} d\Omega + \int_{\Gamma^H} \mathbf{w} \cdot \mathbf{H} d\Gamma + \int_{\Gamma^M} \mathbf{u}_{,\alpha} n^\alpha \cdot \mathbf{A}_3 M d\Gamma. \quad (5.32)$$

In the above equations,  $N^{\alpha\beta}(\mathbf{u}^h)$  are components of the membrane stress tensor,

$$N^{\alpha\beta}(\mathbf{u}^h) = t c^{\alpha\beta\gamma\delta} \alpha_{\gamma\delta}(\mathbf{u}^h), \quad (5.33)$$

$M^{\star\alpha\beta}(\mathbf{u}^h)$  are components of the bending moment tensor,

$$M^{\star\alpha\beta}(\mathbf{u}^h) = \frac{t^3}{12} c^{\alpha\beta\gamma\delta} \beta_{\gamma\delta}^*(\mathbf{u}^h), \quad (5.34)$$

$\mathbf{F}$  is the resultant force acting on the mid-surface,  $\mathbf{H}$  is the resultant force applied on the force boundary  $\Gamma^H$ ,  $M$  is the normal component of the moment vector to the moment boundary  $\Gamma^M$  and  $\eta$  is a dimensionless number. For the case of linear Kirchhoff plates,  $\eta > 0$  ensures stability. The last term on the left-hand side of equation (5.31) is a penalty-like term and is necessary for stability, see the works by (Brezzi et al., 2000) and Arnold et al. (2002) for more discussions. The effect of the penalty parameter for shell problems will be investigated numerically later. Since  $r_{\alpha\beta}^{e,g}$  and  $r_{\alpha\beta}^e$  vanish if the normal rotation is continuous, the presence of the penalty term as well as curvature-like terms  $R_{\alpha\beta}^g$  and  $R_{\alpha\beta}$  do not violate consistency. This procedure allows to use standard  $C^0$  basis functions in terms of the displacement only.

### 5.3 Implementation aspects

Three types of shell elements, which will be used in the numerical examples, are considered in this work: flat shell elements, curved shell elements with an exact geometry and curved shell elements with an interpolated geometry.

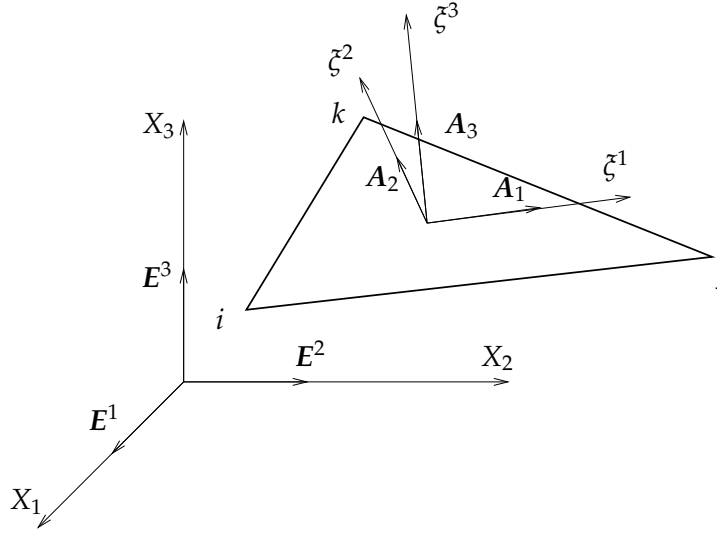


Figure 5.2: Flat shell element.

### 5.3.1 Flat shell elements

Flat shell elements are constructed by combining membrane elements and plate bending elements. Curved mid-surfaces are thus represented approximately by assembled flat elements in a mesh. When the mesh is refined, the approximated mid-surface comes closer to the exact shell mid-surface.

An orthonormal local coordinate system  $(A_1, A_2, A_3)$  is chosen for each element. This local system can be defined in various manners. In this work, the local coordinate system shown in Figure 5.2 is used. It is defined as follows. Let  $i, j$ , and  $k$  denote vertices of a finite element. We first determine the components of vectors  $\vec{ij}$  and  $\vec{ik}$  in the global coordinate system  $(E^1, E^2, E^3)$  via the coordinates of the vertices. The basis vectors  $A_1$  and  $A_3$  are then defined by

$$A_1 = \frac{\vec{ij}}{\|\vec{ij}\|}, \quad (5.35)$$

$$A_3 = \frac{\vec{ij} \times \vec{ik}}{\|\vec{ij} \times \vec{ik}\|}. \quad (5.36)$$

Note that the vector  $A_1$  is parallel to  $\vec{ij}$  and the vector  $A_3$  is normal to the element surface. The basis vector  $A_2$  therefore reads

$$A_2 = A_3 \times A_1. \quad (5.37)$$

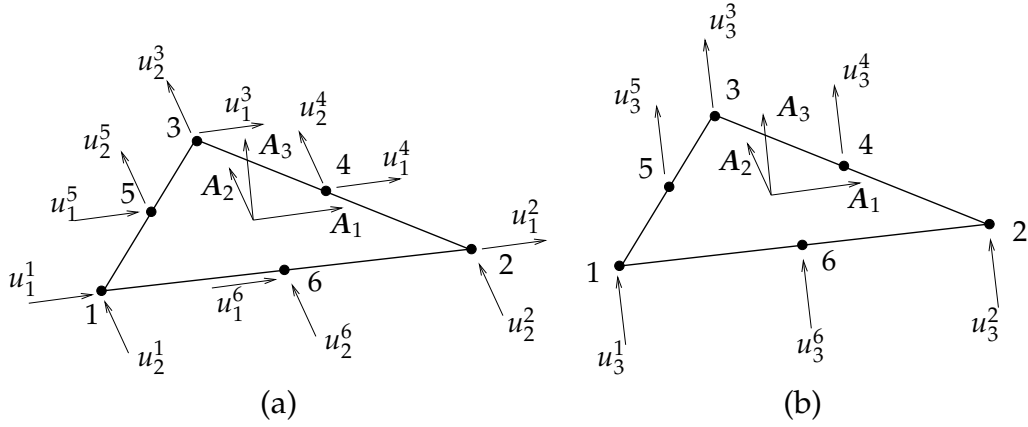


Figure 5.3: Local displacement components for quadratic elements corresponding to: (a) the membrane strain; (b) the bending strain.

Displacement, force, strain and stress components are first expressed with respect to the local coordinate system. As the approximated mid-surface is flat within each finite element, there exists no membrane-bending coupling term in the expressions of strain and stress tensors. Figure 5.3 shows displacement components corresponding membrane and bending strains for quadratic elements, where  $u_i^n$  is the displacement component  $i$  at node  $n$  in the local coordinate system.

In order to construct a global stiffness matrix, the nodal displacement and nodal force components in the local basis are transformed onto the global coordinate system by using a transforming matrix. By means of  $u_{E^j}^n$ , denoting the displacement component following the  $E^j$  direction at node  $n$ , the relationship between the local and global displacement components at node  $n$  reads

$$u_i^n = T_i^j u_{E^j}^n, \quad (5.38)$$

where  $T_i^j = \cos(A_i, E^j)$ ,  $(A_i, E^j)$  denotes the angle between the two vectors  $A_i$  and  $E^j$ . Conversely,

$$u_{E^j}^n = T_j^i u_i^n. \quad (5.39)$$

Note that the nodal displacement vector  $\mathbf{u}^n$  at node  $n$  reads:  $\mathbf{u}^n = u_i^n \mathbf{A}^i = u_{E^j}^n \mathbf{E}^j$ . Here, coefficients  $T_i^j$  form an orthogonal symmetric matrix which does not change within each finite element,  $T_j^i = T_i^j$ . For the implementation of a flat shell element, there is no distinction between the physical domain  $\Omega$  and the parametric domain  $\omega$ .

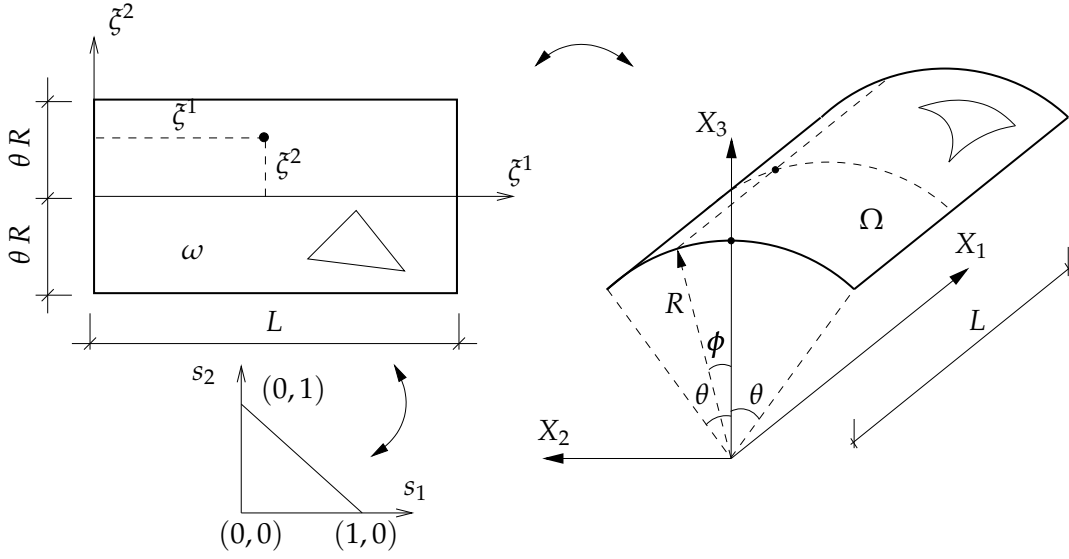


Figure 5.4: Exact mapping for a roof.

### 5.3.2 Curved shell elements with an exact geometry

In some cases, the geometry of shell mid-surfaces can be described exactly using a proper coordinate system. For instance, spherical polar coordinates can be used for the case of a spherical shell and cylindrical polar coordinates can be used for a cylinder. Finite elements then have curvatures which can be determined analytically.

Here, we distinguish between the physical domain  $\Omega$  of the shell mid-surface and the parametric domain  $\omega$  upon which the expressions for the considered problem is built. See Figure 5.4 for an illustration. A mapping from the global coordinates  $X_i$  into the local curvilinear coordinates  $\zeta^\alpha$  associating to  $\omega$  is necessary. Using this mapping, shell equations will be formulated with respect to the parametric coordinates  $\zeta^\alpha$ . For the case of the roof in Figure 5.4, the mappings can be expressed as

$$\zeta^1 = X_1 \quad 0 \leq X_1 \leq L, \quad (5.40)$$

$$\zeta^2 = R\phi \quad -\theta \leq \phi \leq \theta, \quad (5.41)$$

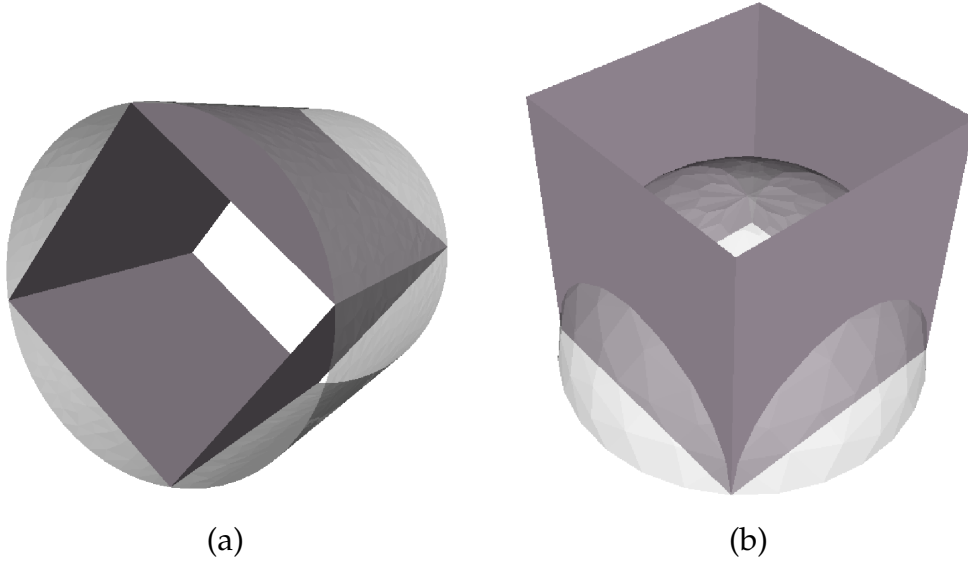


Figure 5.5: Multi-mapping: (a) Cylinder; (b) Hemisphere. Curved surfaces are physical domains and flat surfaces are parametric domains.

where  $\phi = \arcsin (X_2 / R)$ . Conversely,

$$X_1 = \xi^1 \quad 0 \leq \xi^1 \leq L, \quad (5.42)$$

$$X_2 = R \sin \left( \frac{\xi^2}{R} \right) \quad -\theta R \leq \xi^2 \leq \theta R, \quad (5.43)$$

$$X_3 = R \cos \left( \frac{\xi^2}{R} \right) \quad -\theta R \leq \xi^2 \leq \theta R. \quad (5.44)$$

It is important to note that mappings must be injective. This means that a point in the parametric domain corresponds exclusively to a point in the physical domain, and vice versa. In the case of a complex geometry, when the mid-surface can not be defined by a single mapping, a multi-mapping can be used. There, the shell mid-surface can be decomposed into sub-domains such that for each domain, a single mapping can be used. See Figure 5.5 for an example of multi-mappings.

Using an exact mapping, metric quantities on the shell surface regarding to the local curvilinear basis can be quantified exactly and simply. Therefore, the expressions of displacement and force components in a curvilinear coordinate system are often preferable.

### 5.3.3 Curved shell elements with an interpolated geometry

In many cases of thin shells, an exact geometric mapping is difficult to formulate. In such cases, an interpolated geometry is preferable. First, nodal coordinates of finite elements are evaluated, then an interpolation is utilised to approximate coordinates of points on the shell elements.

In order to avoid covariant derivatives of quantities representing the approximated shell mid-surface, displacement components in the global coordinate system are used. Components  $X_i$  of the position vector and components  $u_i$  of the displacement vector at a point on the shell mid-surface can be interpolated in a same form by

$$X_i(\xi^1, \xi^2) = X_i(s_1, s_2, s_3) = \sum_{j=1}^n N_j(s_1, s_2, s_3) X_i^j, \quad (5.45)$$

$$u_i(\xi^1, \xi^2) = u_i(s_1, s_2, s_3) = \sum_{j=1}^n N_j(s_1, s_2, s_3) u_i^j, \quad (5.46)$$

where  $N_j$  denotes the interpolation basis function corresponding node  $j$ ,  $X_i^j$  denotes the  $i$  component of the position vector at node  $j$ ,  $u_i^j$  denotes the  $i$  component of the displacement vector at node  $j$ ,  $n$  is number of nodes for interpolating, and  $(s_1, s_2, s_3)$  denotes area coordinates in the parent triangle. It is reminded that  $s_1 + s_2 + s_3 = 1$ . In this work, the local curvilinear coordinates for each finite element  $(\xi^1, \xi^2)$  have been identified with two components  $(s_1, s_2)$  of the area coordinate system. The parametric domain therefore coincides with the parent triangle within each finite element. See Figure 5.6 for an illustration.

It is important to determine the unit tangent vector  $\mathbf{s}$  and the unit outward normal vector  $\mathbf{n}$  to the curved element boundary. At a point on an edge of the parent element, let  $s$  denote a local coordinate following the tangent direction. Obviously, the relation between  $s$  and  $\xi^\alpha$  is linear. For instance, for the case shown in Figure 5.6,  $s = \sqrt{2}\xi^1 = \sqrt{2}(1 - \xi^2)$ . For the later use, we define two coefficients  $m^1$  and  $m^2$  by

$$m^1 = \frac{\partial \xi^1}{\partial s}, \quad (5.47)$$

$$m^2 = \frac{\partial \xi^2}{\partial s}. \quad (5.48)$$

Within the concept of a parametrised curve (Pressley, 2001), we can evaluate a tangent vector  $\mathbf{s}'$  and a normal vector  $\mathbf{n}'$  by

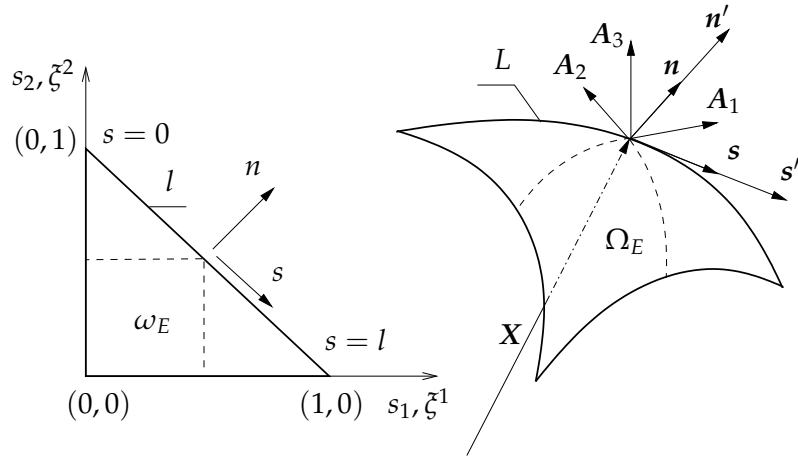


Figure 5.6: Outward normal and tangent vectors.

$$\mathbf{s}' = \frac{\partial \mathbf{X}}{\partial s} = \frac{\partial \mathbf{X}}{\partial \xi^\alpha} \frac{\partial \xi^\alpha}{\partial s} = A_1 m^1 + A_2 m^2, \quad (5.49)$$

$$\mathbf{n}' = A_3 \times \mathbf{s}' = A_3 \times (A_1 m^1 + A_2 m^2), \quad (5.50)$$

where  $(A_1, A_2, A_3)$  is the local curvilinear basis at the considered point on the curved boundary. After normalised, the unit tangent vector  $\mathbf{s}$  reads:

$$\mathbf{s} = \frac{\mathbf{s}'}{\|\mathbf{s}'\|}, \quad (5.51)$$

and the unit normal vector  $\mathbf{n}$  reads

$$\mathbf{n} = \frac{\mathbf{n}'}{\|\mathbf{n}'\|}, \quad (5.52)$$

where  $\|\cdot\|$  denotes the  $L^2$ -norm of a vector. Covariant components of these vector are evaluated by

$$s_\alpha = \mathbf{s} \cdot \mathbf{A}_\alpha, \quad (5.53)$$

$$n_\alpha = \mathbf{n} \cdot \mathbf{A}_\alpha, \quad (5.54)$$

and contravariant components of these vector are evaluated by

$$s^\alpha = \mathbf{s} \cdot \mathbf{A}^\alpha, \quad (5.55)$$

$$n^\alpha = \mathbf{n} \cdot \mathbf{A}^\alpha. \quad (5.56)$$

The length of the curved edge can be calculated by

$$L = \int_{s=0}^{s=l} \|\mathbf{s}'\| ds. \quad (5.57)$$

Note that, vectors  $\mathbf{s}$  and  $\mathbf{n}$  need to be determined at every integration point associated with element boundary integrals.

## 5.4 Numerical examples

The proposed formulation is examined here through some benchmark tests, consisting of the Scordelis-Lo roof, hemisphere and pinched cylinder tests. These problems have been used by other authors, including Belytschko et al. (1985) and Cirak et al. (2000). Here, shell elements with various orders of basis functions in  $P^k(K_i)$  are used. The lifting functions in  $P^l(K_i)$  are chosen such that  $l = k - 2$ . All three types of shell elements which are interpolated, flat and curved triangular elements are tested. Here, the interpolated type is significantly interesting and is focused on. Symmetry is not exploited, and the entire structures are modelled.

### 5.4.1 Scordelis-Lo roof

The Scordelis-Lo roof benchmark is useful for testing the ability of a shell element to represent situations in which membrane effects are considerable. The geometry of the roof is shown in Figure 5.7. The roof is rigidly supported at the two curved boundaries and free along the straight boundaries. The side length  $L = 50$  m, radius  $R = 25$  m, thickness  $t = 0.25$  m, angle  $\phi = 2/9\pi$  rad, Young's modulus  $E = 4.32 \times 10^8$  N/m<sup>2</sup> and Poisson's ratio  $\nu = 0$ . A uniform body load of  $F = -90$  N/m<sup>2</sup> in the  $X_3$ -direction is applied.

The vertical displacement, computed at the mid-point of the free edge, is normalised using a reference solution of 0.3024 m (Belytschko et al., 1985). Figure 5.8 shows the results using quadratic elements with different values of the penalty parameter  $\eta$ . For all values of the penalty parameter, the results converge well to the analytical solution. A particularly large penalty value of  $\eta = 1 \times 10^6$  has also been considered to examine the influence of a large penalty value, in which case one may expect some locking behaviour. It appears however that, even for very large penalty values, the solution converges well. The convergence behaviour for cubic triangular elements is presented in Figure 5.9. Compared to the quadratic

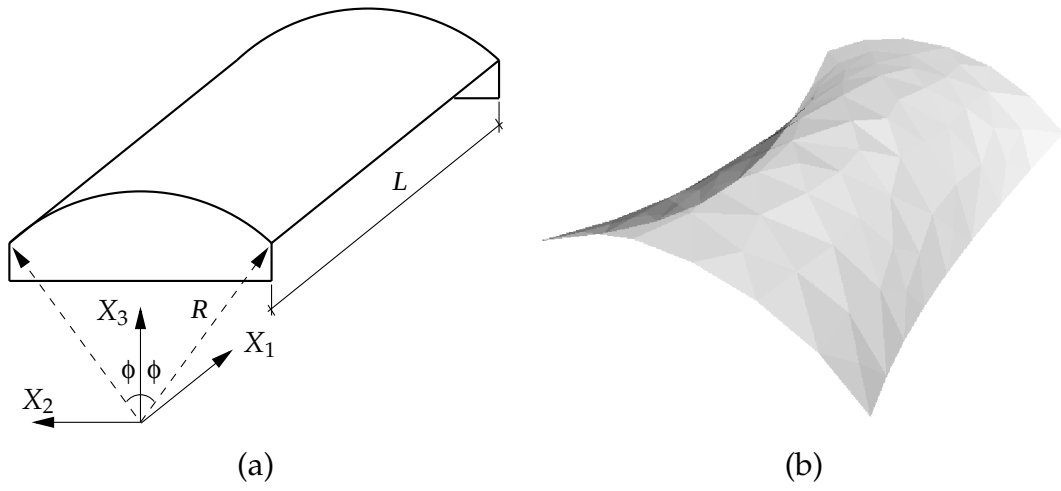


Figure 5.7: Scordelis-Lo roof problem: (a) Geometry; (b) Illustration of deformation.

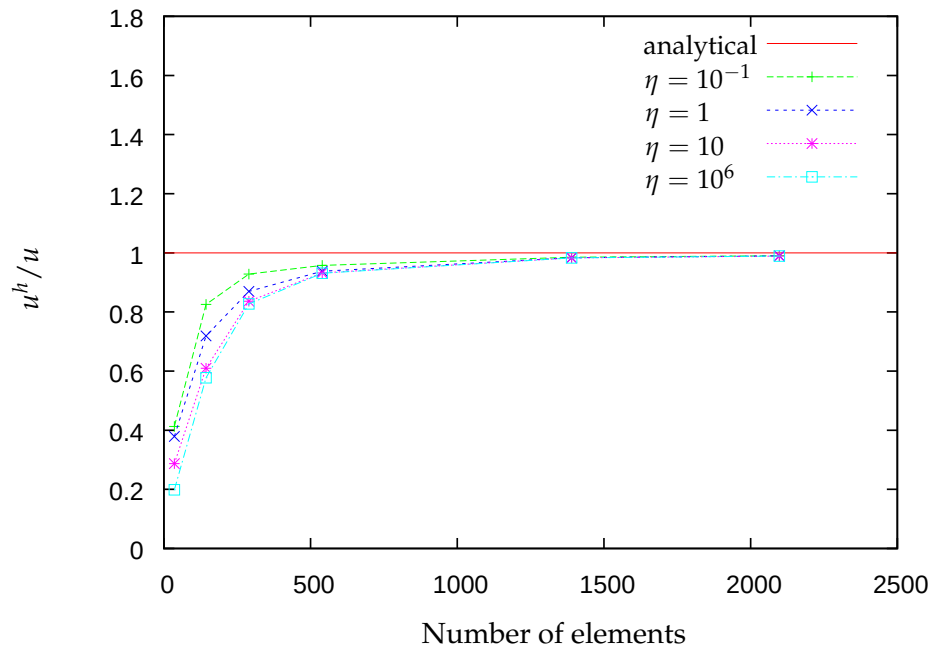


Figure 5.8: Scordelis-Lo roof: normalised displacement the case  $k = 2$  and  $l = 0$  with various penalty values.

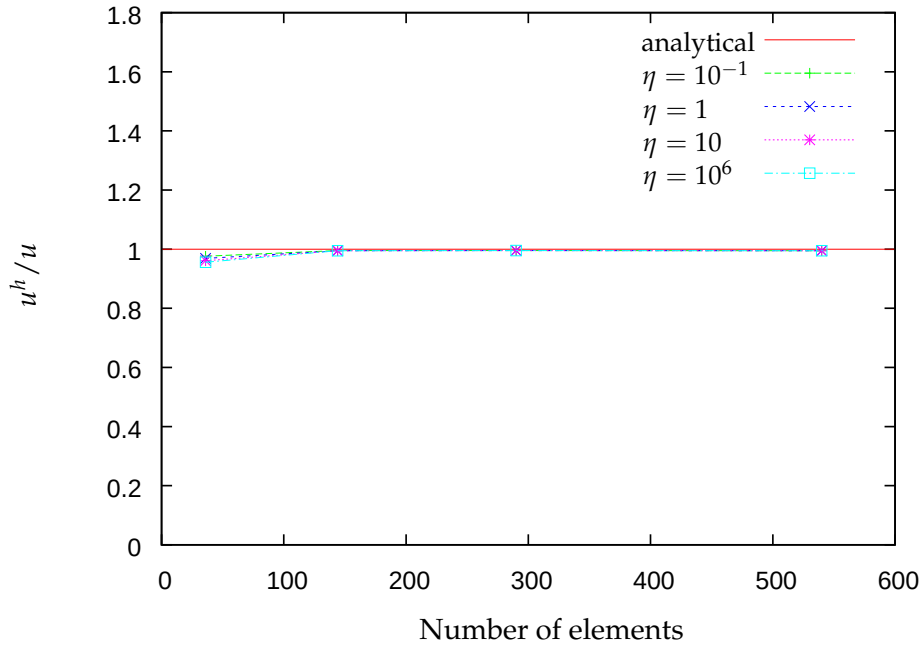


Figure 5.9: Scordelis-Lo roof: normalised displacement for the case  $k = 3$  and  $l = 1$  with various penalty values.

elements, faster convergence and less sensitivity to the penalty parameter are observed.

Shown in Figure 5.10 are results using flat-shell and curved triangular shell elements of order two and three with the penalty parameter  $\eta = 1$ . All results converge well to the exact solution. It is observed that the flat-shell elements are softer. The reason for this may be the lack of membrane-bending coupling effects in the flat-shell model. Again, cubic elements converge faster and show less sensitivity to the penalty parameter  $\eta$  compared to quadratic elements. The results show that the presented elements are able to correctly represent the membrane and bending strains in this test case.

### 5.4.2 Hemispherical shell

The hemispherical shell problem tests the ability of the formulation to deal with in-extensional bending. The geometry of the hemisphere is shown in Figure 5.11, where radius  $R = 10$  m, thickness  $t = 0.04$  m, Young's modulus  $E = 6.825 \times 10^7$  N/m<sup>2</sup> and Poisson's ratio  $\nu = 0.3$ . The applied forces are of magnitude

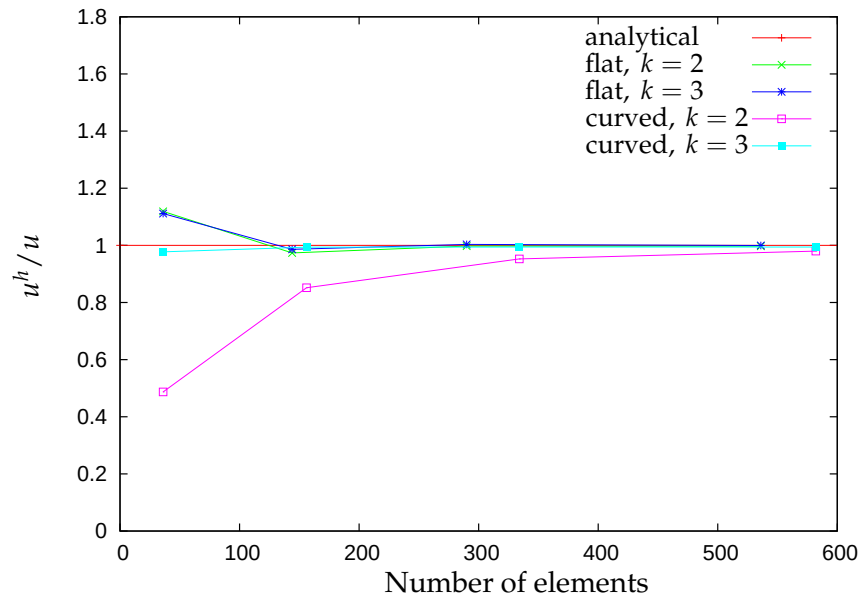


Figure 5.10: Scordelis-Lo roof: normalized displacement with flat and curved elements for the case  $k = 2$  and  $k = 3$  with  $l = k - 2$  and  $\eta = 1$ .

$P = 2 \text{ N}$ .

The computed displacement at the loading points for the hemispherical shell is normalised using the reference solution of 0.0924 m (Belytschko et al., 1985). Figure 5.12 shows the normalised displacement for quadratic elements. The results converge to the analytical solution, although slowly. This can be attributed to membrane locking, which is known to afflict quadratic elements (Stolarski and Belytschko, 1983). Figure 5.13 shows the normalised displacement for cubic elements, which converges rapidly and robustly, and membrane locking is not observed. The magnitude of the penalty parameter has little effect on the computed results.

### 5.4.3 Pinched cylinder

Together with the hemispherical shell, the pinched cylinder benchmark is a useful test of the performance of an element for in-extensional bending. The cylinder, illustrated in Figure 5.14, is of length  $L = 600 \text{ mm}$ , radius  $R = 300 \text{ mm}$  and thickness  $t = 3 \text{ mm}$ , with Young's modulus  $E = 3 \times 10^6 \text{ N/mm}^2$  and Poisson's ratio  $\nu = 0.3$ . Two opposing forces of  $P = 1 \text{ N}$  are applied at opposite sides of

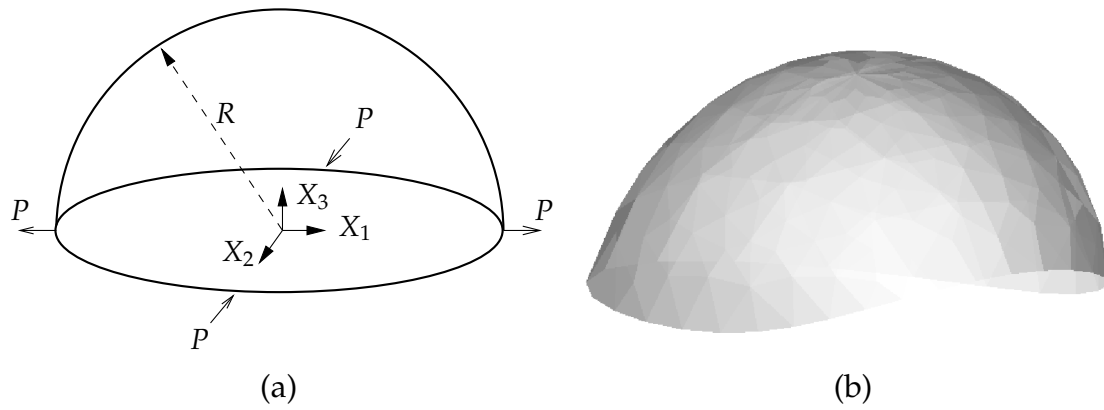


Figure 5.11: Hemispherical shell problem: (a) Geometry; (b) Illustration of deformation.

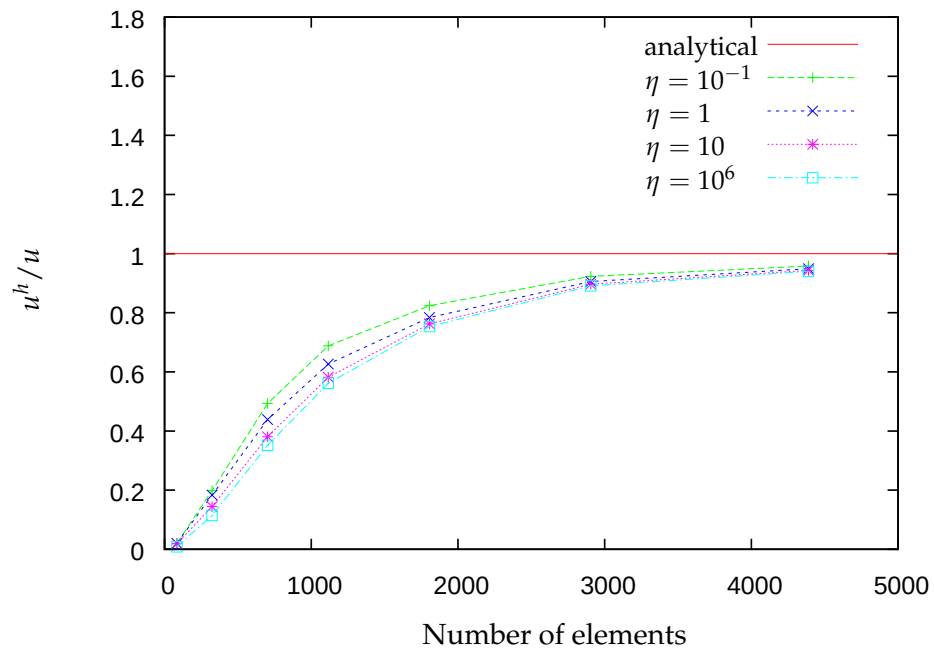


Figure 5.12: Hemispherical shell: normalised displacement for the case  $k = 2$  and  $l = 0$  with various penalty values.

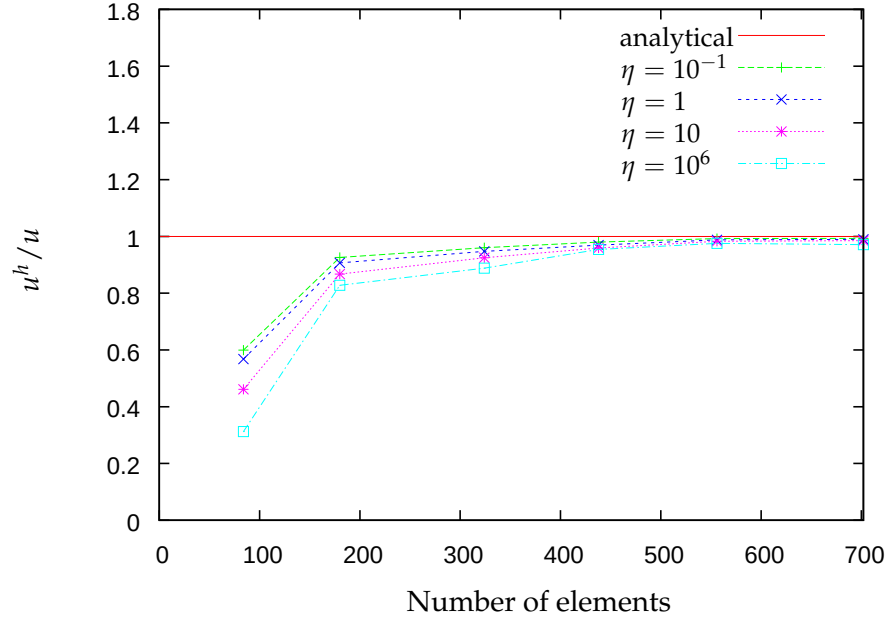


Figure 5.13: Hemispherical shell: normalised displacement for the case  $k = 3$  and  $l = 1$  with various penalty values.

the cylinder. Two cases are considered; one in which the ends of the cylinder are free, and a second in which the ends of the cylinder are constrained by a rigid diaphragm. The rigid diaphragm case is particularly testing. Again, we do not exploit symmetry, but discretise the entire cylinder.

### Free-end case

For the free-end case, a reference solution of  $4.520 \times 10^{-4}$  mm (Cirak et al., 2000) for the displacement at the loading points has been used to normalise the computed results. The reference solution is derived for an in-extensional cylinder under the assumption that all strain components at the mid-surface vanish. Figure 5.15 shows the normalised displacement for quadratic elements. The results converge to the reference solution, although not particularly rapidly, and a degree of sensitivity to the penalty parameters is observed. The rather slow convergence can be attributed to membrane locking. The results for cubic elements are presented in Figure 5.18. Rapid convergence and minimal sensitivity to the penalty term are observed and, as expected, membrane locking is not apparent. For the finer meshes, the computed displacement is larger than the analytical

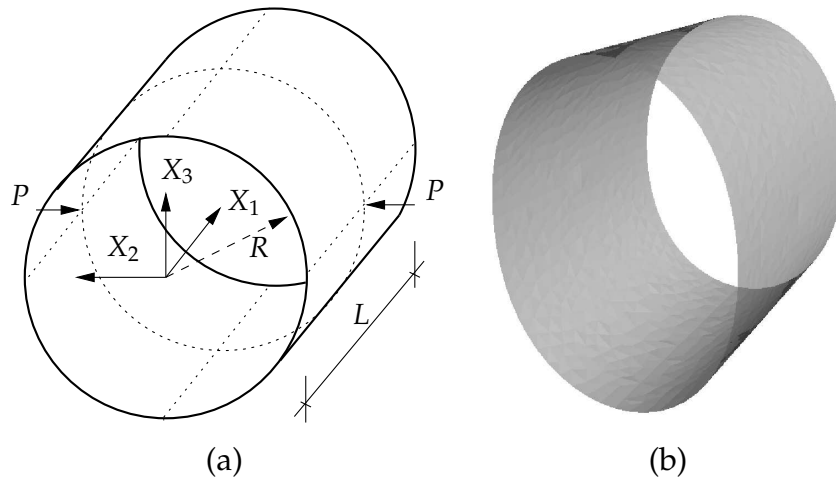


Figure 5.14: Pinched cylinder problem: (a) Geometry; (b) Illustration of deformation for the free boundary case.

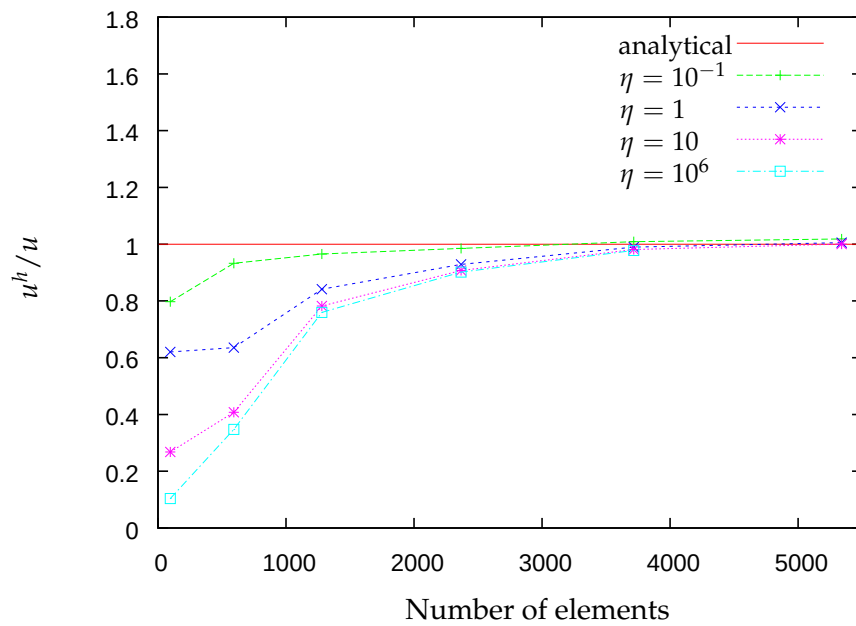


Figure 5.15: Pinched cylinder with free ends: normalised displacement for the case  $k = 2$  and  $l = 0$  with various penalty values.

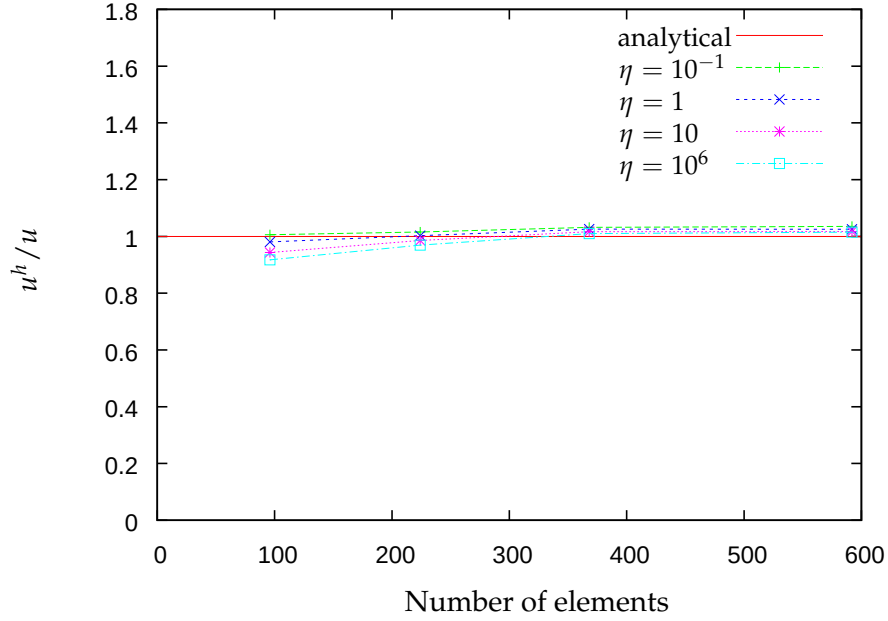


Figure 5.16: Pinched cylinder with free ends: normalised displacement for the case  $k = 3$  and  $l = 1$  with various penalty values.

solution, due to assumptions in the derivation of the analytical solution.

### Rigid diaphragm case

For the rigid diaphragm case, a reference solution of  $1.8248 \times 10^{-5}$  mm (Belytschko et al., 1985) for the displacement at the loading points has been used to normalise the computed displacement. The normalised displacement, computed using quadratic elements, is shown in Figure 5.17. For a sufficiently large penalty value, the computed solution converges to the analytical result. For smaller penalty values, the response is less reliable, which is consistent with observed results for thin plates using quadratic elements in Chapter 3. Somewhat curiously, using a low penalty parameter leads to a response which is softer than the analytical solution. The normalised displacement using cubic elements is shown in Figure 5.18. As for the quadratic elements, low penalty parameters lead to a response which is softer than the analytical solution. For higher values of the penalty term, the computed results converge well to the analytical solution.

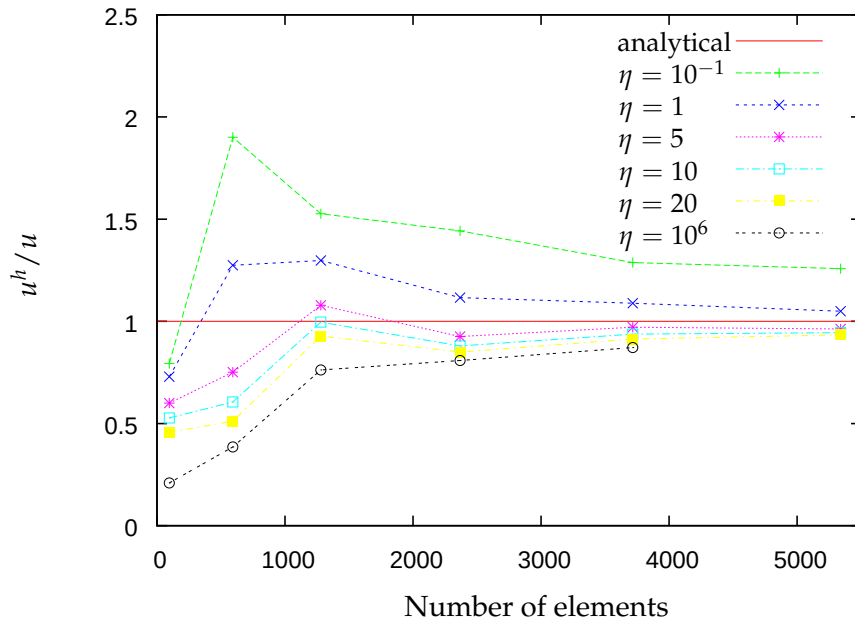


Figure 5.17: Pinched cylinder with rigid diaphragms: normalised displacement for the case  $k = 2$  and  $l = 0$  with various penalty values.

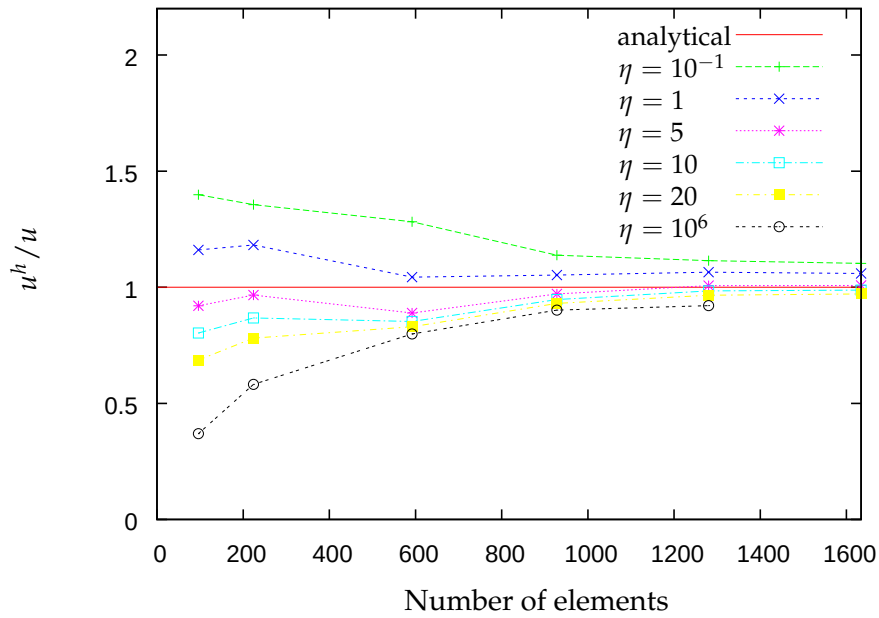


Figure 5.18: Pinched cylinder with rigid diaphragms: normalised displacement for the case  $k = 3$  and  $l = 1$  with various penalty values.

# Chapter 6

## $C^0$ formulation for geometrically nonlinear thin shells\*

The formulation presented here is an extension of the finite element formulation presented in Chapter 5 to the case of large deformation thin shells (the governing equations were introduced in Chapter 4). Again, the formulation utilises standard Lagrange polynomials as basis functions, and involves displacement unknowns only. Lifting operations are used to control the jumps in normal rotation which appear on inter-element edges. The absence of rotation degrees-of-freedom avoids complications in the geometrically nonlinear case. The formulation allows a non-smooth shell mid-surface can be simply dealt with and exact linearisation of the approach is tractable. For numerical implementation, the total Lagrange approach is employed. Since thin shell theory has been addressed directly, shear locking is completely avoided.

### 6.1 Lifting-type formulation

The governing equations (4.1) to (4.54) of the geometrically nonlinear thin shells are now considered. We adopt the finite element discretisation of the shell mid-surface defined in Section 5.1, the finite element spaces  $V^h$  and  $W^h$  given in equations (5.1) and (5.2) and the function space  $\mathcal{R}^h$  defined in equation (5.4). Similar to the case of linear thin shells, it can be seen that the problematic part is a term

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\*This chapter is based on Dung and Wells (2008).

in curvature tensor  $\beta_{\alpha\beta}$  (equation (4.39)). That is

$$\beta'_{\alpha\beta} = -\mathbf{u}_{,\alpha\beta}^h \cdot \mathbf{a}_3. \quad (6.1)$$

It is the part of the second-order derivative of the displacement vector which is projected onto the normal direction to the mid-surface in the current configuration. The corresponding problematic part in the variation  $\delta\beta_{\alpha\beta}$  reads

$$\delta\beta'_{\alpha\beta} = -\delta\mathbf{u}_{,\alpha\beta}^h \cdot \mathbf{a}_3 - \mathbf{u}_{,\alpha\beta}^h \cdot \delta\mathbf{a}_3. \quad (6.2)$$

### 6.1.1 Lifting approach for the rotation term

For the derivation of the lifting formulation for the problem of interest, we utilise a similar procedure to that has been done for linear thin shell (see Section 5.2.1). Note that the current configuration is now being considered. After some manipulations, the variational form of equation (6.1) reads: find  $\hat{\beta}_{\alpha\beta} \in \mathcal{R}^h$  such that

$$\begin{aligned} \int_{\bar{\Omega}} \tau^{\alpha\beta} \hat{\beta}_{\alpha\beta} d\Omega = & - \int_{\bar{\Omega}} \tau^{\alpha\beta} \mathbf{u}_{,\alpha\beta}^h \cdot \mathbf{a}_3 d\Omega + \int_{\bar{\Gamma} \cup \Gamma^\theta} \langle \tau_{nn} \rangle \left[ \left[ \mathbf{u}_{,n}^h \cdot \mathbf{a}_3 \right] \right] d\Gamma \\ & - \int_{\Gamma^\theta} \tau_{nn} g^\theta d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h. \end{aligned} \quad (6.3)$$

where  $\hat{\beta}_{\alpha\beta}$  denotes the part of the problematic term  $\beta'_{\alpha\beta}$  which corresponds to the normal contributions of the inter-element integral term. The reader is reminded that  $\tau_{nn} = \tau^{\alpha\beta} n_\alpha n_\beta$ ,  $\mathbf{u}_{,n}^h = \mathbf{u}_{,\alpha}^h n^\alpha \Xi$ ,  $\Xi$  is the sign function defined in equation (2.16),  $g^\theta$  is the normal rotation to the rotation boundary, and  $\mathbf{a}_3$  is the unit normal vector to the shell mid-surface in the current configuration.

A lifting operation which is applied to the two last terms of equation (6.3) is defined by: find  $R_{\alpha\beta} \in \mathcal{R}^h$  such that

$$\int_{\bar{\Omega}} \tau^{\alpha\beta} R_{\alpha\beta} d\Omega = \int_{\bar{\Gamma} \cup \Gamma^\theta} \langle \tau_{nn} \rangle \left[ \left[ \mathbf{u}_{,n}^h \cdot \mathbf{a}_3 \right] \right] d\Gamma - \int_{\Gamma^\theta} \tau_{nn} g^\theta d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h. \quad (6.4)$$

A jump in the normal derivative, projected onto  $\mathbf{a}_3$ , has been transformed into a curvature-like term  $R_{\alpha\beta}$  on element interiors. A version of the lifting operation for dealing with the problematic part  $\delta\beta'_{\alpha\beta}$  of  $\delta\beta_{\alpha\beta}$  in equation (6.2) reads: for a given displacement variation  $\delta\mathbf{u}^h$  about the current configuration, find  $\delta R_{\alpha\beta} \in \mathcal{R}^h$  such that

$$\begin{aligned} \int_{\bar{\Omega}} \tau^{\alpha\beta} \delta R_{\alpha\beta} d\Omega = & \int_{\bar{\Gamma} \cup \Gamma^\theta} \langle \tau_{nn} \rangle \left[ \left[ \delta\mathbf{u}_{,n}^h \cdot \mathbf{a}_3 \right] \right] d\Gamma \\ & + \int_{\bar{\Gamma} \cup \Gamma^\theta} \langle \tau_{nn} \rangle \left[ \left[ \mathbf{u}_{,n}^h \cdot \delta\mathbf{a}_3 \right] \right] d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h. \end{aligned} \quad (6.5)$$

Taking the lifting operations into account, the total curvature  $\beta_{\alpha\beta}^*$  defined on the mid-surface is given by

$$\beta_{\alpha\beta}^* = \mathbf{a}_{\alpha,\beta} \cdot \mathbf{a}_3 - \mathbf{a}_{\alpha,\beta} \cdot \mathbf{a}_3 - \mathbf{u}_{,\alpha\beta}^h \cdot \mathbf{a}_3 + R_{\alpha\beta}, \quad (6.6)$$

and a variation of the total curvature is given by

$$\delta\beta_{\alpha\beta}^* = -\mathbf{a}_{\alpha,\beta} \cdot \delta\mathbf{a}_3 - \delta\mathbf{u}_{,\alpha\beta}^h \cdot \mathbf{a}_3 - \mathbf{u}_{,\alpha\beta}^h \cdot \delta\mathbf{a}_3 + \delta R_{\alpha\beta}. \quad (6.7)$$

If  $\mathbf{u}^h$  and  $\delta\mathbf{u}^h$  are constructed such that the normal rotation and the variation of the normal rotation projected onto  $\mathbf{a}_3$  are continuous,  $R_{\alpha\beta}$  and  $\delta R_{\alpha\beta}$  vanish.

In practice, the terms  $R_{\alpha\beta}$  and  $\delta R_{\alpha\beta}$  are computed edge-wise. For  $e \in \tilde{\Gamma}$ ,

$$\int_{K^+ \cup K^-} \tau^{\alpha\beta} r_{\alpha\beta}^e d\Omega = \int_e \langle \tau_{nn} \rangle \left[ \left[ \mathbf{u}_{,n}^h \cdot \mathbf{a}_3 \right] \right] d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h, \quad (6.8)$$

$$\begin{aligned} \int_{K^+ \cup K^-} \tau^{\alpha\beta} \delta r_{\alpha\beta}^e d\Omega &= \int_e \langle \tau_{nn} \rangle \left[ \left[ \delta \mathbf{u}_{,n}^h \cdot \mathbf{a}_3 \right] \right] d\Gamma \\ &\quad + \int_e \langle \tau_{nn} \rangle \left[ \left[ \mathbf{u}_{,n}^h \cdot \delta \mathbf{a}_3 \right] \right] d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h, \end{aligned} \quad (6.9)$$

and for  $e \in \Gamma^\theta$ ,

$$\int_{K^+} \tau^{\alpha\beta} r_{\alpha\beta}^e d\Omega = \int_e \tau_{nn} \left( \mathbf{u}_{,n}^h \cdot \mathbf{a}_3 - g^\theta \right) d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h, \quad (6.10)$$

$$\int_{K^+} \tau^{\alpha\beta} \delta r_{\alpha\beta}^e d\Omega = \int_e \tau_{nn} \delta \mathbf{u}_{,n}^h \cdot \mathbf{a}_3 d\Gamma + \int_e \tau_{nn} \mathbf{u}_{,n}^h \cdot \delta \mathbf{a}_3 d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h, \quad (6.11)$$

where  $K^+$  and  $K^-$  are elements sharing the edge  $e$ . The lifting terms then read

$$R_{\alpha\beta} = \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} r_{\alpha\beta}^e, \quad (6.12)$$

$$\delta R_{\alpha\beta} = \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} \delta r_{\alpha\beta}^e. \quad (6.13)$$

Since the functions  $\tau_{\alpha\beta}$  and  $r_{\alpha\beta}^e$  are discontinuous across element boundaries, their computation is local to an element edge and the connected elements, in the current configuration.

### 6.1.2 Galerkin variational problem

Substituting the expressions for the total curvature (6.6) and its variation (6.7) into equation (4.52) and adding a penalty-like term to enhance stability, the equilibrium of virtual work at the current configuration for the Galerkin problem

reads: find  $\mathbf{u}^h \in V^h$  such that

$$\delta \Pi^{\text{int}} = \delta \Pi^{\text{ext}} \quad \forall \delta \mathbf{u}^h \in W^h, \quad (6.14)$$

where

$$\begin{aligned} \delta \Pi^{\text{int}} = & \int_{\Omega} \delta \alpha_{\alpha\beta} N^{\alpha\beta} d\Omega + \int_{\tilde{\Omega}} \delta \beta_{\alpha\beta}^* M^{\star\alpha\beta} d\Omega \\ & + \sum_{e \in \tilde{\Gamma} \cup \Gamma^{\theta}} \int_{\Omega} \frac{\eta t^3}{12} \delta r_{\alpha\beta}^e c^{\alpha\beta\gamma\delta} r_{\gamma\delta}^e d\Omega, \end{aligned} \quad (6.15)$$

$$\delta \Pi^{\text{ext}} = \int_{\Omega} \delta \mathbf{u}^h \cdot \mathbf{F} d\Omega + \int_{\Gamma^H} \delta \mathbf{u}^h \cdot \mathbf{H} d\Gamma + \int_{\Gamma^M} \delta \mathbf{a}_3 \cdot \mathbf{n} M d\Gamma. \quad (6.16)$$

$N^{\alpha\beta}$  are components of the membrane stress tensor,

$$N^{\alpha\beta} = t c^{\alpha\beta\gamma\delta} \alpha_{\gamma\delta}, \quad (6.17)$$

and  $M^{\star\alpha\beta}$  are components of the bending moment tensor,

$$M^{\star\alpha\beta} = \frac{t^3}{12} c^{\alpha\beta\gamma\delta} \beta_{\gamma\delta}^*. \quad (6.18)$$

The reader is reminded that  $\mathbf{F}$ ,  $\mathbf{H}$  and  $M$  are the resultant force vector acting on the mid-surface, the resultant force vector applied on the force boundary and the normal component of the moment vector to the moment boundary, respectively. The last term on the LHS of equation (6.15) is a penalty-like term and is necessary for stability. The effect of this term for linear thin shell problems has been investigated numerically in the previous chapter.

## 6.2 Linearisation of the variational problem

The efficient solution of the nonlinear variational problem using a Newton-Raphson approach relies on linearisation of the nonlinear problem. The linearisation of  $\delta \Pi^{\text{int}}$ , the LHS of equation (6.14), about the current configuration reads

$$\begin{aligned} \mathcal{D}_{\Delta \mathbf{u}^h} \delta \Pi^{\text{int}} = & \int_{\Omega} \delta \alpha_{\alpha\beta} \mathcal{D}_{\Delta \mathbf{u}^h} (N^{\alpha\beta}) d\Omega + \int_{\tilde{\Omega}} \delta \beta_{\alpha\beta}^* \mathcal{D}_{\Delta \mathbf{u}^h} (M^{\star\alpha\beta}) d\Omega \\ & + \int_{\tilde{\Omega}} \frac{\eta t^3}{12} \delta r_{\alpha\beta}^e c^{\alpha\beta\gamma\delta} \mathcal{D}_{\Delta \mathbf{u}^h} (r_{\gamma\delta}^e) d\Omega + \int_{\Omega} \mathcal{D}_{\Delta \mathbf{u}^h} (\delta \alpha_{\alpha\beta}) N^{\alpha\beta} d\Omega \\ & + \int_{\tilde{\Omega}} \mathcal{D}_{\Delta \mathbf{u}^h} (\delta \beta_{\alpha\beta}^*) M^{\star\alpha\beta} d\Omega, + \sum_e \int_{\tilde{\Omega}} \frac{\eta t^3}{12} \mathcal{D}_{\Delta \mathbf{u}^h} (\delta r_{\alpha\beta}^e) c^{\alpha\beta\gamma\delta} r_{\gamma\delta}^e d\Omega, \end{aligned} \quad (6.19)$$

where  $\mathcal{D}_{\Delta u^h}(\cdot)$  denotes the directional derivative with respect to the displacement. Linearisation of the membrane stress (equations (4.38) and (6.17)) and the variation of the membrane strain (equation (4.45)) is straightforward, and reads

$$\mathcal{D}_{\Delta u^h} N^{\alpha\beta} = \frac{t}{2} c^{\alpha\beta\gamma\delta} \left( \mathcal{D}_{\Delta u^h} \mathbf{u}_{,\alpha}^h \cdot \mathbf{a}_\beta + \mathbf{a}_\alpha \cdot \mathcal{D}_{\Delta u^h} \mathbf{u}_{,\beta}^h \right), \quad (6.20)$$

$$\mathcal{D}_{\Delta u^h} \delta \alpha_{\alpha\beta} = \frac{1}{2} \left( \delta \mathbf{a}_\alpha \cdot \mathcal{D}_{\Delta u^h} \mathbf{u}_{,\beta}^h + \mathcal{D}_{\Delta u^h} \mathbf{u}_{,\alpha}^h \cdot \delta \mathbf{a}_\beta \right). \quad (6.21)$$

Linearisation of the curvature and moment terms is more complicated, and reads

$$\begin{aligned} \mathcal{D}_{\Delta u^h} \delta \beta_{\alpha\beta}^* = & -\mathbf{A}_{\alpha,\beta} \cdot \mathcal{D}_{\Delta u^h} \delta \mathbf{a}_3 - \delta \mathbf{u}_{,\alpha\beta}^h \cdot \mathcal{D}_{\Delta u^h} \mathbf{a}_3 - \mathcal{D}_{\Delta u^h} \mathbf{u}_{,\alpha\beta}^h \cdot \delta \mathbf{a}_3 \\ & - \mathbf{u}_{,\alpha\beta}^h \cdot \mathcal{D}_{\Delta u^h} \delta \mathbf{a}_3 + \mathcal{D}_{\Delta u^h} \delta R_{\alpha\beta}, \end{aligned} \quad (6.22)$$

$$\mathcal{D}_{\Delta u^h} M^{\alpha\beta} = \frac{t^3}{12} c^{\alpha\beta\gamma\delta} \left( -\mathcal{D}_{\Delta u^h} \mathbf{u}_{,\gamma\delta}^h \cdot \mathbf{a}_3 - \mathbf{a}_{\gamma,\delta} \cdot \mathcal{D}_{\Delta u^h} \mathbf{a}_3 + \mathcal{D}_{\Delta u^h} R_{\gamma\delta}^g \right), \quad (6.23)$$

where

$$\begin{aligned} \mathcal{D}_{\Delta u^h} \mathbf{a}_3 = & \frac{1}{\sqrt{a}} \left( \mathcal{D}_{\Delta u^h} \mathbf{u}_{,1}^h \times \mathbf{a}_2 + \mathbf{a}_1 \times \mathcal{D}_{\Delta u^h} \mathbf{u}_{,2}^h \right) \\ & - \frac{\mathbf{a}_3}{\sqrt{a}} \left( \mathcal{D}_{\Delta u^h} \mathbf{u}_{,1}^h \cdot (\mathbf{a}_2 \times \mathbf{a}_3) + \mathcal{D}_{\Delta u^h} \mathbf{u}_{,2}^h \cdot (\mathbf{a}_3 \times \mathbf{a}_1) \right). \end{aligned} \quad (6.24)$$

Here, the complicated term relates to  $\delta \mathbf{a}_3$ . Its linearisation is given by

$$\begin{aligned} \mathcal{D}_{\Delta u^h} \delta \mathbf{a}_3 = & \frac{1}{\sqrt{a}} \left( \delta \mathbf{u}_{,1}^h \times \mathcal{D}_{\Delta u^h} \mathbf{u}_{,2}^h + \mathcal{D}_{\Delta u^h} \mathbf{u}_{,1}^h \times \delta \mathbf{u}_{,2}^h \right) \\ & - \frac{\mathcal{D}_{\Delta u^h} a}{2a^{3/2}} \left( \delta \mathbf{u}_{,1}^h \times \mathbf{a}_2 + \mathbf{a}_1 \times \delta \mathbf{u}_{,2}^h \right) \\ & + \left( \frac{\mathcal{D}_{\Delta u^h} a}{2a^{3/2}} \mathbf{a}_3 - \frac{\mathcal{D}_{\Delta u^h} \mathbf{a}_3}{\sqrt{a}} \right) \left\{ \delta \mathbf{u}_{,1}^h \cdot (\mathbf{a}_2 \times \mathbf{a}_3) + \delta \mathbf{u}_{,2}^h \cdot (\mathbf{a}_3 \times \mathbf{a}_1) \right\} \\ & - \frac{\mathbf{a}_3}{\sqrt{a}} \left( \delta \mathbf{u}_{,1}^h \cdot \left( \mathcal{D}_{\Delta u^h} \mathbf{u}_{,2}^h \times \mathbf{a}_3 + \mathbf{a}_2 \times \mathcal{D}_{\Delta u^h} \mathbf{a}_3 \right) \right) \\ & - \frac{\mathbf{a}_3}{\sqrt{a}} \left( \delta \mathbf{u}_{,2}^h \cdot \left( \mathcal{D}_{\Delta u^h} \mathbf{a}_3 \times \mathbf{a}_1 + \mathbf{a}_3 \times \mathcal{D}_{\Delta u^h} \mathbf{u}_{,1}^h \right) \right), \end{aligned} \quad (6.25)$$

where

$$\mathcal{D}_{\Delta u^h} a = 2\sqrt{a} \mathbf{a}_3 \cdot \left( \mathcal{D}_{\Delta u^h} \mathbf{u}_{,1}^h \times \mathbf{a}_2 + \mathbf{a}_1 \times \mathcal{D}_{\Delta u^h} \mathbf{u}_{,2}^h \right). \quad (6.26)$$

Linearised lifting operations for  $e \in \tilde{\Gamma}$  read

$$\begin{aligned} \int_{K^+ \cup K^-} \tau^{\alpha\beta} \mathcal{D}_{\Delta u^h} r_{\alpha\beta}^e d\Omega = & \int_e \langle \tau_{nn} \rangle \left[ \mathcal{D}_{\Delta u^h} \mathbf{u}_{,n}^h \cdot \mathbf{a}_3 \right] d\Gamma \\ & + \int_e \langle \tau_{nn} \rangle \left[ \mathbf{u}_{,n}^h \cdot \mathcal{D}_{\Delta u^h} \mathbf{a}_3 \right] d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h, \end{aligned} \quad (6.27)$$

$$\begin{aligned} \int_{K^+ \cup K^-} \tau^{\alpha\beta} \mathcal{D}_{\Delta u^h} \delta r_{\alpha\beta}^e d\Omega &= \int_e \langle \tau_{nn} \rangle \left[ \delta \mathbf{u}_{,n}^h \cdot \mathcal{D}_{\Delta u^h} \mathbf{a}_3 \right] d\Gamma \\ &+ \int_e \langle \tau_{nn} \rangle \left[ \mathbf{u}_{,n}^h \cdot \mathcal{D}_{\Delta u^h} \delta \mathbf{a}_3 \right] d\Gamma + \int_e \langle \tau_{nn} \rangle \left[ \mathcal{D}_{\Delta u^h} \mathbf{u}_{,n}^h \cdot \delta \mathbf{a}_3 \right] d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h, \end{aligned} \quad (6.28)$$

and for  $e \in \Gamma^\theta$  read

$$\begin{aligned} \int_{K^+} \tau^{\alpha\beta} \mathcal{D}_{\Delta u^h} r_{\alpha\beta}^e d\Omega &= \int_e \tau_{nn} \mathcal{D}_{\Delta u^h} \mathbf{u}_{,n}^h \cdot \mathbf{a}_3 d\Gamma \\ &+ \int_e \tau_{nn} \mathbf{u}_{,n}^h \cdot \mathcal{D}_{\Delta u^h} \mathbf{a}_3 d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h, \end{aligned} \quad (6.29)$$

$$\begin{aligned} \int_{K^+} \tau^{\alpha\beta} \mathcal{D}_{\Delta u^h} \delta r_{\alpha\beta}^e d\Omega &= \int_e \tau_{nn} \delta \mathbf{u}_{,n}^h \cdot \mathcal{D}_{\Delta u^h} \mathbf{a}_3 d\Gamma + \int_e \tau_{nn} \mathbf{u}_{,n}^h \cdot \mathcal{D}_{\Delta u^h} \delta \mathbf{a}_3 d\Gamma \\ &+ \int_e \tau_{nn} \mathcal{D}_{\Delta u^h} \mathbf{u}_{,n}^h \cdot \delta \mathbf{a}_3 d\Gamma \quad \forall \tau^{\alpha\beta} \in \mathcal{R}^h, \end{aligned} \quad (6.30)$$

which leads to expressions for  $\mathcal{D}_{\Delta u^h} R_{\alpha\beta}$  and  $\mathcal{D}_{\Delta u^h} \delta R_{\alpha\beta}$ ,

$$\mathcal{D}_{\Delta u^h} R_{\alpha\beta} = \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} \mathcal{D}_{\Delta u^h} r_{\alpha\beta}^e, \quad (6.31)$$

$$\mathcal{D}_{\Delta u^h} \delta R_{\alpha\beta} = \sum_{e \in \tilde{\Gamma} \cup \Gamma^\theta} \mathcal{D}_{\Delta u^h} \delta r_{\alpha\beta}^e. \quad (6.32)$$

The force and moment boundary values appearing in the expression of the external virtual work (6.16) may depend on the current configuration of the shell, so if a nonlinear force and moment are prescribed on part of the boundary, these terms should be linearised also.

When the fixed orthonormal basis  $(\mathbf{E}^1, \mathbf{E}^2, \mathbf{E}^3)$  is used to express displacement components, the increment displacement  $\mathcal{D}_{\Delta u^h}$  and their directional derivatives are expressed by

$$\mathcal{D}_{\Delta u^h} \mathbf{u}^h = \Delta u_i^h \mathbf{E}^i, \quad (6.33)$$

$$\mathcal{D}_{\Delta u^h} \mathbf{u}_{,\alpha}^h = \Delta u_{i,\alpha}^h \mathbf{E}^i, \quad (6.34)$$

$$\mathcal{D}_{\Delta u^h} \mathbf{u}_{,\alpha\beta}^h = \Delta u_{i,\alpha\beta}^h \mathbf{E}^i, \quad (6.35)$$

$$\mathcal{D}_{\Delta u^h} \mathbf{u}_{,n}^h = \Delta u_{i,\alpha}^h n^\alpha \Xi \mathbf{E}^i. \quad (6.36)$$

The terms involving to the variation displacement  $\delta \mathbf{u}^h$  can be expressed similarly. When the curvilinear basis  $(\mathbf{A}^1, \mathbf{A}^2, \mathbf{A}^3)$  is used, expressions of the above quantities become more complicated as they involve covariant derivative components of the local basis vectors. This approach is not considered here.

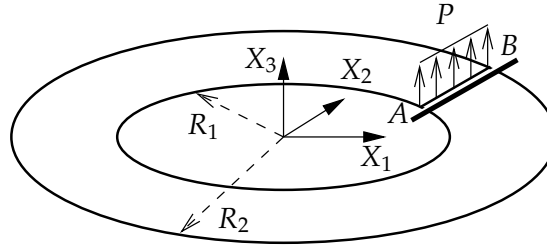


Figure 6.1: Slit annular plate problem.

## 6.3 Numerical examples

All examples use cubic triangular elements, although the method is applicable for any shape element. Gauss quadrature is applied, and for quadratic elements six points are used for membrane terms, three points are used for bending terms and two points are used on element edges. For cubic elements, twelve points are used for membrane terms, six points are used for bending terms and three points are used on element edges. Unstructured triangular meshes are used, and the order  $l$  of the lifting functions is chosen such that  $l = k - 2$ , where  $k$  is the polynomial order of the displacement components. Note that various combinations of  $k$  and  $l$  have been investigated for thin plates previously. A penalty value of  $\eta = 5$  is adopted in all tests. The shell geometry is interpolated using the finite element basis functions, leading to a  $C^0$  mid-surface surface.

### 6.3.1 Slit annular plate under line force

The problem has been considered in a number of works, including those by Bařar and Ding (1992), Buechter and Ramm (1992), Barut et al. (1997), Sansour and Kollmann (2000) and Sze et al. (2004). The geometry of the slit annular plate is shown in Figure 6.1 with the inner radius  $R_1 = 6$  m, outer radius  $R_2 = 10$  m, thickness  $t = 0.03$  m, Young's modulus  $E = 21 \times 10^6$  N/m<sup>2</sup> and Poisson's ratio  $\nu = 0$ . One end of the slit of the plate is clamped and the other end is loaded by an uniform force of the maximum magnitude  $P_{\max} = 0.8$  N/m.

The mesh is composed of 642 elements. The undeformed and deformed slit annular plate is shown in Figure 6.2. The deformation is not magnified, illustrating the large displacement. The load–displacement response at the points A and B for cubic elements is shown in Figure 6.3. In this figure, the reference solution was given in Sze et al. (2004) using 800 elements with four nodes per element



Figure 6.2: Undeformed and deformed slit annular plate.

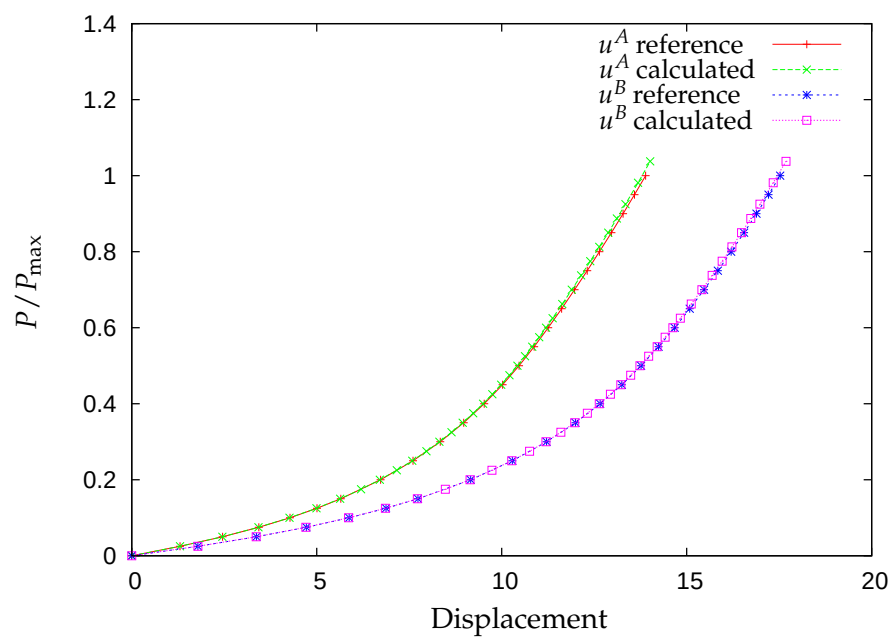


Figure 6.3: Slit annular plate under line force: Load–vertical displacement response with cubic elements.

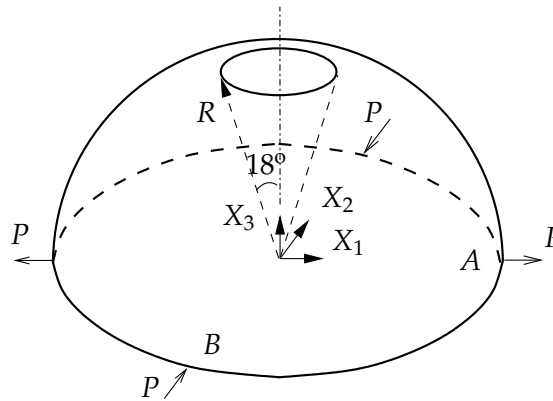


Figure 6.4: Pinched hemispherical shell problem.

and six degrees-of-freedom per node (namely S4R elements). It is shown that the computed result matches well the reference solution.

### 6.3.2 Hemisphere under radial forces

This is one of the most demanding tests to investigate the ability of the formulation for simulating an inextensional bending and rigid-body motions. This test has been considered in many works, including Simo et al. (1990), Saleeb et al. (1990), Buechter and Ramm (1992), Sansour and Kollmann (2000) and Sze et al. (2004). The geometry of the hemisphere with a  $18^\circ$  circular hole at the top is shown in Figure 6.4, where radius  $R = 10$  m, thickness  $t = 0.04$  m, Young's modulus  $E = 6.825 \times 10^7$  N/m<sup>2</sup> and Poisson's ratio  $\nu = 0.3$ . Two pairs of opposite forces are of the maximum magnitude  $P_{\max} = 4 \times 10^2$  N.

The undeformed and deformed hemisphere is shown in Figure 6.5. Here, the mesh is composed of 1444 elements. The deformation is not magnified, illustrating the large displacement. The load–displacement response at the loaded points for cubic elements is shown in Figure 6.6. Note that since it is geometrically non-linear, displacement values at the points A and B are different from each other. In Figure 6.6, the reference solution using 1024 S4R elements was given in Sze et al. (2004). Again, it is shown that the computed result matches well the reference solution and membrane locking is not observed using cubic elements.

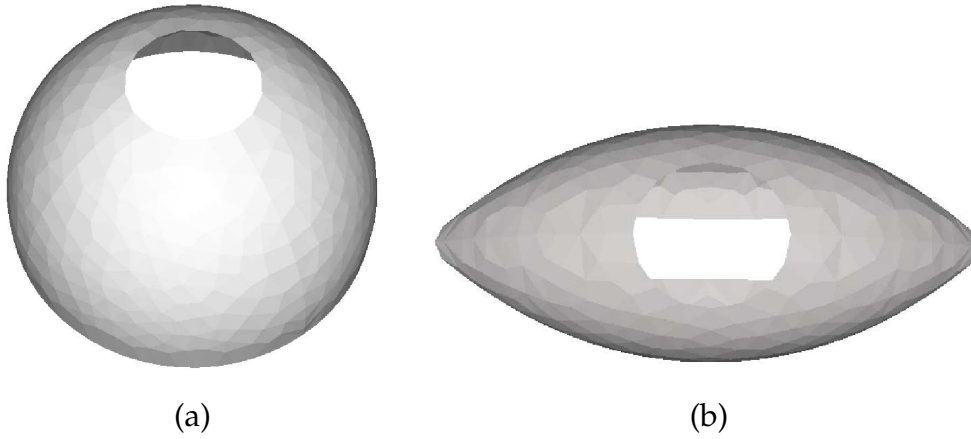


Figure 6.5: Hemisphere under in (a) undeformed and (b) deformed configurations. The displacement is not magnified.

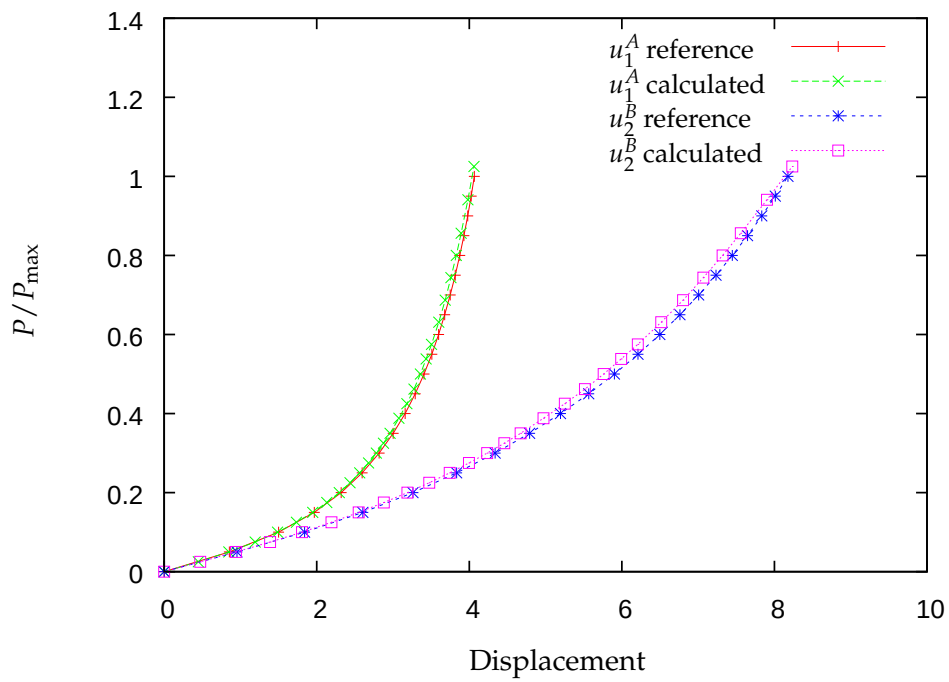


Figure 6.6: Hemisphere under radial forces: Load–displacement response with cubic elements.

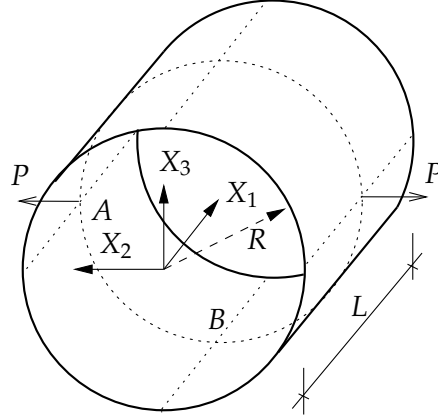


Figure 6.7: Stretched free-end cylinder problem.



Figure 6.8: Nonlinear stretched cylinder problem in (a) undeformed and (b) deformed configurations. The displacement is not magnified.

### 6.3.3 Stretched free-end cylinder

We consider a stretched cylinder benchmark problem from Gruttmann et al. (1989), Park et al. (1995), Sansour and Kollmann (2000) and Sze et al. (2004). For the cylinder shown in Figure 6.7, length  $L = 10.35$  m, radius  $R = 4.953$  m, thickness  $t = 0.094$  m, Young's modulus  $E = 10.5 \times 10^6$  N/m<sup>2</sup> and Poisson's ratio  $\nu = 0.3125$ . The ends of the cylinder are free. The applied load for this problems is outward relative to the shell surface, where  $P_{\max} = 4 \times 10^4$  N.

The mesh is composed of 1256 elements, while the reference solution (Sze et al., 2004) uses 3072 S4R elements. The undeformed and deformed cylinder is shown in Figure 6.8. The deformation is not magnified. The load–displacement response at points A and B for cubic elements is shown in Figure 6.9. Again, the lifting formulation matches the reference solution well. Nonlinear benchmarks

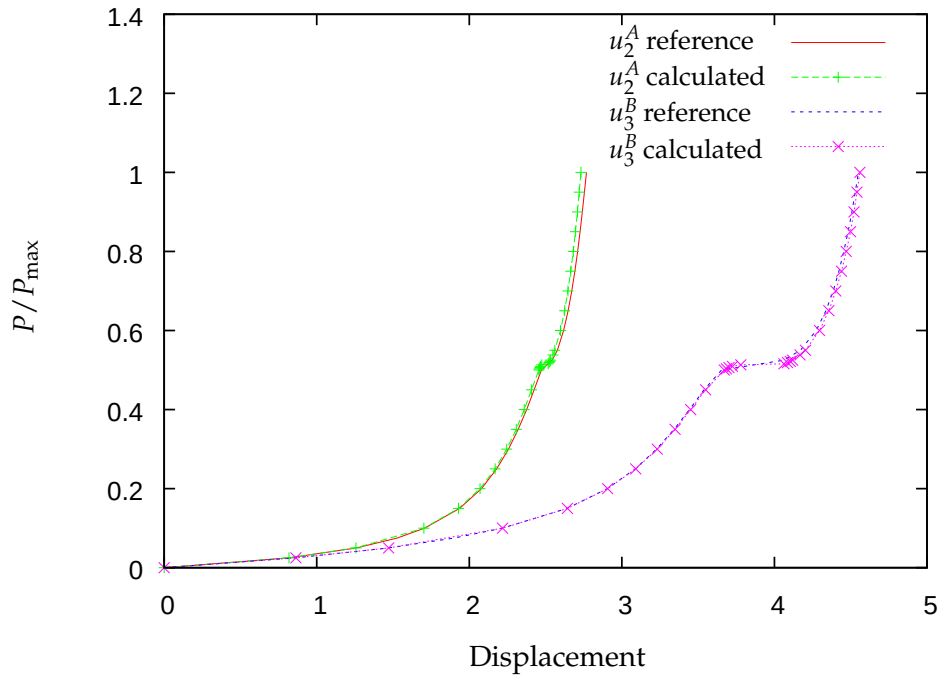


Figure 6.9: Nonlinear stretched cylinder problem with free ends: Load–displacement response with cubic elements.

show that cubic elements in particular perform very well under a variety of loading conditions. Membrane locking has not been observed and convergence is rapid when using cubic basis functions.

# Chapter 7

## Conclusions

The simulation of thin bending problems remains a challenging task. Difficulties are primarily related to the required  $C^1$  continuity of the displacement field. In this work, discontinuous Galerkin method have been exploited to solve this issue. Since the method is built on discontinuous piece-wise functions to construct an approximation solution, it allows to relax continuity requirements on inter-element boundaries.

Lifting-type  $C^0$  Galerkin formulations for thin bending problems possess attractive features. The formulations utilise conventional  $C^0$  Lagrange basis functions for displacement unknowns only. Lifting operations transform jumps in the normal derivative across element boundaries into a field defined on element interiors. *A priori* analysis for thin plate bending problems, has shown that the proposed formulations are consistent. Importantly, the penalty parameter is concretely quantifiable therefore stability of the formulations is simply guaranteed. Error estimate analysis has shown that the formulations converge optimally in energy-like norms for basis functions of polynomial orders higher than or equal to two. In the  $L^2$  norm, convergence of the formulations is optimal for polynomial orders greater than or equal to three. The eigenvalue problems associated with plate buckling and vibration have also been investigated numerically. The results indicate that the penalty parameter has little influence on the obtained eigenvalues.

The successful application of the lifting approach for thin shells is important as it has shown the capability of the approach to deal with difficult geometries. A series of testing geometrically linear and nonlinear benchmarks show that cubic elements in particular perform well under a variety of loading conditions.

As was the case for plate bending, shear locking is completely avoided as a thin shell theory has been addressed directly. Membrane locking has not been observed when using cubic elements. Unfortunately, there exists membrane locking phenomena when using quadratic elements. This results in a slow convergence in some obstacle benchmark tests such as the rigid-end cylinder and the hemisphere. For geometrically nonlinear problems, the absence of rotation degrees-of-freedom avoids complications in implementation and exact linearisation of the approach is tractable. The proposed formulation is general in the sense that it can be used for curved elements with a non-smooth shell mid-surface.

More work can be done to investigate the presented formulations, improve the performance of finite elements and apply to more interesting problems. *A priori* analyses on convergence behavior of linear and nonlinear thin shells should be examined more elaborately. Especially, the cause of membrane locking phenomena for low-order finite elements needs to be explored. Regarding implementation aspects, the use of quadrilateral elements may offer some advantages. Note that the lifting formulations are general for all type of finite elements. Naturally, buckling and vibration problems are often involved in a geometrically nonlinear regime of thin shells and need to be investigated.

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# List of Symbols

## Roman Symbols

|                      |                                                                                          |
|----------------------|------------------------------------------------------------------------------------------|
| $a_{\alpha\beta}$    | covariant components of metric tensor on deformed mid-surface                            |
| $a^{\alpha\beta}$    | contravariant components of metric tensor on deformed mid-surface                        |
| $\mathbf{a}_i$       | covariant basis vector on deformed mid-surface                                           |
| $\mathbf{a}^i$       | contravariant basis vector on deformed mid-surface                                       |
| $A$                  | determinant of matrix $(A_{\alpha\beta})$                                                |
| $A_{\alpha\beta}$    | covariant components of metric tensor in reference mid-surface                           |
| $A^{\alpha\beta}$    | contravariant components of metric tensor in reference mid-surface                       |
| $\mathbf{A}_i$       | covariant basis vector on reference mid-surface                                          |
| $\mathbf{A}^i$       | contravariant basis vector on reference mid-surface                                      |
| $B_{\alpha\beta}$    | covariant components of second fundamental form in reference configuration               |
| $B_{\alpha}^{\beta}$ | covariant-contravariant components of second fundamental form in reference configuration |
| $B(\cdot, \cdot)$    | bilinear form                                                                            |
| $c_{ijkl}, c^{ijkl}$ | components of elastic constitutive tensor                                                |
| $d$                  | space dimension                                                                          |
| $d_{ij}$             | stiffness tensor on inter-element edges                                                  |
| $e$                  | sharing edge of neighbouring elements in a partition                                     |
| $\mathbf{E}^i$       | fixed orthonormal basis vector                                                           |
| $E$                  | Young's modulus                                                                          |
| $\mathbf{E}$         | Green-Lagrange strain tensor                                                             |
| $\mathbb{E}^d$       | Euclidean physical space of dimension $d$                                                |
| $\mathbf{f}$         | body force vector                                                                        |
| $\mathbf{F}$         | resultant force vector on mid-surface                                                    |
| $g_{ij}$             | covariant components of metric tensor in deformed configuration                          |
| $g^{ij}$             | contravariant components of metric tensor in deformed configuration                      |
| $\mathbf{g}_i$       | covariant basis vector in deformed body                                                  |
| $\mathbf{g}^i$       | contravariant basis vector in deformed body                                              |

|                                              |                                                                      |
|----------------------------------------------|----------------------------------------------------------------------|
| $G_{ij}$                                     | covariant components of metric tensor in reference configuration     |
| $G^{ij}$                                     | contravariant components of metric tensor in reference configuration |
| $\mathbf{G}_i$                               | covariant basis vector in reference body                             |
| $\mathbf{G}^i$                               | contravariant basis vector in reference body                         |
| $g^u, \mathbf{g}^u$                          | displacement boundary value                                          |
| $g^\theta$                                   | normal rotation boundary value                                       |
| $h$                                          | mesh size parameter                                                  |
| $\mathbf{h}, \mathbf{H}$                     | force boundary value                                                 |
| $H^m(\Omega)$                                | Sobolev space of order $m$ defined in $\Omega$                       |
| $K$                                          | open bounded finite element                                          |
| $\partial K$                                 | finite element boundary                                              |
| $L(\cdot)$                                   | linear form                                                          |
| $L^2(\Omega)$                                | square integrable space defined in $\Omega$                          |
| $m_{\alpha\beta}, M^{\alpha\beta}$           | moment tensors                                                       |
| $\hat{m}_{\alpha\beta}$                      | numerical moment tensor                                              |
| $m_{\alpha\beta}^*, M_{\star}^{\alpha\beta}$ | total moment tensors                                                 |
| $M$                                          | normal moment boundary value                                         |
| $M_t$                                        | torsion moment boundary value                                        |
| $\mathbf{n}$                                 | unit outward normal vector to $\partial\Omega$                       |
| $\hat{\mathbf{n}}$                           | sharing unit normal vector to $e$                                    |
| $N$                                          | number of finite elements                                            |
| $N_e$                                        | number of edges belonging to an element                              |
| $P^k(K)$                                     | polynomial function space of order $k$ on $K$                        |
| $Q$                                          | shear-force boundary value                                           |
| $r_{ij}^e, r_{ij}^{e,g}$                     | local lifting strain tensors                                         |
| $R_{ij}, R_{ij}^g$                           | lifting strain tensors                                               |
| $\mathbb{R}^d$                               | real number mathematical space of dimension $d$                      |
| $s_i$                                        | area coordinates                                                     |
| $\mathbf{s}$                                 | unit tangent vector to $\partial\Omega$                              |
| $t$                                          | thickness                                                            |
| $u, \mathbf{u}$                              | trial displacement solution                                          |
| $\hat{\mathbf{u}}$                           | numerical displacement solution                                      |
| $w, \mathbf{w}$                              | test displacement solution                                           |
| $V^h$                                        | finite element space of trial functions                              |
| $V_{\star}^h$                                | broken Sobolev space                                                 |
| $W^h$                                        | finite element space of test functions                               |
| $\mathbf{x}$                                 | position vector in deformed configuration                            |
| $\mathbf{X}$                                 | position vector in reference configuration                           |

## Greek symbols

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| $\alpha_{\alpha\beta}$                  | membrane strain tensor                                            |
| $\beta_{\alpha\beta}$                   | bending strain tensor                                             |
| $\mathcal{B}$                           | boundary surface of reference shell body                          |
| $\mathcal{B}_t$                         | top surface of reference shell body                               |
| $\mathcal{B}_b$                         | bottom surface of reference shell body                            |
| $\mathcal{B}_s$                         | side surface of reference shell body                              |
| $\mathcal{B}_h$                         | force-boundary surface of reference shell body                    |
| $\mathcal{C}_{\alpha\beta\gamma\delta}$ | elastic constitutive tensor for plate bending                     |
| $\delta_{ij}, \delta_i^j$               | Kronecker symbol                                                  |
| $\epsilon_{ij}$                         | strain tensor                                                     |
| $\eta$                                  | penalty parameter                                                 |
| $\Gamma$                                | union of all finite element boundaries in reference configuration |
| $\tilde{\Gamma}$                        | interior boundary in reference configuration                      |
| $\Gamma^h, \Gamma^H$                    | force boundary in reference configuration                         |
| $\Gamma^M$                              | moment boundary in reference configuration                        |
| $\Gamma^u$                              | displacement boundary                                             |
| $\Gamma^\theta$                         | rotation boundary                                                 |
| $\Gamma_{\beta\gamma}^\alpha$           | surface Christoffel symbols                                       |
| $\lambda, \mu$                          | Lam'e coefficients                                                |
| $\nu$                                   | Poisson's ratio                                                   |
| $\omega$                                | open bounded two-dimensional parametric domain                    |
| $\bar{\omega}$                          | closure of domain $\omega$                                        |
| $\Omega$                                | shell mid-surface in reference configuration                      |
| $\Omega'$                               | shell mid-surface in deformed configuration                       |
| $\bar{\Omega}$                          | closure of domain $\Omega$                                        |
| $\tilde{\Omega}$                        | interior domain in reference configuration                        |
| $\partial\Omega$                        | exterior boundary in reference configuration                      |
| $\Phi$                                  | geometric mapping for reference mid-surface                       |
| $\phi$                                  | geometric mapping for deformed mid-surface                        |
| $\mathcal{P}^h$                         | finite element partition                                          |
| $\mathcal{R}^h$                         | finite element space of lifting functions                         |
| $\sigma_{ij}, \Sigma_{ij}$              | stress tensors                                                    |
| $\hat{\sigma}_{ij}$                     | numerical stress tensor                                           |
| $\Sigma^h$                              | finite element space of stress tensors                            |

|               |                              |
|---------------|------------------------------|
| $\tau_{ij}$   | trial strain tensor          |
| $\mathcal{V}$ | three-dimensional shell body |
| $\xi^i$       | curvilinear coordinates      |
| $\Xi$         | sign function                |

## Operations and abbreviations

|                                                          |                                                    |
|----------------------------------------------------------|----------------------------------------------------|
| $\langle \varphi \rangle _{e \in \tilde{\Gamma}}$        | $\frac{1}{2} (\varphi^+ + \varphi^-)$              |
| $\langle \varphi \rangle _{e \in \partial \Gamma}$       | $\varphi$                                          |
| $\llbracket \varphi \rrbracket _{e \in \tilde{\Gamma}}$  | $\varphi^+ - \varphi^-$                            |
| $\llbracket \varphi \rrbracket _{e \in \partial \Gamma}$ | $\varphi$                                          |
| $\varphi_{,n}$                                           | $\varphi_{,\alpha} n^\alpha \Xi$                   |
| $\varphi_n$                                              | $\varphi_\alpha n^\alpha \Xi$                      |
| $\varphi_{nn}$                                           | $\varphi_{\alpha\beta} n^\alpha n^\beta$           |
| $\varphi_{(ij)}$                                         | $\frac{1}{2} (\varphi_{ij} + \varphi_{ji})$        |
| $\boldsymbol{\varphi} \otimes \boldsymbol{\phi}$         | $\varphi_i \phi_j$                                 |
| $\boldsymbol{\varphi} \cdot \boldsymbol{\phi}$           | $\varphi_i \phi_i$                                 |
| $\nabla \varphi$                                         | $\varphi_{,i}$                                     |
| $\nabla \boldsymbol{\varphi}$                            | $\varphi_{i,j}$                                    |
| $\varphi^h$                                              | finite element approximation of quantity $\varphi$ |
| $\ \varphi\ _{m,\Omega}$                                 | Sobolev norm                                       |
| $ \varphi _{m,\Omega}$                                   | Sobolev semi-norm                                  |
| $   \varphi   $                                          | mesh dependent energy-like norm                    |
| LHS                                                      | left-hand side                                     |
| RHS                                                      | right-hand side                                    |

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# Summary

A structural thin bending problem is essentially associated with a fourth-order partial differential equation. Within the finite element framework, the numerical solution of thin bending problems demands the use of  $C^1$  continuous shape functions. Elements using these functions are challenging and difficult to construct.

A particular discontinuous Galerkin method has been used to deal with thin bending problems. It exploits standard Lagrange finite element basis functions with displacement degrees-of-freedom only. The method relies on a lifting operation to transform jumps in the normal derivative across element boundaries to a field defined on element interiors. By introducing special integrals over element boundaries, continuity requirements associated with thin bending problems are met weakly while consistency of the Galerkin problem is preserved and the stability of the formulation is controlled.

The approach is formulated for a range of bending problems. Firstly, a finite element formulation for linear Kirchhoff plates is proposed. The formulation is analysed in terms of stability and convergence. *A priori* error estimates are supported by a range of numerical examples, including static plate bending, plate buckling and vibration. The method has also been developed for geometrically linear and nonlinear thin shell models. For geometrically nonlinear problems, an exact linearisation of the proposed formulation is presented. Using the approach, a non-smooth shell geometry can be dealt with. The performance of the approach is demonstrated via a range of numerical benchmark tests for both geometrically linear and nonlinear thin shells.

Based on the obtained results, it can be concluded that the proposed formulation is robust, accurate and relatively simple. As a thin shell theory has been addressed directly, shear locking is completely avoided. Although membrane locking exists, it can be alleviated when high-order basis functions are used. Cubic elements perform particularly well in various challenging benchmark tests.

The method is general in sense that it does not rely on a particular element type or shape.

# Samenvatting

Constructieve problemen aangaande de beschrijving van buiging in dunwandige constructies zijn gerelateerd aan een vierde orde partiële differentiaalvergelijking. Voor de numerieke oplossing van dunwandige constructies met behulp van de eindige-elementenmethode is het gebruik van  $C^1$  continue vormfuncties vereist. Het formuleren van elementen met zulke functies is uitdagend en moeilijk.

Een aangepaste discontinue Galerkin methode wordt gebruikt om dunwandige constructies met behulp te behandelen. Er wordt gebruik gemaakt van standaard Lagrange basisfuncties met alleen verplaatsingsvrijheidsgraden. De methode omvat een *lifting* bewerking om discontinuïteiten in de afgeleide in normaalrichting over elementgrenzen te transformeren tot een veld gedefinieerd over de binnenzijde van het element. Door de introductie van aparte integralen over elementgrenzen, worden de continuïteitseisen inherent aan dunwandige constructies met behulp zwak voldaan, terwijl de consistentie van het Galerkin probleem behouden blijft en de stabiliteit van de formulering beheerst wordt.

De benadering wordt opgesteld voor verschillende buigproblemen. Ten eerste wordt een eindige-elementenformulering voor lineaire Kirchhoff-platen voorgesteld. Deze formulering wordt geanalyseerd betreffende stabiliteit en convergentie. A priori schattingen van de fout worden ondersteund door verschillende numerieke voorbeelden, waaronder statische buiging van platen, knik van platen en trillingen. Verder wordt de methode ontwikkeld voor geometrisch lineaire en niet-lineaire modellen voor dunne schalen. Voor geometrisch niet-lineaire problemen wordt een exacte linearisatie van de voorgestelde formulering gepresenteerd. Deze benadering is geschikt voor schalen met geknikte geometrie. De mogelijkheden van de benadering worden getoond met verschillende numerieke benchmark testen voor zowel geometrisch lineaire als niet-lineaire dunne schalen.

Op basis van de verkregen resultaten kan geconcludeerd worden dat de voorgestelde formulering robuust, nauwkeurig en betrekkelijk eenvoudig is. Daar een theorie voor dunne schalen direct is toegepast, wordt *shear locking* volledig vermeden. Hoewel *membrane locking* wel voorkomt, kan dit verminderd worden door het gebruik van hogeregraads basisfuncties. Derdegraads elementen presteren bijzonder goed in verschillende veeleisende benchmark testen. De methode is generiek in de zin dat deze niet afhankelijk is van type of vorm van de elementen.

# Propositions

## *Stellingen*

1. The shorter, the simpler.

*Hoe korter, hoe eenvoudiger.*

2. There are three important ingredients for research: reading, wondering and trying.

*Er zijn drie belangrijke ingrediënten voor onderzoek: lezen, verwonderen en proberen.*

3. One of the most important questions a research student must ask him- or herself is: "What should I do next?".

*Eén van de belangrijkste vragen die een onderzoeksstudent zichzelf moet stellen is: "Wat moet ik hierna doen?".*

4. A westerner often prefers to decompose a problem into sub-problems, while an oriental often prefers to compose sub-problems into a problem. A researcher needs to become both of them.

*De Westerse mens geeft er vaak de voorkeur aan een probleem in deelproblemen te verdelen, terwijl de Oosterse vaak liever een probleem uit deelproblemen samenstelt. Een onderzoeker dient beide te worden.*

5. A paradox: discontinuous Galerkin methods are suitable for problems which require a high-order of continuity of the solution.

*Een paradox: discontinue Galerkin methoden zijn geschikt voor problemen die een hoge orde continuïteit van de oplossing vereisen.*

6. "You can not get something for nothing!", said the Law of Conservation of

Energy. A proposed method will therefore always have disadvantages, but these are tolerable if understood.

*“Het is onmogelijk iets voor niets te krijgen!”, zegt de Wet van Behoud van Energie. Een voorgestelde methode zal daarom altijd nadelen hebben, maar deze zijn aanvaardbaar indien begrepen.*

7. Computers can do what you tell them and only can do what you tell them. From them, a researcher in computational mechanics is empowered and exposed to risk.

*Computers kunnen doen wat je ze zegt en slechts dat wat je ze zegt. Aldus wordt een onderzoeker in numerieke mechanica bekrachtigd en aan risico blootgesteld.*

8. Beneath a good result is a correct mathematical foundation.

*Onder een goed resultaat ligt een correct wiskundig fundament.*

9. Conflicts between ideas in a good research environment will always exist and are necessary.

*Conflicten tussen ideeën zullen in een goede onderzoeksomgeving altijd bestaan en zijn noodzakelijk.*

10. Happiness and sadness are coupled elements. Without one, the other no longer exists.

*Geluk en verdriet zijn gekoppelde elementen. Zonder het een, bestaat het ander niet meer.*

**These propositions are considered opposable and defensible and as such have been approved by the supervisor Prof. dr. ir. L. J. Sluys and the adjunct supervisor Dr. G. N. Wells.**

*Deze stellingen worden opponeerbaar en verdedigbaar geacht en zijn als zodanig goedgekeurd door de promotor Prof. dr. ir. L. J. Sluys en de toegevoegd promotor Dr. G. N. Wells.*

# Curriculum Vitae

|                       |                                                                                                                                        |
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